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Paper Machine Auditing

Many paper machine clothing manufacturers offer services to paper companies related to operation and performance of fabrics. The main purposes of a paper machine audit are to gather and analyze operational data that may be used by the paper company to benchmark operations, troubleshoot operational problems and find opportunities to improve machine performance. A typical paper machine survey may take a few days and usually consists of several stages as explained below (Figure 5.1). Examples from actual measurements are given throughout the chapter to demonstrate the type of information that can be gathered from paper machine auditing. It should be emphasized that paper machine auditing is machine condition specific and therefore each audit may be different than the other. Grade structure may also affect the audit results.

5.1 Stock Approach and Pulsation Studies

A pulsation study is performed to evaluate the stability and performance of stock approach piping and rotating elements. This evaluation is then compared to the on-line basis weight signal to determine if any approach piping instability is detrimental to the overall basis weight.

The more accurate and precise the information provided, the more accurate the pulsation

study will be. To insure accurate results, several key pressure tap locations must be provided. These locations must be available before a detailed pulsation study can be performed.

- before and after cleaner supply pump
- before and after headbox supply pump
- before and after pressure screens
- before and after attenuator
- headbox backside and front side
- headbox recirculation
- couch vacuum

It is possible to measure two locations from one tap, such as after headbox supply pump and before pressure screens.

Figure 5.2 shows pressure tap locations. Taps should be located at least one pipe diameter away from any elbows, pump, screens, valves or other pipe line obstructions if possible and should have a minimum of 12–18 inches of clearance beyond the end of the valve. The preferable tap size is 1 inch NPT; however 1/2 or 3/4 inch is also acceptable (Figure 5.3). Larger diameter taps can also be utilized if reduced to 1 inch.

Pump and screen speeds as well as any gearbox ratios will be needed during the analysis. The number of vanes on the pump impellers and the number of foils on the screen rotors will also be needed. It would also be helpful to know if either the pump or screen design is specified as low pulse.

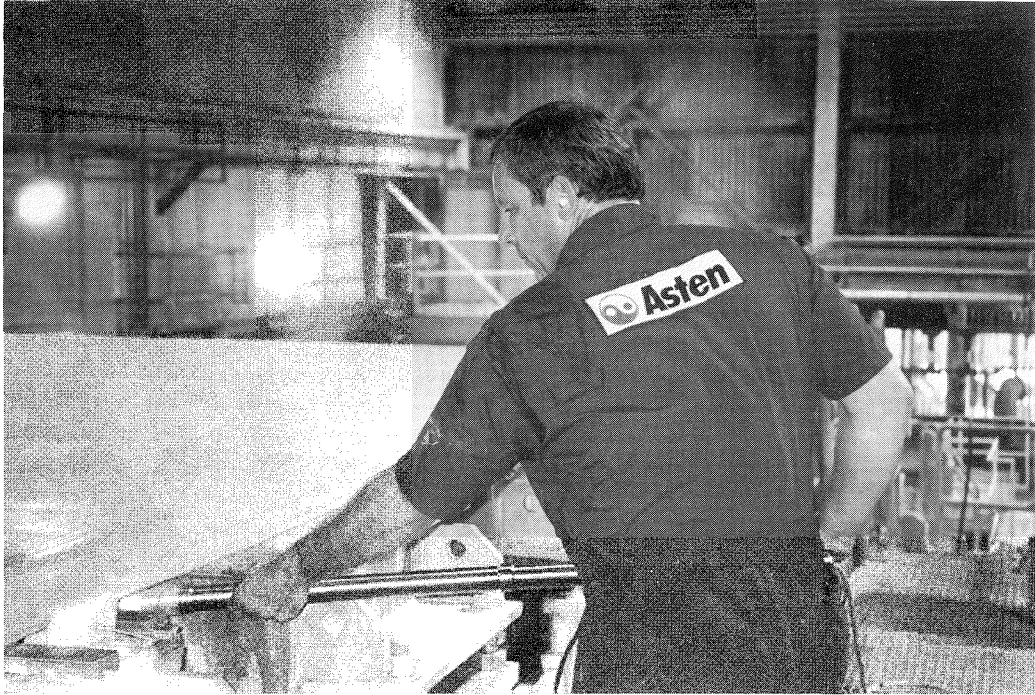


FIGURE 5.1. A service engineer conducting wet-end survey.

Access to the raw analog basis weight signal is required during the study. This signal is acquired before the system computer and will involve no proprietary information. The signal is interfaced with an instrument that has several gig ohm input impedance and will not affect the measurement signal.

Using a real time Fast Fourier Transform (FFT) analyzer and other special instrumentation, the following are analyzed during stock approach and pulsation survey.

- high frequency analysis: 1–200 cycles per second
- low frequency analysis: DC—2 cycles per second
- thin stock consistency variation
- cross correlation between stock approach pulsation/vibration and consistency versus basis weight are checked for both frequency ranges
- fan pump and pressure screen speed stability
- fan pump and pressure screen mechanical performance evaluation
- head box vibration
- accepted piping practice recommendations
- attenuator performance evaluation (where applicable)
- stream flow valve position control (where applicable)

Pulsation studies are applicable to any paper grade on a particular paper machine. However, it is recommended that a critical paper grade be selected for the study on that machine.

Figure 5.4 shows the low frequency signal correlation between the basis weight backside, fan pump in, screen #1 out, screen #2 out, screen inlet and fan pump out locations of a typical stock approach system shown in the figure. These signals are conditioned through a 0.25 Hz low pass filter and recorded over an eight minute running time. The information

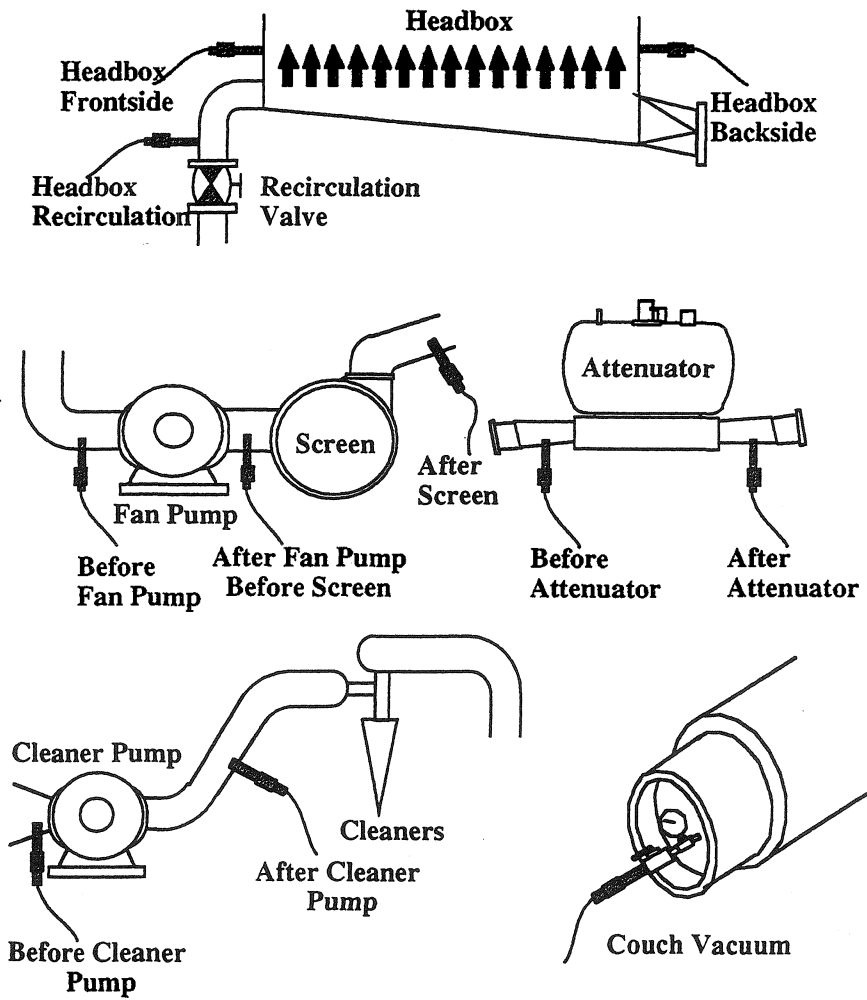


FIGURE 5.2. Pressure tap locations.

on the left side of the graphs presents the mean value and engineering units and the information on the right side of the graphs presents the peak to peak information in the same engineering units. All of the pressure signal scales are presented on the same scale of ± 7 inches of H_2O with the exception of the fan pump out. The scale on the fan pump out is ± 15 inches of H_2O .

Figure 5.5 shows the raw basis weight front side spectra and the synchronous time averaged basis weight front side spectra averaged against the pressure screens and fan pump for the system shown in Figure 5.4. Table 5.1 lists the measured pressure levels at different locations for the same stock approach system.

5.2 Forming Section Analysis (Wet-End Performance)

The purpose of this audit is to evaluate wet-end performance to highlight any conditions

that may be detrimental to sheet formation, water removal or clothing performance. Headbox delivery, table activity, trim squirt efficiency and couch sheet release are among the items that are inspected.

Utilizing a Doppler laser sensing device, it is possible to measure the headbox jet velocity readings across the profile of the slice to highlight variations that may be contributing to fiber polar angle anomalies and hence promote curl, formation and profile concerns.

Using a gamma gauge, data are collected to determine table drainage profile and wet-end water balance. Other tests may be conducted as needed to inspect fabric drive loads and fabric wear.

High speed video camera and/or still photography are used to photograph the following:

- headbox/slice delivery to check for bubbles or flow irregularities

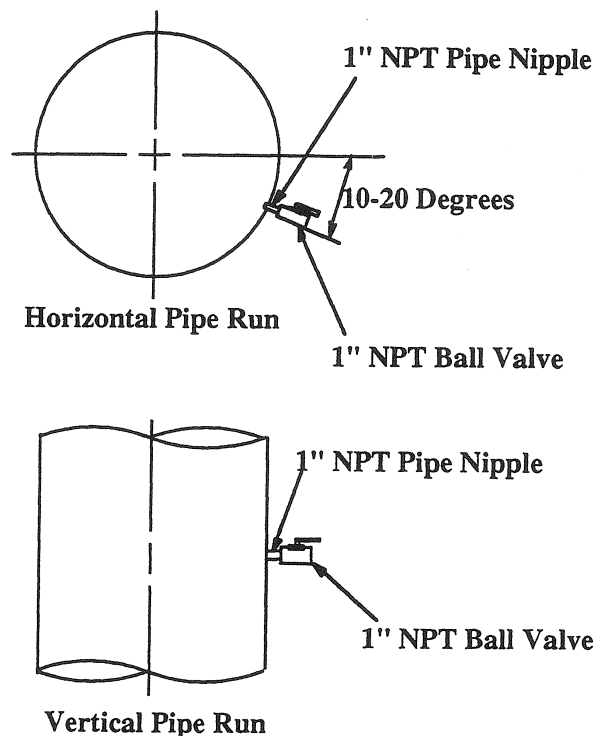


FIGURE 5.3. Specifications for pulsation pressure taps.

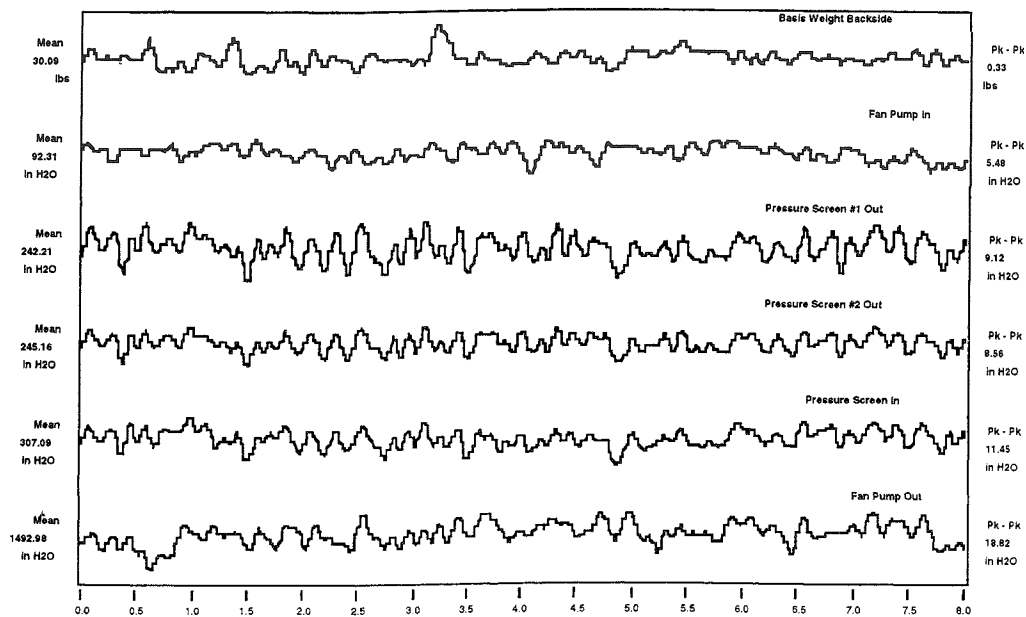
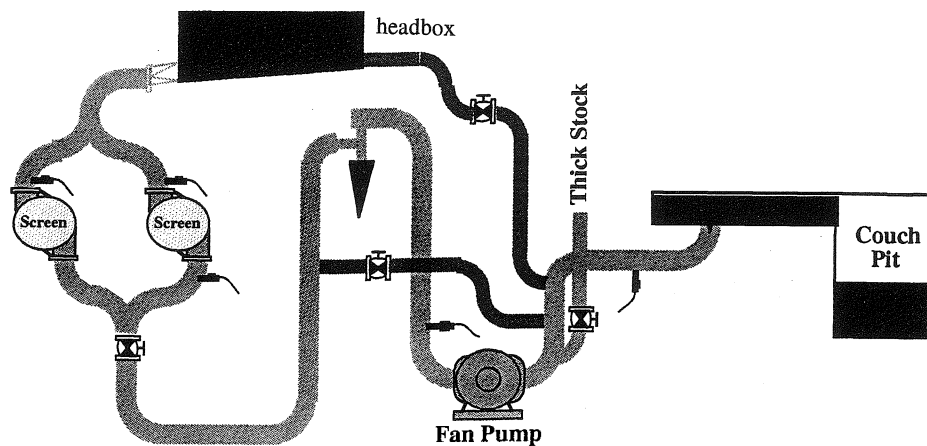


FIGURE 5.4. Low frequency signal correlation of a typical stock approach system.

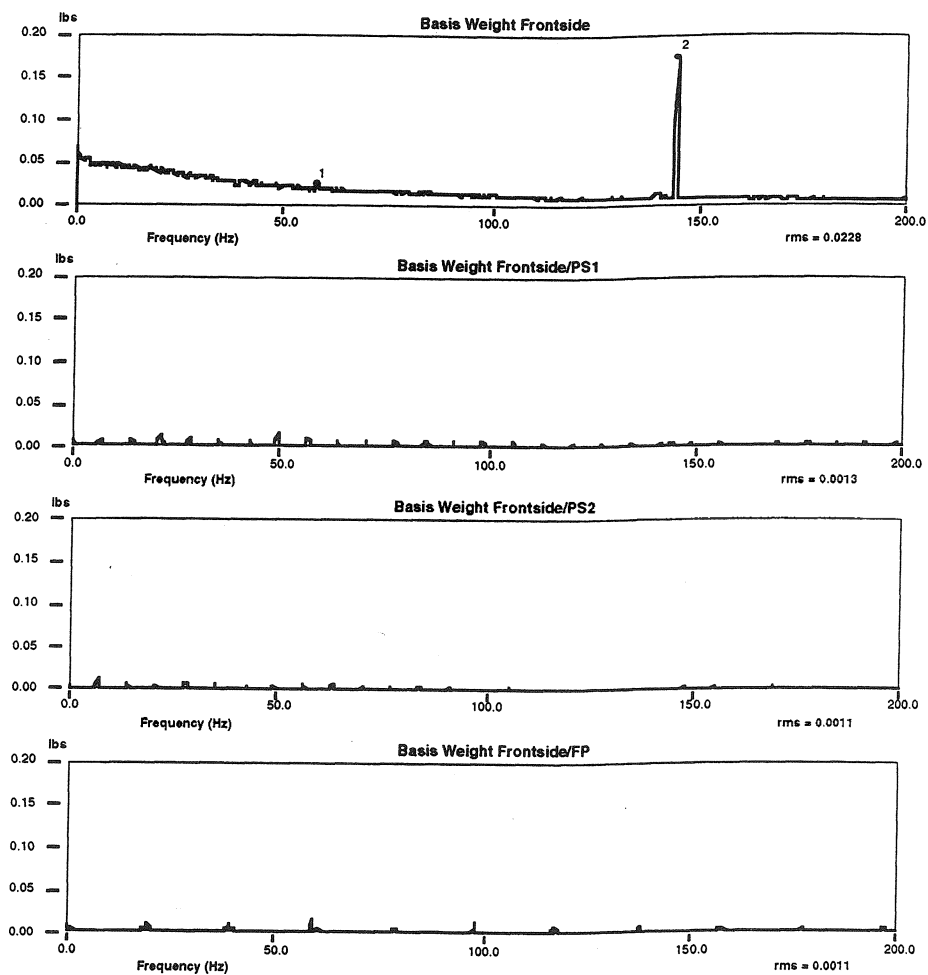


FIGURE 5.5. Basis weight front side spectra.

TABLE 5.1. Stock Approach Pressures of the System Shown in Figure 5.4.

Tap Location	Pressure (psi)	Pressure (H ₂ O)
Fan pump in	3.3	92
#1 Screen out	8.7	242
#2 Screen out	8.8	245
Screen in	11.1	307
Fan pump out	53.86	493

- elemental table activity looking for stock jump or dead areas
- trim squirts to determine cleanness of cut
- sheet release from couch

In addition to these tests, white water consistency tests and first pass fiber and filler retention tests may also be required. It is preferred that a critical paper grade is selected for the wet-end performance test of any paper machine.

Figures 5.6–5.8 show measurements of percent headbox flow drained, stock flow versus drainage and percent consistency for a fourdrier machine, respectively.

TAPPI Technical Information Sheet, TIS 0502-12 Drainage Evaluation by Mass Measurement, describes the method of consistency measurement and calculations needed to develop an accurate drainage profile.

5.3 Press Section Evaluation

In press section, vibration and press operation analysis are done for any grade of paper.

5.3.1 Vibration Analysis

Vibration analysis is done to determine press section roll and fabric operational health to highlight potential operating concerns. Using a real time FFT analyzer, the following data are collected:

- time signature of each press section roll

- frequency spectrum of each press section roll
- time signature of each wet felt

Velocity sensors measuring speed (in inch/second) and Linear Voltage Displacement Transducers (LVDT) measuring displacement (in mils, 1 mil = 1/1000 inch) along with a real time FFT analyzer are combined to gather data. The Synchronous Time Averaging (STA) technique utilizes once per revolution trigger signals to isolate the contributions to vibration caused by individual components. Photo-optic sensors are used as triggers on the fabrics and the LVDTs are used to measure actual nip displacement. These tools can aid in isolating roll problems such as corrugation, cover anomalies, misalignment, bent shafts, looseness and unbalance. Correcting these problems can assist in gaining better fabric life, improved paper quality, uniformity and improved operating efficiency.

From the data collected, the following analysis is done:

- critical bearing frequencies (potential failures)
- roll alignment and balance
- roll surface conditions
- fabric related vibration components

The following information from the paper mill is essential for this evaluation:

- roll diameter and bearing information
- bearing listing (bearing type, complete manufacturer code, location of bearings)
- list of driven rolls and their drive train architecture (two gears, four gears, etc.)
- dates of most recent equipment and supply changes on presses (rolls, fabrics, etc.)
- reflective tape on critical roll journals to use as a “trigger”

The nomenclature of Table 5.2 is used in the discussion of the vibration data. An industry accepted practice of 2.0 mils or 0.3 inch/second peak-peak vibration at a specific frequency is considered significant.

Front Side

Basis Wt: 30 lb / 3300 sq ft
Grade: Bond
Wire Speed: 1100 fpm
Reel Speed: 1147 fpm
Machine Draw: 47 fpm
Draw @ Press: 16 fpm
HP Shower : 230 psi
Furnish: 40% Softwood
50% Hardwood
10% Central Broke
100# Clay
Fabric:
Date On (Life): 95
Tension:
Retention: 82.62% Solids

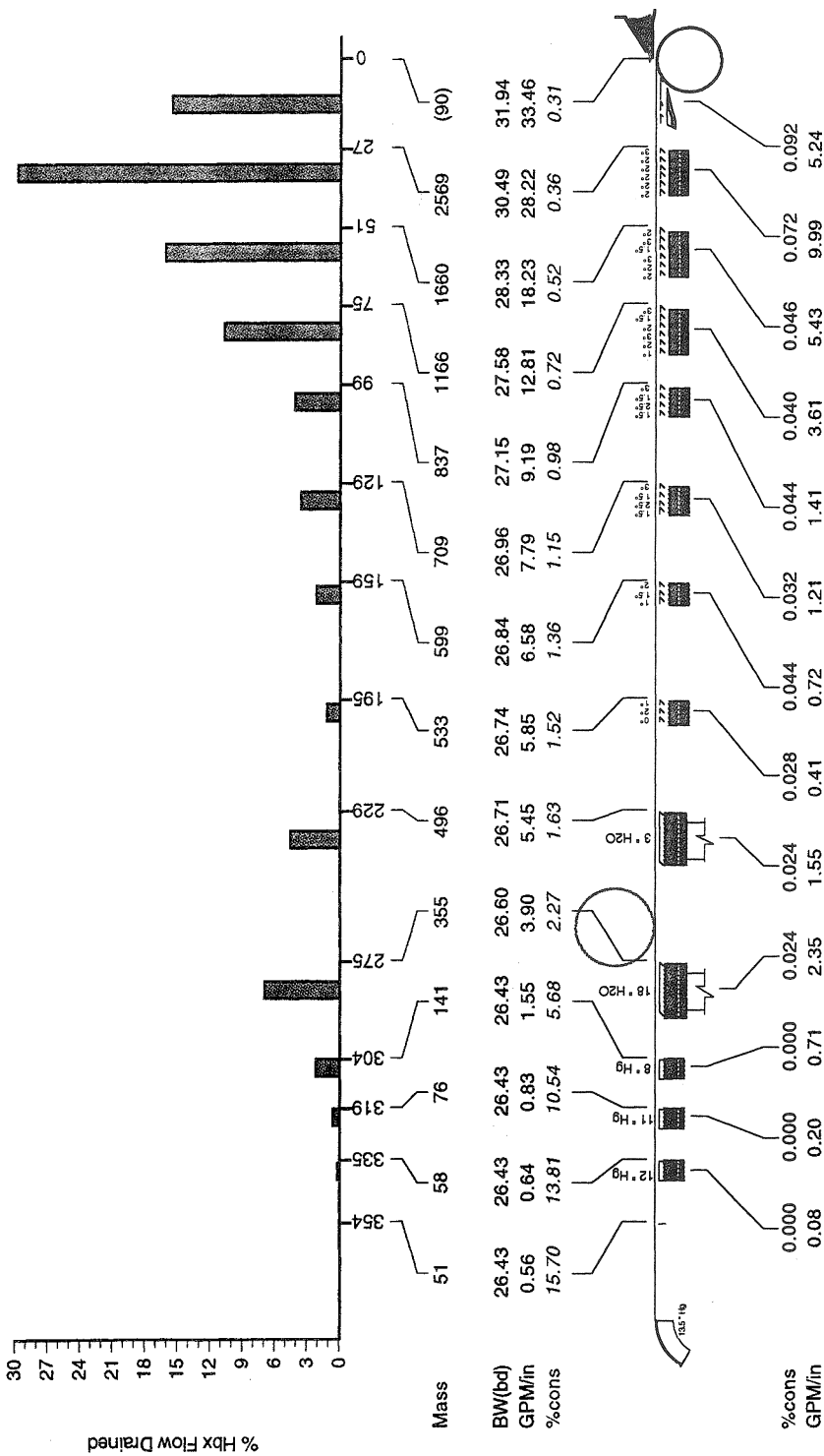


FIGURE 5.6. Percent headbox flow drained on a fourdrinier machine.

Front Side

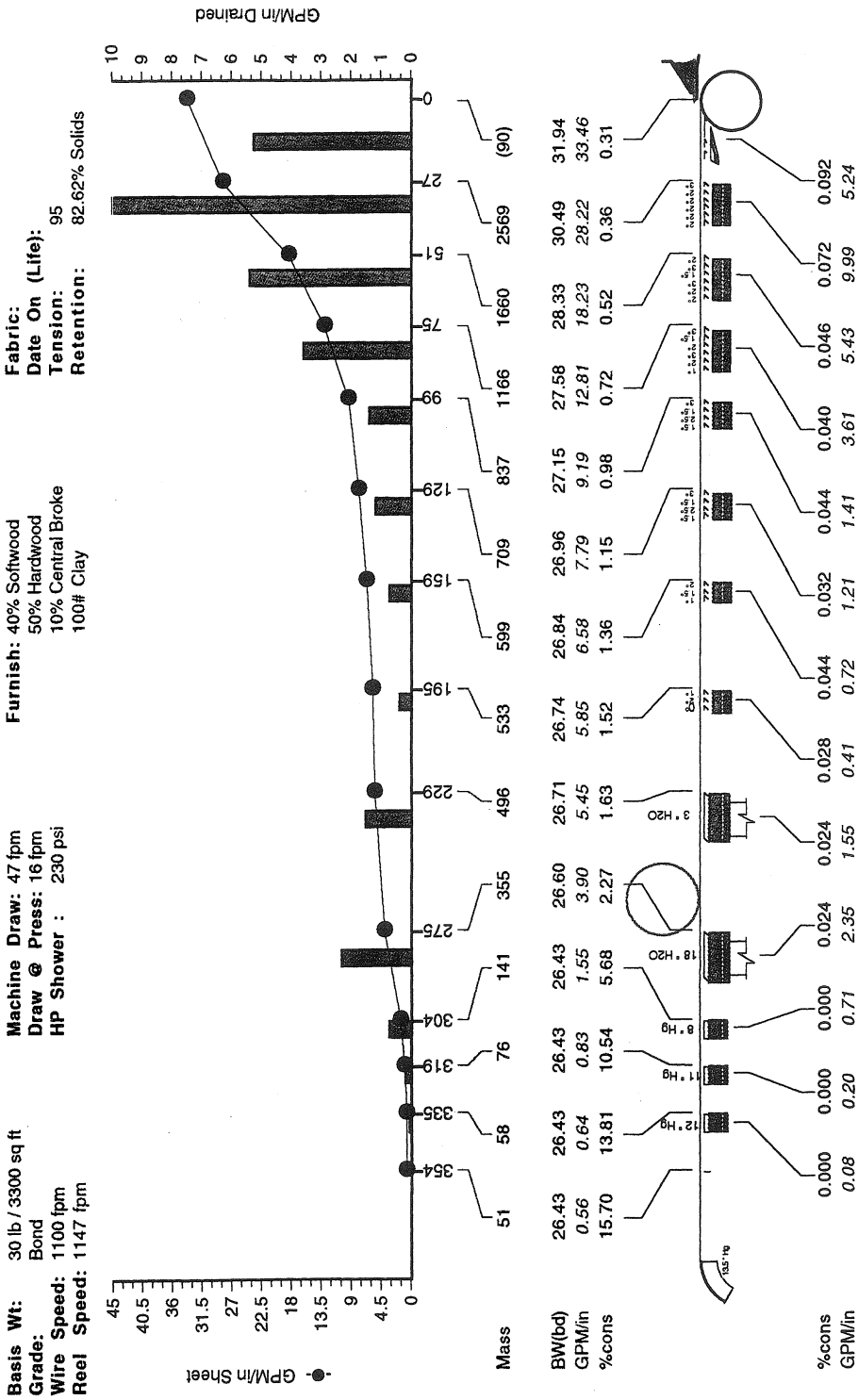


FIGURE 5.7. Stock flow versus drainage on a fourdrinier machine.

Front Side

Basis Wt: 30 lb / 3300 sq ft
Grade: Bond
Wire Speed: 1100 fpm
Reel Speed: 1147 fpm
Machine Draw: 47 fpm
Draw @ Press: 16 fpm
HP Shower : 230 psi
Furnish: 40% Softwood
 50% Hardwood
 10% Central Broke
 100# Clay
Fabric:
Date On (Life): 95
Tension:
Retention: 82.62% Solids

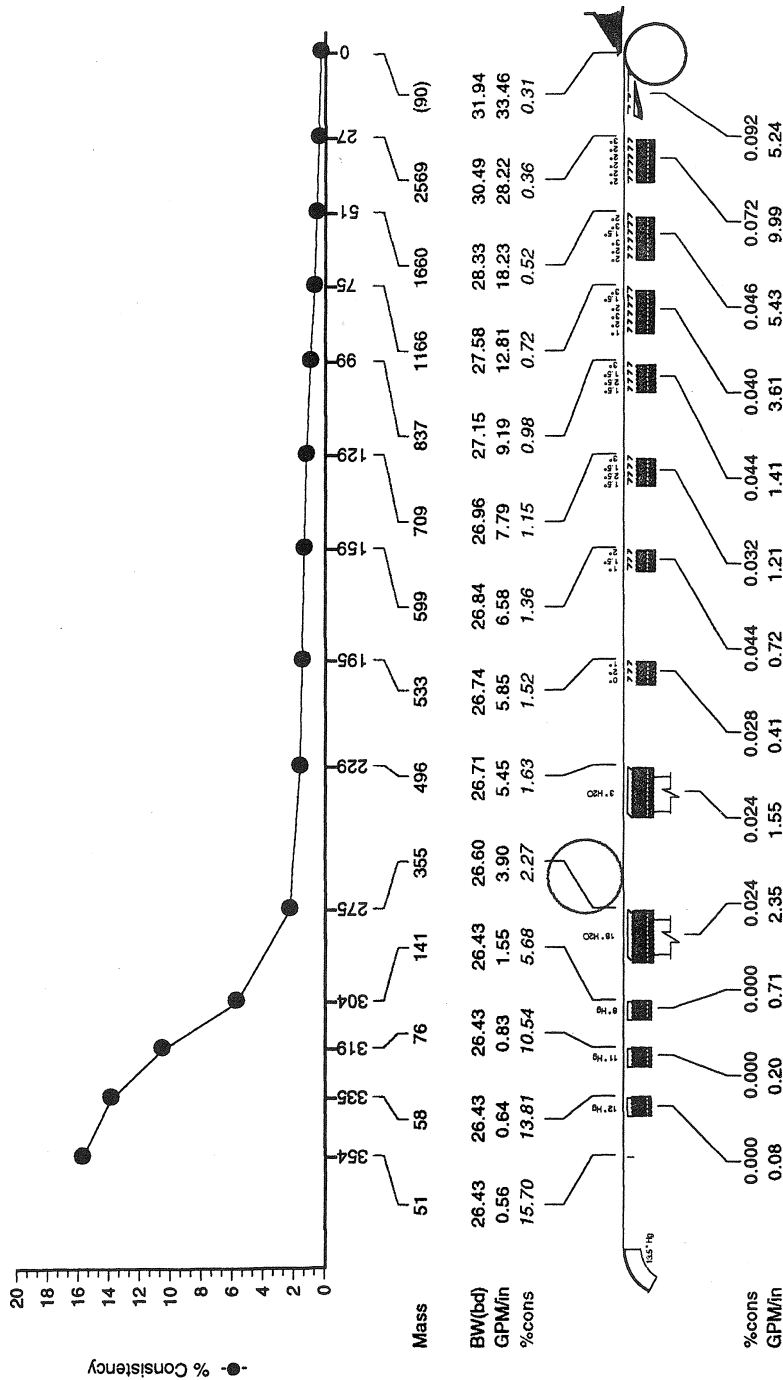


FIGURE 5.8. Percent consistency on a fourdrinier machine.

TABLE 5.2. *Vibration Nomenclature.*

Severity	Explanation	Amplitude
Acceptable		Less than 0.25 mils
Minor	No action needed	0.25–0.75 mils
Moderate	Should be monitored for degradation in condition	0.75–2.0 mils
Significant	Should be removed or repaired during next down	2.0–4.0 mils
Excessive	Should be removed or repaired at earliest opportunity	over 4.0 mils

Press Roll Trigger Placement

During the vibration study, physical movement is collected from sensors mounted on the roll bearing housings and press frame and converted to computer signals for analysis. For the analysis to be meaningful, the specific vibration contribution of each roll and felt in a nip must be known. The process for isolating these signals relies on the ability to sense each revolution of the roll or fabric (known as triggering). Currently, light-based sensors on the fabrics are used to pick up the tradeline. For a roll, reflective tape needs to be placed on the head for the sensors to effectively trigger.

Table 5.3 lists the acceptable locations for the placement of reflective tape. It should be noted that extra sections (coaters and calenders especially) take extra time and should be scheduled only if the mill has a specific vibration problem in that area. For example, tracing a vibration problem in a coater usually takes longer than completing an entire press vibration survey.

Figure 5.9 shows the acceptable locations

TABLE 5.3. *Locations for the Placement of Reflective Tape.*

1. Press vibration survey
 - major press rolls
2. In conjunction with machine audit
 - major press rolls
 - wire turning, lumpbreaker and couch rolls
 - breaker stack and size press rolls
3. Other areas (per mill request)
 - coater backing roll and paper rolls in coater section
 - calender stack rolls

for placement of reflective tape. The tape needs to be placed in only *one* of these locations. This tape needs to be only about 1/2 inch square and should be placed in a location that is away from bolt heads and dirty areas as much as possible. The tape needs to be stuck on with SuperGlue. Experience has shown that anything less than SuperGlue will not hold up and even SuperGlue will only stay on the machine for a couple of weeks.

Figures 5.10–5.12 show the importance of triggering roll revolutions. These graphs show a vibration signal plotted against time for press rolls in a nip. The rolls are rotating close to the same speeds (210 rpm versus 213 rpm) which makes the resulting signal in Figure 5.10 very confusing. Figure 5.11 contains the same data triggered against the top roll and shows that this roll is in “poor” condition. Figure 5.12 contains the data triggered against the bottom roll and shows that this roll has moderate problems without much significance.

5.3.2 Press Operations

This analysis is done to characterize fabric performance and general conditions of the press section. Using Scan Pro, high speed photography, hot wire anemometer, caliper gauge and gamma gauge, the following data are collected:

- felt water load
- felt moisture profile
- felt caliper
- uhle box air flows
- relative sheet consistency into and out of press

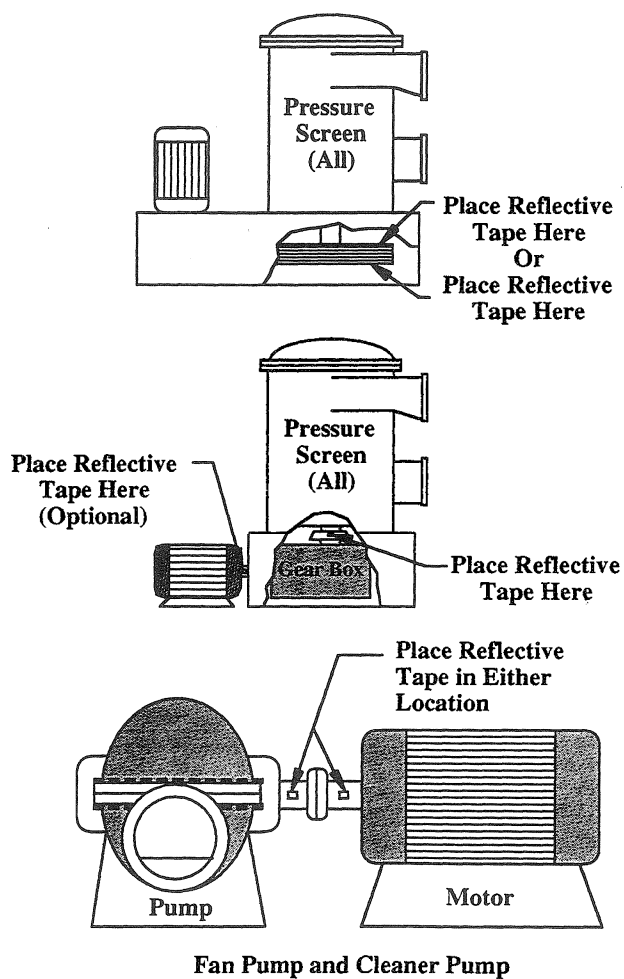


FIGURE 5.9. Reflective tape placement.

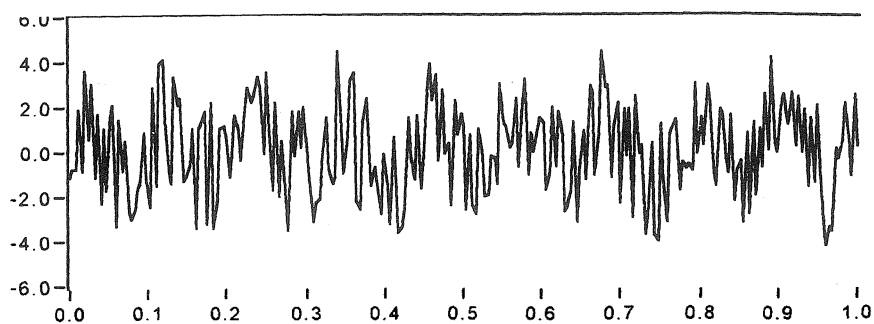


FIGURE 5.10. Raw signal.

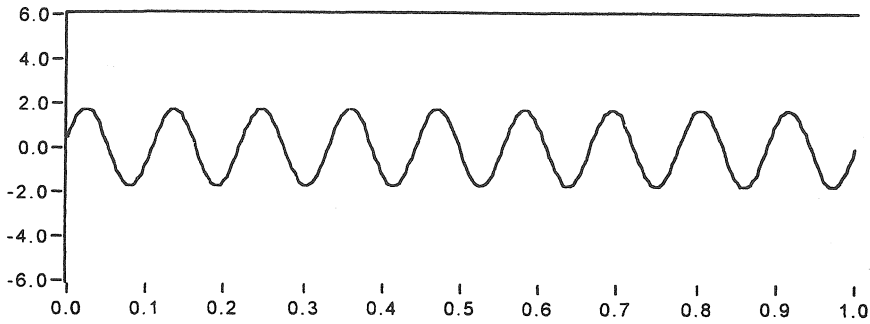


FIGURE 5.11. Top roll signal.

- proper positioning and penetration of high pressure showers.

Felt water removal and press nip anomalies such as skewed press loadings, crown/load concerns and section performance trends are evaluated using a Scan Pro. Vacuum system survey is performed to evaluate the performance of pumps and uhle boxes. Press water balance is constructed using a gamma gauge.

Figure 5.13 shows the typical moisture profiles of the first and second presses of a paper machine. Table 5.4 shows the press section water balance analysis. Figure 5.14 shows the total mass profile after the second press for the same machine.

5.4 Dry-End Performance

Dry-end analysis generally consists of four major parts.

5.4.1 Determining Balance of the Hood and the Effect of Exhaust Loading and Air Filtration

Using a vane anemometer and relative humidity analyzers, mass flows of hood exhaust and supply plenums are measured. In addition, all equipment conditions are inspected.

Air temperature, relative humidity and barometric pressure measurements are recorded for the hood exhaust fans. A calibrated, digital readout pitot tube assembly is used to measure static and dynamic air pressures in all exhaust plenums. These static and dynamic pressure readings are then converted to air velocity readings. The data are calculated to determine the actual air permeability (cubic feet per minute, cfm), mass of dry air per minute, air density, absolute humidity, grains, dew point and mass of water vapor per minute. The dew point is the temperature at which water will condense from the air. Grains is a

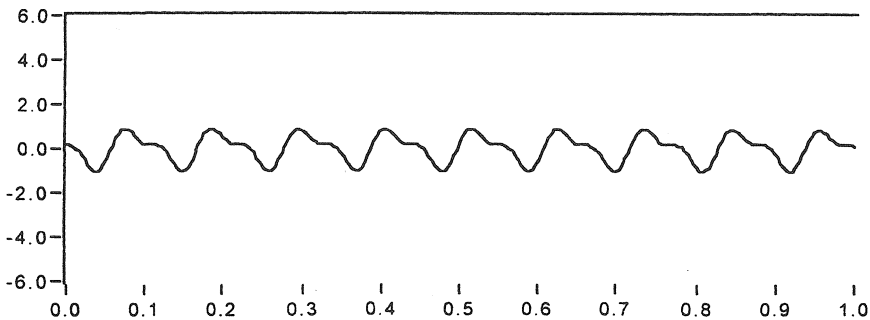


FIGURE 5.12. Bottom roll signal.

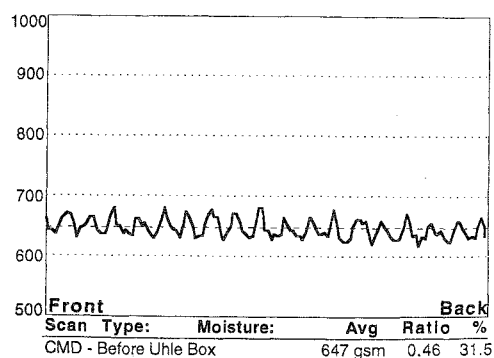
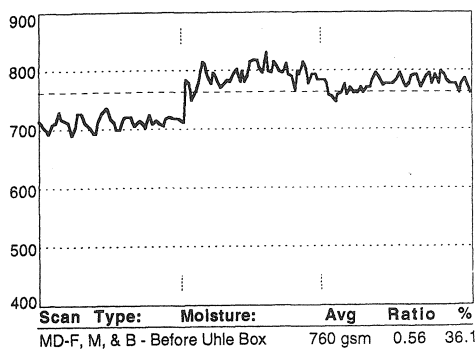
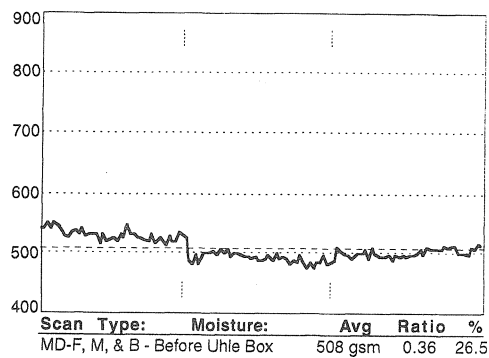
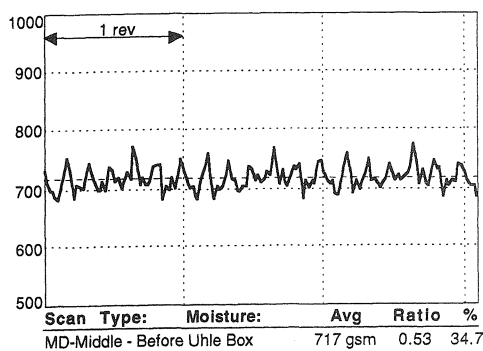
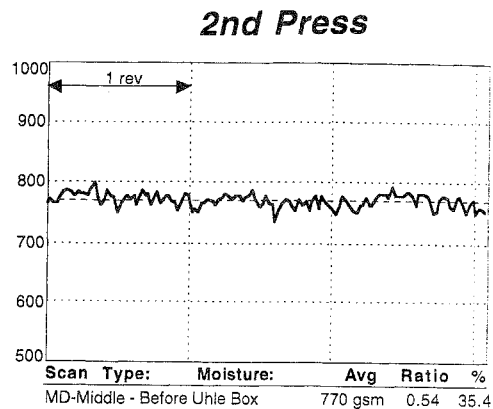
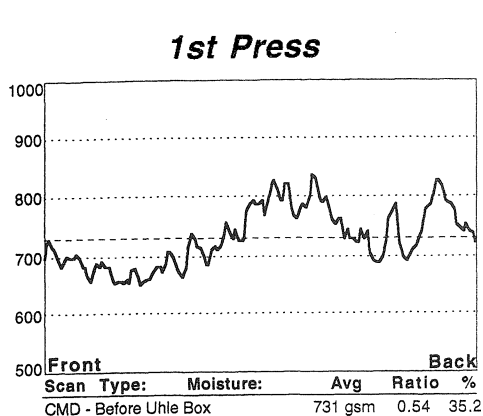


FIGURE 5.13. Moisture profiles of the first and second presses of a typical press section.

TABLE 5.4. Press Section Water Balance Analysis.

NDC Gamma Gauge				
Mill:		Location:		
Machine:		Date:		
Grade:	30# Bond			Vacuum - In. Hg
Wire Speed:	1100	Press Loading		Roll Uhle
1st Press	1116	1st Press:	410 410	16 10
Reel Speed:	1147	2nd Press:	400 440	9
Basis Wt.	Size Addition	Moisture	Dry Weight	
29.7	6.00%	5.30%	26.44	
Mass Readings				
	#/1000	Sq.Ft.	Mass Change % Solid	
Dry weight of sheet	8.01		#/1000	
Weight On Wire	51	Before Couch (Drainage Study)	15.7%	
		Water Removed at Couch	0	
1st felt Before Press	331			
1st w/Sheet Before	382	Sheet & Water	51	15.7%
1st Felt after Press	350	Water Gained by 1st	19	
		Water Removed From Sheet	22	
Sheet After 1st	29			27.6%
2nd Felt Before	317			
2nd Felt After	325	Water Gained by 2nd	8	
		Sheet Water loss	8	
Sheet After 2nd	21			38.2%
Results:				
Consistency		Water Removed		
Before Couch	15.7%		Gal/Min	
		at Couch	0.0	
Off Couch	15.7%			Press Water Flow
		at 1st Press	33.4	Gal/Min
Out Of 1st	27.6%			Felt 28.8 63.3%
		at 2nd Press	12.1	Rolls 4.6 10.0%
Out of 2nd	38.2%			Felt 12.1 26.7%
				Rolls 0.0 0.0%
		Press Total	45.50	

measurement of moisture in the air. There are 7000 grains per pound of water. The grain loading is calculated by multiplying the absolute humidity by 7000. ACFM is the actual cubic feet per minute of air in each duct. The air flow rate (in acfm) is calculated from air velocity readings and the duct size corrected for the air density at a given barometric pressure. This value is used to calculate the hood balance. Table 5.5 shows an example of typical hood performance results.

5.4.2 Determining Dryer Section Performance and Efficiency

Using a platinum RTD probe and process readings the following is established:

- TAPPI drying rate
- U-factor
- drawdowns
- steam consumption
- dryer temperature

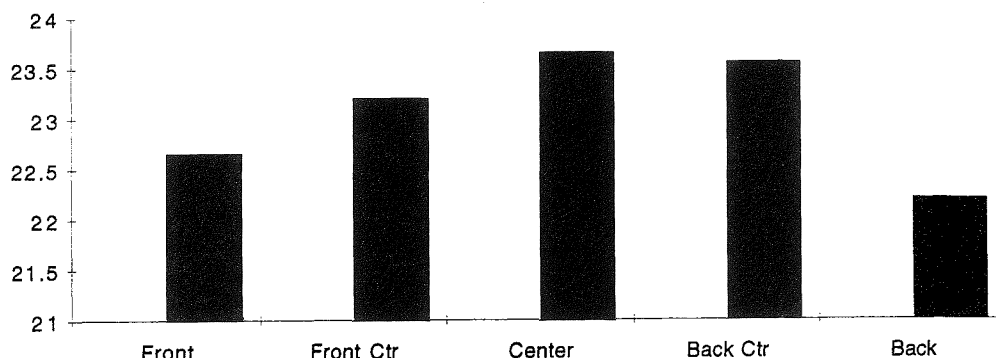


FIGURE 5.14. Total mass profile after the second press (%).

The dryer can surface temperature can be measured with a noncontact pyrometer. Figure 5.15 shows an example of the difference between the dryer surface temperature and saturated steam temperature (ΔT) at a given steam pressure in the dryer can. This difference should not be greater than 50–60°F. Although $\Delta T = 60^\circ$ is not uncommon throughout the industry, there are dryers running at the 40–50°F ΔT levels. It is recommended that dryers with ΔT s greater than 60° be investigated. High temperature differences point out that there may be a problem with the heat transfer from the dryer can to the sheet. Some of these problems may be condensate removal, excessive blow through (i.e., too high a differential pressure), buildup on dryer cans, internal scale buildup, siphon clearance or broken siphons, or too low a differential pressure.

From the humidity surveys and fabric condition surveys, it has been shown that there is an increased potential for dryer fabric hydrolysis when the conditions of the exhaust from a dryer hood possess air temperatures of at least 180°F and an absolute humidity level of 0.12 lb H₂O/lb dry air or higher. Figure 5.16 shows a hypothetical potential hydrolysis graph for a dryer section in which the air temperature and absolute humidity of the exhaust are well below the critical levels.

Steam and condensate studies can be performed to analyze the condensing rates of various steam sections to compare them to theoretical and standard values by grade. This is useful to highlight the drying inefficiencies related to the physical condition of drying

elements such as scale buildup, siphon size/clearance inadequacies and undersized steam lines.

5.4.3 Determining Drying Impacts of Pocket Humidities and Air Velocities

With pocket humidity equipment, the relative humidity, temperature and air velocity of each dryer pocket are measured. The air temperature and percent relative humidity are usually converted into absolute humidity results for each location. The absolute humidities are used as the basis for the evaluation of the drying conditions in each of the pockets.

Dryer pocket humidity measurements can be used to evaluate evaporation from the sheet, as well as the performance of the pocket ventilation system and the effect of the dryer fabric permeability. The humidity results are an indirect measure of the drying rate in a section and are a direct measure of the pocket ventilation efficiency.

It is generally accepted that the absolute humidities in the dryer pockets should be kept at a level below 0.20 lb of water per lb of dry air. If the absolute humidity rises above this critical level, the efficiency of the drying process is reduced. The pocket humidity levels are affected by several factors including the basis weight, steam pressure, dryer fabric tension, ventilation in the pockets, machine speed and air permeability of the dryer fabrics. Figure 5.17 shows an example of humidity measurements in a typical dryer section. As can

TABLE 5.5. Example of Hood Performance Results.

Main Hood											
Duct ID	Air Temp. (°F)	Rel. Hum. (%)	Abs. Hum.	Humidity (Grains)	Dew Point Temp. (°F)	Air Velocity (fpm)	Duct X-Sect. Area (Sq.Ft.)	Air Flow Volume (acfm)	Mix Density (lb/cu ft)	Lbs. D.A./min	Lbs. W.V./min
Exhaust #1	125.00	63.20	0.058	409.42	108.65	3096.50	21.229	65,736	0.063	3,948	230
Exhaust #2	127.00	49.10	0.047	330.14	101.92	2776.80	21.229	58,949	0.063	3,588	169
Averages	126.00	56.15	0.053	370	105.3			62,343	0.063		
After Hood											
Exhaust #3	118.00	51.00	0.037	264.22	95.00	2744.80	18.189	49,928	0.065	3,132	188

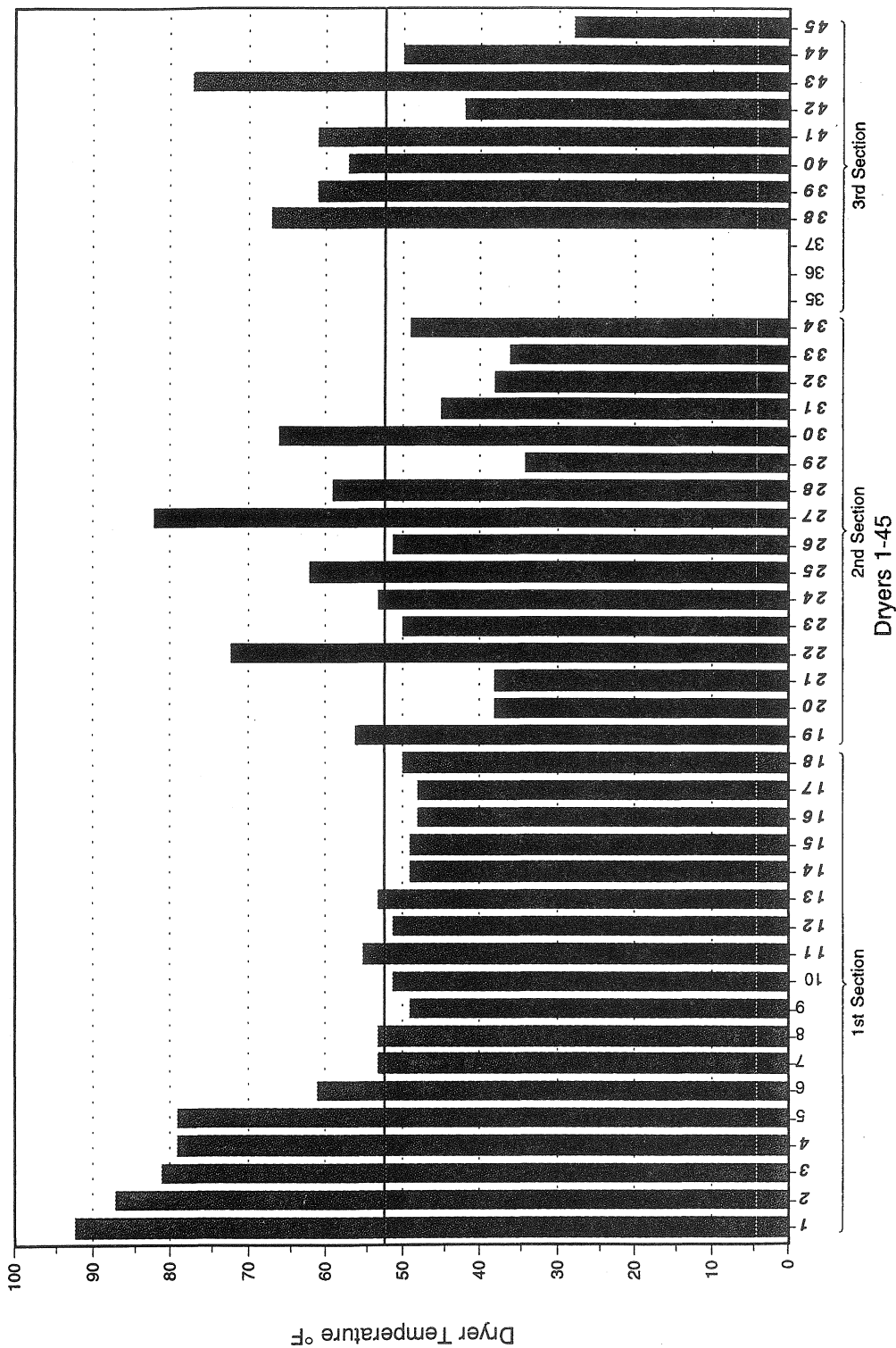


FIGURE 5.15. Example of the difference between dryer temperature and steam temperature (ΔT).

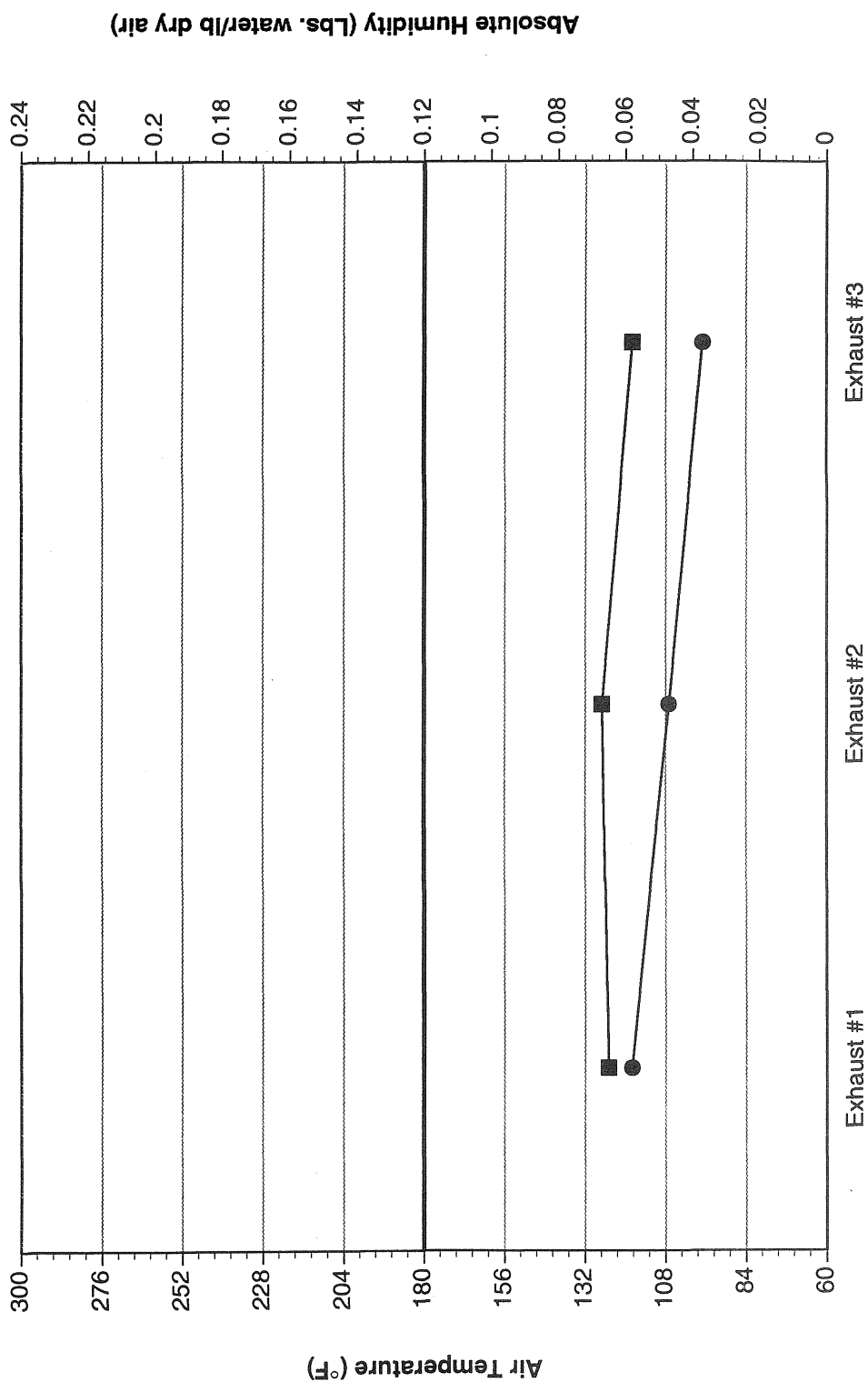


FIGURE 5.16. Example of potential hydrolysis conditions: exhaust air temperature versus absolute humidity.

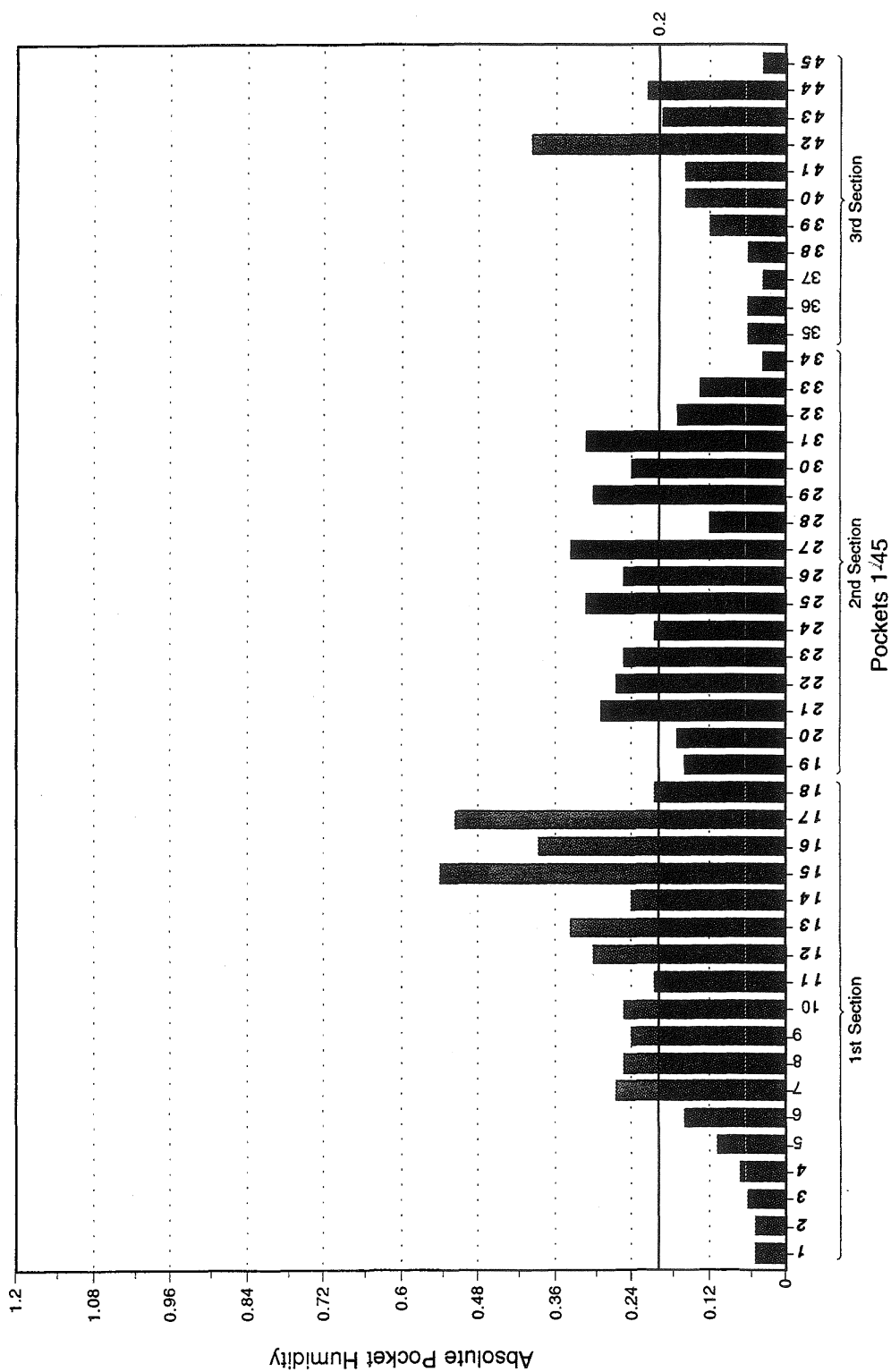


FIGURE 5.17. Example of a typical pocket absolute humidity.

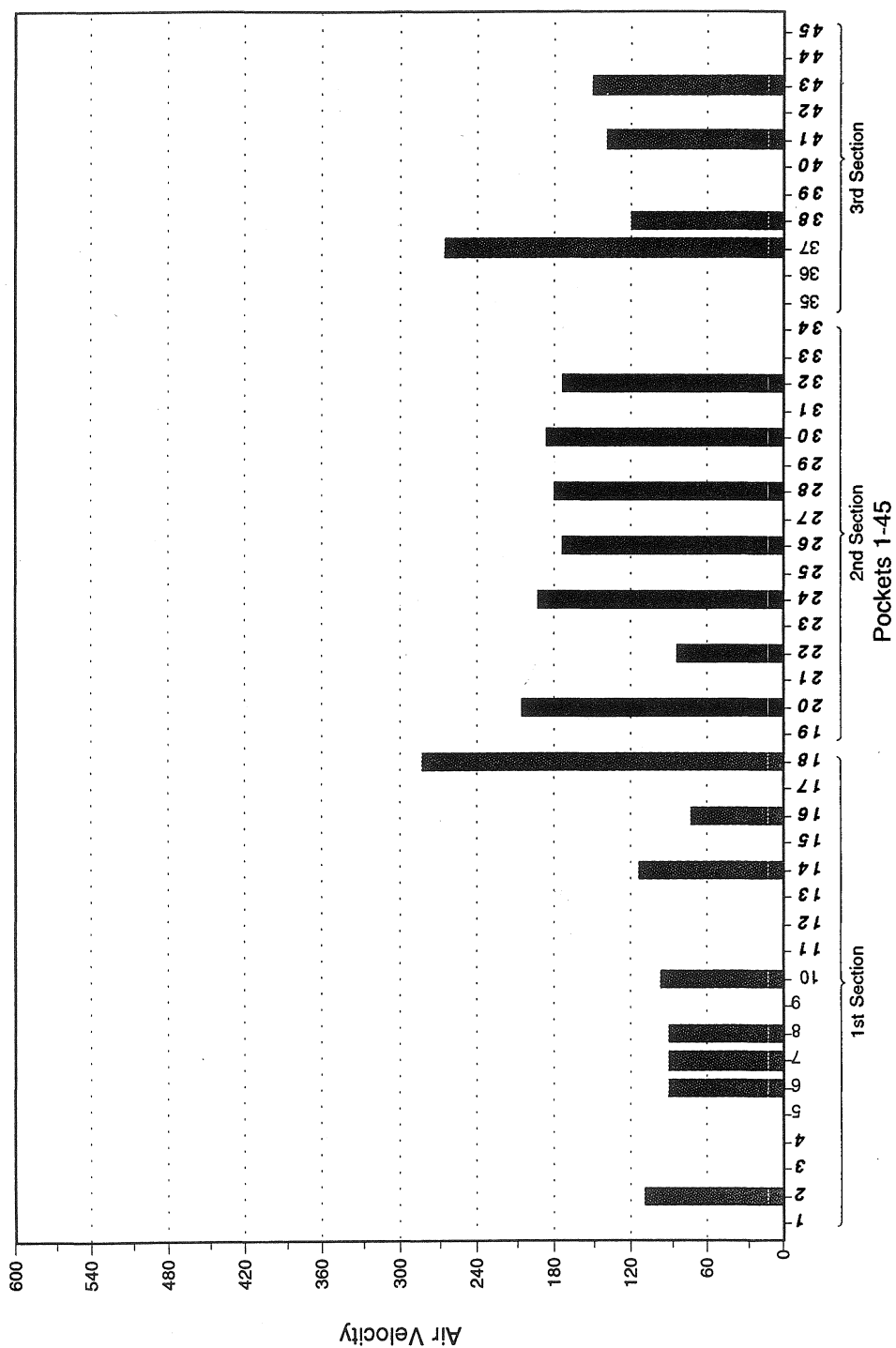


FIGURE 5.18. Pocket air velocity in a dryer section (ft/min).

be seen from the graph, the pocket humidities in the first section rise above the critical humidity levels starting at dryer number 7. The humidity levels remain above the critical humidity level of 0.20 lb water per lb of dry air throughout the remainder of the section until dryer number 31. The only exception is at the section transfer and at dryer number 28.

A closer review of the pocket humidity data reveals a trend that is commonly seen on conventionally clothed machines. The pockets formed above the bottom dryers nearly all had higher humidity levels than the pockets below the top dryers due to less air movement.

In order to ensure evacuation of the moisture filled air from the centers of the dryer pockets, it is generally necessary to have air velocities of 250 fpm or more exiting the machine. In areas of especially high pocket humidities, the air velocity should be kept as high as possible without affecting runnability due to edge flutter. Figure 5.18 shows the pocket air velocity distribution in a dryer section.

5.4.4 Infrared Thermography

Infrared cameras are used in the paper industry for trouble shooting. Characterization of any cross machine sheet or dryer can temperature differences that may highlight a machine moisture problem is possible with this technique.

Using an infrared camera, the sheet is photographed at reel, section breaks and before and after the press. Conditions of dryer fabrics and dryer cylinders are examined. Moisture variations in the paper can be determined. Infrared thermography is also used to examine paper structure [4].

5.5 References

- 1 *Process Study*, Asten Inc., 1996.
- 2 Carlson, D., "From Headbox to Couch—Consistency Measurement under the Wire Provides Benefits to the Papermaker," *PaperAge*, June 1996.
- 3 *TAPPI Test Methods*, 1996-1997, TAPPI Press, Atlanta, GA.
- 4 Kiiskinen, H.T., et al., "Infrared Thermography Examination of Paper Structure," *TAPPI Journal*, April 1997.

5.6 Review Questions

- 1 What is the purpose of paper machine auditing? Explain how the audit results can be used to improve the papermaking process.
- 2 What is FFT? How is it used in paper machine auditing?
- 3 How does vibration affect the press section operation? What can be done to reduce/eliminate vibration?
- 4 Explain the following:
 - pulsation
 - synchronous time averaging
 - pocket air velocity and humidity
 - U-factor
- 5 Find out the working principle of the following devices:
 - gamma gauge
 - noncontact pyrometer
 - LVDT
 - hot wire anemometer
 - pitot tube
 - platinum RTD probe
 - infrared camera

Paper Structure, Properties, and Testing

Since the purpose of paper machine clothing is to help make paper and paperboard, it is proper to overview the structure and properties of paper and paperboard in this chapter. Better understanding of paper structure and properties will help the textile engineer with the design and manufacturing of forming, press and dryer fabrics.

Paper has a fibrous structure. The basic building block of paper is the individual fiber. Various fillers are added to this structure to obtain certain properties such as improved strength, printability, liquid absorption, etc., depending on the end use. Although paper's structure is continuous, it is in fact a heterogeneous network of these discrete fibrous particles and fillers. The fibers and fillers, which are the main components of the paper structure, have different distributions, shapes and chemical compositions. Nevertheless, the paper is expected to behave in a homogeneous manner. In today's technology, this is achieved by evening out the variations in the structure to a certain extent during the manufacturing process. Scientific attempts to relate the paper characteristics and structure to final properties began in the 1960s [1]. The product quality in paper machine clothing is usually measured by the performance and life of the fabric. For paper, quality may require fulfillment of several requirements which sometime may conflict. The papermaker, converter and consumer have various analytical tests available to judge the quality and performance of

the paper. The uniformity of quality may be as important as the quality itself. The uniformity of quality depends on uniformity of structure.

In laboratories and research centers, paper is usually made using handsheet molds for testing and analysis purposes. It is reported that strength of handmade paper is generally better than comparable papers made on commercial paper machines [2].

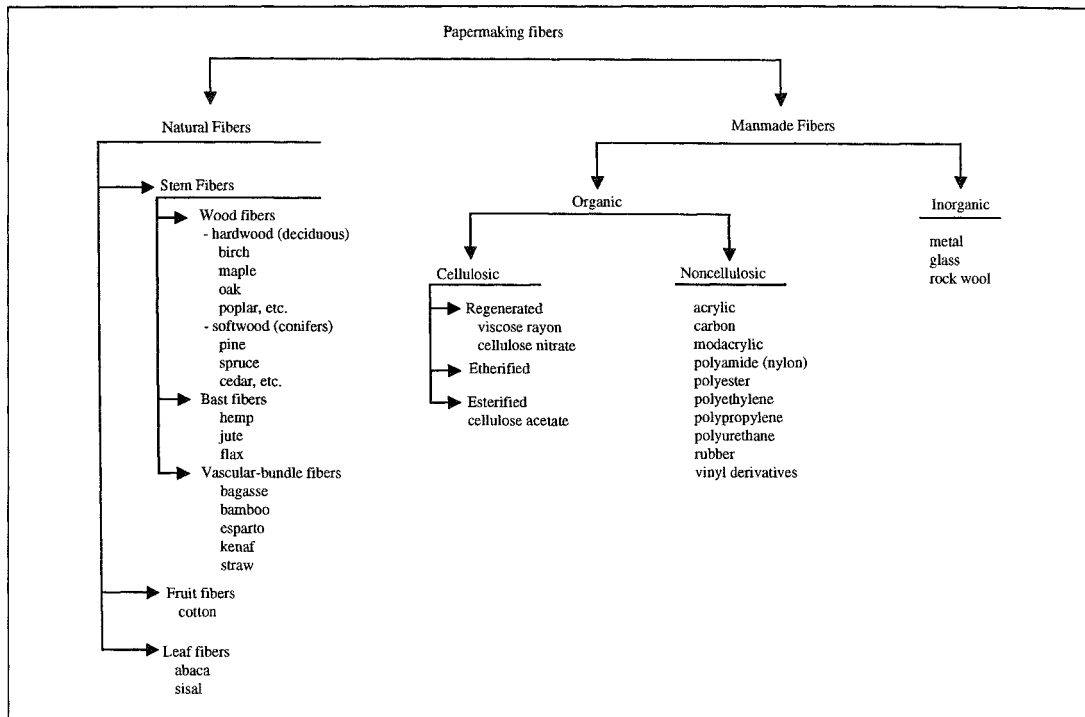
6.1 Fibers Used in Papermaking

Fiber can be defined as "a thread-like body or filament, many times longer than its diameter" [3].

Both natural and synthetic fibers are used for manufacturing of paper and paper related products. Plants, which are the most abundant organic materials on earth, are the major source of natural fibers for papermaking. Although different plant fibers can be used for papermaking, wood pulp fibers are the most widely used because of their cellulose content. The use of synthetic fibers in paper and related products is increasing due to exceptional strength properties they provide. Sometimes, animal fibers such as wool or silk are also used for special types of papers. Table 6.1 shows the fibers that are used in papermaking [2,4].

Important properties of papermaking fibers include length, coarseness, intrinsic strength,

TABLE 6.1. Classification of Major Fibers Used in Papermaking.



cohesiveness and flexibility. Initial and maximum fiber length, coarseness and fiber strength are determined by the genetic code of the tree. Simplified structure of a typical wood fiber is shown in Figure 6.1. The fiber has an outer primary wall and three layers of

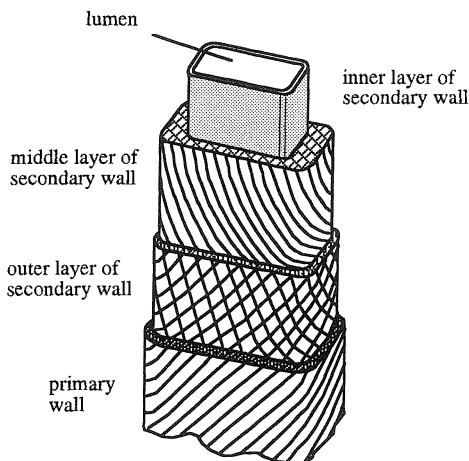


FIGURE 6.1. Schematic structure of a typical wood fiber.

inner secondary wall. The inner layers which are rich in cellulose constitute the main body of the fiber. In the center of the fiber, there is an empty space called lumen. The individual fibers are held together with lignin. Typical wood fiber lengths vary from 1 mm to 6 mm. The fiber's cross-sectional area is approximately 1/100th of its length. Processes during papermaking usually reduce the length, coarseness and strength of fibers but increase the cohesiveness and flexibility.

6.1.1 Wood Fibers

The main ingredient of wood fiber is cellulose material. Although cellulose fibers can be obtained from different plants, wood is the most widely used material in papermaking because it is abundant and relatively inexpensive. Figure 6.2 shows the structure of cellulose. The main ingredient of paper is cellulose which is a white, hygroscopic material. It has good water absorption which affects its dimensional stability. Cellulose can form hydrogen bonds. Tensile strength and flexibility of

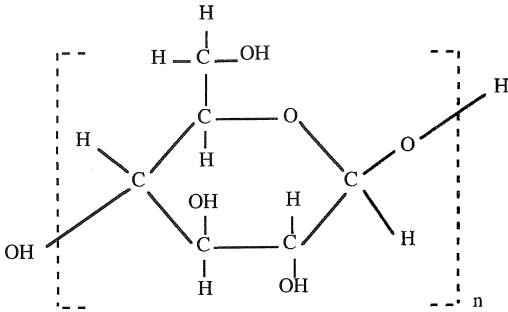


FIGURE 6.2. Structure of cellulose.

cellulose are high. It is also combustible. Since paper is made of cellulosic materials, it has all the properties that cellulose has. In addition, the properties of paper can be modified by different pulping and papermaking processes [2].

Trees are classified as hardwoods and softwoods. They have different amount of lignin in them. Another difference is that softwood fibers are longer than hardwood fibers. The length of softwood fibers is 3-6 mm while the length of hardwood fibers is 1-2 mm. A combination of hardwood and softwood is used in most papers. Short hardwood fibers provide bulk and long softwood fibers provide strength.

Hardwood Trees

Hardwood trees have broad leaves and are usually deciduous (lose their leaves annually). Examples of hardwood trees are aspen, birch, maple, oak, eucalyptus and poplar. They have short, slender, medium-walled fibers. Eucalyptus (blue gum) has good softness properties for tissue and smoothness properties for fine paper. Hardwood stem is composed of about 50% cellulose and 20% lignin. Figures 6.3 and 6.4 show photomicrographs of different hardwood fibers.

Softwood Trees

Softwood trees are also called conifers or evergreens. They have needle-like or scale-like leaves. Examples of softwood trees are

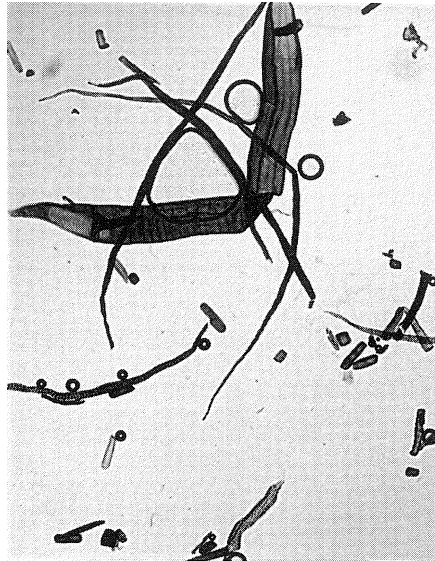


FIGURE 6.3. Southern hardwood bleached kraft (sweet gum) fibers. Magnification: 50x. Circular shapes are air bubbles (courtesy of Integrated Paper Services).



FIGURE 6.4. Northern hardwood bleached kraft (aspen) fibers. Magnification: 50x (courtesy of Integrated Paper Services).

pine, spruce, and cedar. Softwood stem consists of 50% cellulose and 30% lignin. Figures 6.5–6.8 show photomicrographs of various softwood fibers.

Southern pine, which includes Loblolly pine, shortleaf, longleaf and yellow pine, is

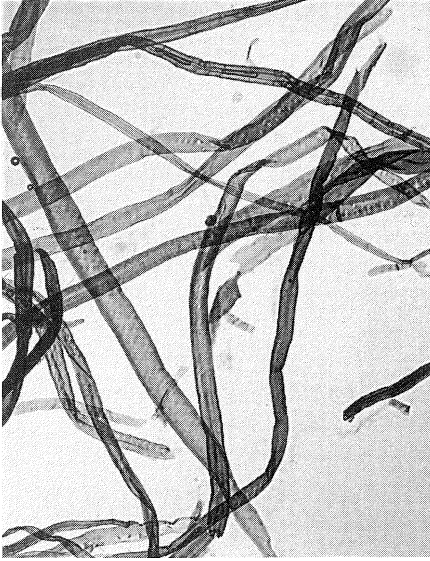


FIGURE 6.5. Southern yellow pine softwood bleached kraft fibers. Magnification: 50× (courtesy of Integrated Paper Services).

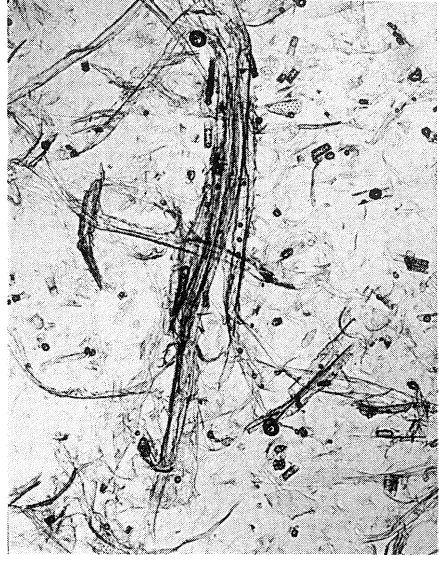


FIGURE 6.7. Northern softwood stone groundwood fibers (spruce and fir with trace of hardwood). Magnification: 50× (courtesy of Integrated Paper Services).

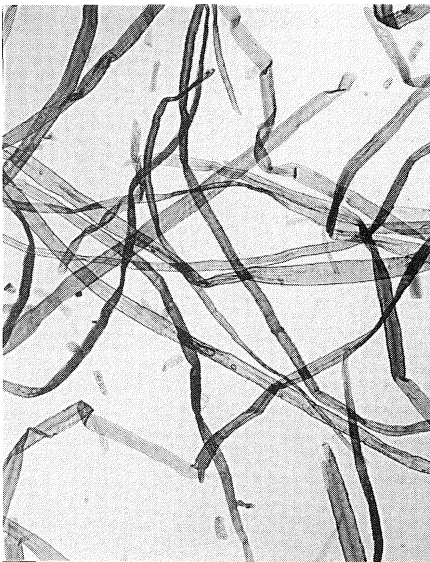


FIGURE 6.6. Northern softwood bleached kraft fibers (mostly spruce but mixed with some pine and others). Magnification: 50× (courtesy of Integrated Paper Services).

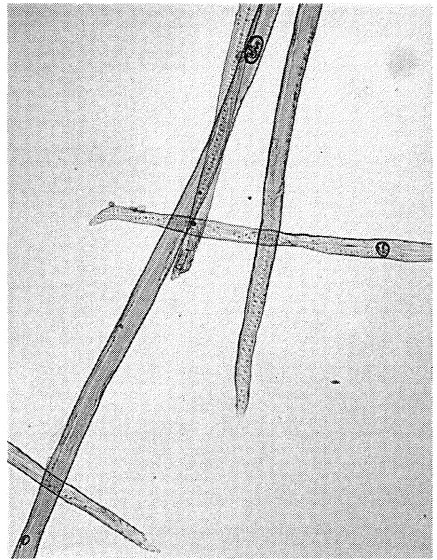


FIGURE 6.8. Northern softwood thermomechanical pulp fibers (spruce). Magnification: 50× (courtesy of Integrated Paper Services).

widely used in papermaking. They are very fast growing trees with long coarse, thick walled fibers. These fibers give excellent tear properties in the paper [2]. Another coniferous or softwood tree is spruce which has a long,

medium slender, medium walled fiber. It is widely used for newsprint/groundwood and sulfite pulping due to its brightness. Western red cedar, another softwood tree, has long, medium slender, thin-walled fibers.

6.1.2 Nonwood Cellulosic Fibers

Nonwood plants are also suitable to make paper since they contain cellulose. As a result, different types of plants may be used for different types of papers. An example is cotton plant. Cotton fibers are longer (10–30 mm) and have more cellulose (90%) than wood fibers. However, cotton is more expensive than wood fibers therefore it is used in special business papers alone or in combination with wood fiber.

Other plants that are used to make paper include sugar cane, hemp and flax, barley, rye, oat, bamboo, kenaf, abaca, sisal and rice straw. It should be emphasized that wood, by far, is the major source of papermaking due to proper fiber length, ease of processability, availability and cost.

6.1.3 Synthetic Fibers

Synthetic fibers such as polypropylene, polyester, glass, etc., are mostly used in the nonwoven manufacturing (Section 2.6.3).

6.2 Properties of Single Pulp Fibers

The important mechanical properties of single wood pulp fibers are determined in tension, bending or torsion with destructive or non-destructive test methods. The tensile elastic properties are the most important properties which are measured in the range of mN (milliNewton) and μm (micrometer) [5–7].

6.2.1 Fiber Length

Fiber length is an important property that affects several sheet properties. It has been reported that for a relatively unbeaten pulp, coarseness of the fibers being equal, fiber length affects the tensile strength, modulus, burst strength, fold and tear strength of the sheet. For heavy papers with good bonding between the fibers, effect of fiber length on

modulus and tensile strength is small. However, for lightweight papers, fiber length has more effect on modulus and tensile strength [8,9].

The fibers in a paper have a certain length distribution which may be due to the type of fiber. Recycling, method of pulping and degree of beating also affect fiber length distribution. TAPPI test method T-233 is designed to measure the weighted average fiber length of a pulp which is given as [10]:

$$L = \frac{\sum wl}{\sum w} \quad (6.1)$$

where

L = weighted average fiber length (mm)

l = fiber length (mm)

w = fiber weight (mg)

Fibers that are less than 0.2 mm are considered to be fillers, debris or fines. Fibers that are narrower than 1 mm are considered to be too weak as strength bearing members. Other techniques to measure fiber length are fiber length of pulp by projection (TAPPI T-232, Fiber Length of Pulp by Projection), electronic sequential fiber length analysis (Kajanni, developed by Finnish Central Laboratory) and image analysis.

6.2.2 Fiber Coarseness

Coarseness is defined as the weight (mg) per unit length (100 m) of fibers which is called a decigrex (dg). It is 11.1 times larger than denier which is widely used to indicate the density of textile fibers (Section 2.2.3). Fibers that have a coarseness value of less than 10 dg are considered fine fibers. Above 20 dg indicates a coarse fiber. Coarseness affects several paper properties. For example, a finer pulp results in a stronger, smoother paper with better folding ability. Sheet density also increases with fine fibers. Paper made of coarse fibers has more porosity. Methods to measure coarseness include microscopic method, image analysis and Kajanni fiber analyzer [11].

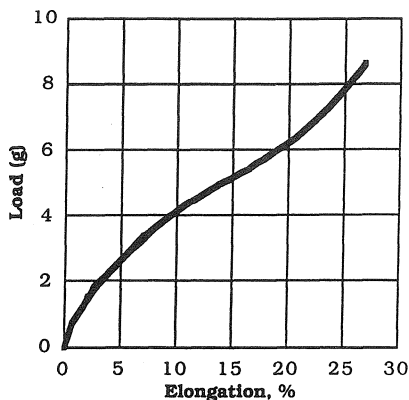


FIGURE 6.9. Load-elongation curve of a bleached-kraft, southern pine fiber [2].

6.2.3 Intrinsic Fiber Strength

Intrinsic strength of a fiber is, first of all, determined by the type of tree. For example, an unbeaten, unbleached, sulfate sweet gum fiber has a breaking strength of 53.6 kg/mm^2 . A birch fiber from the same type of pulp has 97.5 kg/mm^2 breaking strength. Fiber treatment also has an effect on intrinsic fiber strength. Fiber modulus is increased by beating due to increased crystallite orientation. Zero-span tensile test may be used to measure intrinsic fiber strength. A typical load-elongation curve of a bleached-kraft, southern pine fiber is shown in Figure 6.9.

6.2.4 Fiber Flexibility (Wet Fiber Compactability)

Flexibility is the ability of fiber to conform when pressed against other wet fibers and to keep its form during drying due to the cohesiveness of the surface. Fiber diameter and wall thickness, level of fibrillation during refining and fiber cross-sectional shape are the main factors that affect fiber flexibility. Fiber flexibility can be measured in various ways such as single fiber methods and wet and dry specific volume procedures.

6.3 Paper Structure

Paper consists of a three-dimensional pore system and a skeletal fiber system. There are

approximately one million wood fibers in one gram of a typical paper made by chemical pulp [12,13]. In a paper structure, a fraction of the volume is occupied by fibers and fillers and the rest is void volume.

$$V = V_f + V_v \quad (6.2)$$

where

V = total volume

V_f = fiber volume

V_v = void volume

Table 6.2 shows typical void volume of different paper grades. The strength and stiffness of the paper structure comes from the network of fibers. This network can be considered to be of infinite dimensions in the plane of the sheet. The dimensions of the network in the direction perpendicular to the sheet plane are limited [14].

Due to the void volume in the structure, paper is a compressible material. For this reason, converting processes consolidate the sheet even further and give a more compact surface.

6.3.1 Fiber Orientation and Directionality in Paper

The travel direction of fabric and sheet on the paper machine is called machine direction (MD). Cross-machine direction (CD) is the direction at right angle to the MD. Machine direction is also called the grain direction. Because of the moving paper web during manufacturing, the orientation of the fibers in the plane of paper is not totally random. Due to moving direction of the slurry and forming fabric, the fibers tend to orient more in the

TABLE 6.2. Typical Air Volume of Some Paper Grades [2].

Paper Type	% Air Volume
Groundwood	63.1
News	53.1
Greaseproof	43.2
Bond	34.2
Glassine	13.0

machine direction during forming. As a result of this orientation preference, paper strength is usually greater in MD direction than the CD direction. Another reason for the strength difference is the higher tension exerted in the MD direction during drying. The ratio of MD/CD strength is 1.5-2.0 for fourdrinier machines and can be higher for cylinder machines. For this reason, test direction should be indicated in physical and optical tests. Since MD/CD fiber orientation may cause curl in the paper, it is desirable for most paper grades that this ratio is close to 1 [2].

The degree of fiber orientation is defined as the angle that fibers make relative to a particular direction. The fiber orientation distribution $f(\theta)$ is given by:

$$f(\theta) = \frac{1}{\pi} \sum_{i=0}^{\infty} a_i \cos 2i\theta \quad (6.3)$$

where $a_i = 1$ and θ is the direction of a particular fiber relative to the machine direction. Other functions are also used to describe fiber orientation of fibers in paper such as the relative deviation from 45° of the average angle that the fibers make with the MD [15].

Fiber orientation has a big influence on paper tensile properties. It was reported that changing the degree of fiber orientation from 0.5 to 2.5 increased the strength three times. Increasing fiber orientation in MD increases the strength and elasticity in that direction and decreases them correspondingly in CD. It was shown that the tensile strength MD/CD ratio represents a good correlation with the degree of fiber orientation. As a result, it was suggested that the tensile strength MD/CD ratio in paper be used as a measure of the degree of fiber orientation for a given pulp furnish. Htun and Fellers proposed that the tensile strength MD/CD ratio for sheets dried under biaxial restraint be used as an index of fiber orientation, with the term fiber orientation ratio being adapted for this ratio. Figure 6.10 shows the MD/CD ratio for the specific elastic modulus, the tensile index and the compression index of sheets made of bleached sulfate pulp and dried under biaxial restraint against the fiber orientation ratio dried. It was also

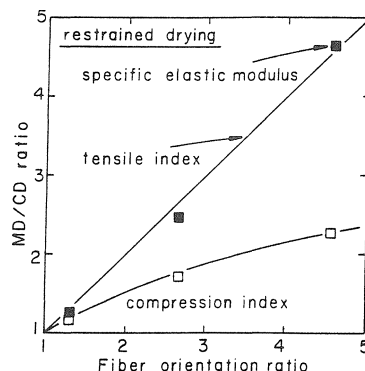


FIGURE 6.10. The MD/CD ratio for the specific elastic modulus, the tensile index and the compression index of sheets made of bleached sulfate pulp and dried under biaxial restraint against the fiber orientation ratio [19].

reported that the elastic properties are directly related to shrinkage during drying. The ratios of elastic modulus MD/CD and the tensile index MD/CD increases if the paper is freely dried. The draws (speed differentials) by the various parts of the machine may also lead to an MD stretching of the sheet. Uneven fiber distribution results in uneven printing [16–18].

The techniques that have been used to quantify the degree of fiber orientation includes averaging fiber segment orientations, X-ray diffraction, light scattering, regular and zero-span tensile strength and Fourier transform methods [20,21]. Figure 6.11 shows a typical fiber orientation distribution.

Paper has an anisotropic structure: its mechanical properties depend on the direction. Fiber orientation and shrinkage during drying

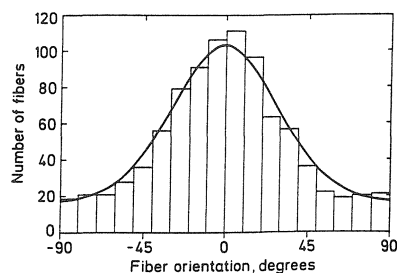


FIGURE 6.11. Typical fiber orientation distribution for a sheet of 80 g/m² [22].

(drying restraints) are the main causes for anisotropy. The anisotropic structure of sheet leads to mechanical properties that are different than those of homogeneous isotropic materials. However, in some respects and on a macroscale, paper can be considered as an orthotropic structure in which there are three mutually perpendicular directions of material symmetry. The material properties are different in each direction. For paper, these three directions are machine direction (MD), cross machine direction (CD) and the thickness direction (Z) as shown in Figure 6.12 [23–25]. The majority of fibers are aligned approximately parallel to the plane of the sheet mostly in machine direction. This alignment depends on the machine configuration, pulp and velocity difference between the jet speed and forming fabric. The biased fiber orientation in MD makes the paper stiffer and stronger in MD than in CD. Moreover, the sheet is subjected to restraining forces in MD in pressing and drying. The paper dries with little restraining force in CD. The strength and stiffness in the Z direction is the weakest: generally 1/10–1/50 of the inplane properties. In general, the in-plane and out-of-plane properties are not related. Some paper products are manufactured in layers. Each layer may have different properties [19].

The variation of paper properties is generally more abrupt in the CD compared to MD. In other words, variations occur more slowly in the MD. The reasons for more sudden variations of properties in the CD direction are possibility of uneven flow of stock from the headbox, uneven wet pressing, uneven tension during drying and edge effects [2].

The directionality of paper can be determined in various ways in addition to tensile

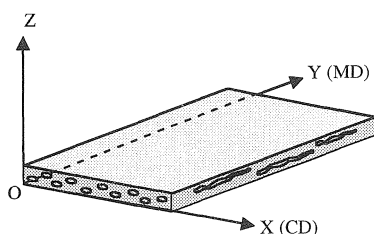


FIGURE 6.12. Three major directions of paper.

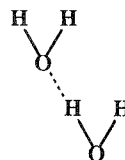


FIGURE 6.13. Hydrogen bonding.

test. If one side of the sheet is wetted, the expansion of the wetted side causes a curl. The expansion is more in the CD and the axis of the curl is almost always parallel to the machine direction. Stiffness in MD is greater than CD. During the bursting test, MD has less stretch and rupture line will be at right angle to MD.

6.3.2 Fiber Bonding and Cohesiveness

The main bonding of fibers in a sheet structure is provided by hydrogen bonds between fibers. Figure 6.13 shows schematic of hydrogen bonding. Cohesion is the hydrogen bonding force that holds the fibers together in the sheet structure which gives the paper its strength and integrity. Cohesion increases the frictional sliding resistance between adjacent fibers against a shearing action as shown in Figure 6.14. Fiber cohesiveness depends on the surface chemistry and area as well as sheet density.

There are two types of cohesion: inplane cohesion (the force is in the plane of the sheet) and transverse cohesion (the force is perpendicular to the sheet plane). It is neither practical nor meaningful to measure individual fiber cohesiveness; therefore, cohesiveness is measured empirically as a sheet property. Methods to measure cohesiveness include percent cohesion test, Clark shear test, split cohesion test and peel cohesion method.

It is a well known fact that wet sheet after pressing does not have the strength of a dry



FIGURE 6.14. Cohesiveness is the resistance to shear provided by the hydrogen bonding of fibers during drying.

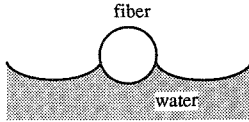


FIGURE 6.15. Surface tension on a fiber in forming section.

paper. This fact indicates that fiber bond in paper develops during drying. It is also worthy to note that if an organic solvent instead of water is used in papermaking, the resulting sheet would have very little strength after drying. This fact proves that hydrogen bonding is the major force that binds the fibers together in sheet structure. For example, glass fibers can not produce a strong sheet (although glass fibers are a lot stronger than wood fibers) due to lack of hydrogen bonding between glass fibers after water removal (for this reason, bonding agents are used to improve fiber bonding in glass sheets).

It should also be noted that unlike textile structures, there is no interlacing (as in weaving and braiding), interlooping (as in knitting) or forced entanglement (as in needlepunching) of wood fibers during paper sheet formation. Instead, fibers are brought together close enough to produce hydrogen bonds which is called the Campbell effect. At a solid content of around 8% during formation on the forming fabric (i.e. approximately where the wet line is on a fourdrinier), the surface tension force acting on each fiber is given by:

$$F = \gamma l \quad (6.4)$$

where γ is the water surface tension and l is the fiber length (Figure 6.15).

In the press section, more water is removed from the web such that water remains only between two fibers. Since there are two menisci between the fibers (Figure 6.16), the surface tension acting on the fibers becomes:

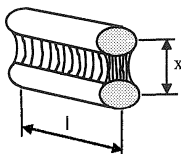


FIGURE 6.16. Surface tension between two fibers in press section.

$$F = 2\gamma l \quad (6.5)$$

This force pulls the fibers together. After drying, more water is removed from the sheet; however, the menisci remain continuous and pull the fibers even closer together. Around 25% solid content, the compacting pressure between the fibers becomes very high:

$$\Delta P = \frac{2\gamma}{x} \quad (6.6)$$

where x is the distance between two fibers. The pressure at this stage can reach to 100–200 atmospheres. Highly flexible fibers require lower pressures for bonding [2].

Important factors affecting the bond strength in a sheet structure are fiber areas in contact, number of bonds in contact area and strength of each bond. The level of bonding can be determined by light scattering techniques, conductivity tests and optical techniques.

6.3.3 Fiber Curl

In general, the fibers or fiber segments are not straight in a typical paper even if the shrinkage during drying is prevented. The fibers may be kinked or curled in or out of the plane of the paper which causes a decrease in paper stiffness. There are two types of curl: fiber curl (which is the curl of the entire fiber) and segment curl. In addition, fibers may partially be wrapped around other fibers. Some twisting of fibers also takes place [22,26–29]. Fiber curl affects the elastic properties of the sheet. Fiber curl can be measured with image analysis techniques.

6.3.4 Internal Stresses

Various stresses develop in the sheet internally during manufacturing. Swollen fibers, which are molded into a sheet in forming and pressing sections, are dried at different degrees of tension in the dryer section. Various parameters during manufacture of paper influence the level of internal stresses in the paper. Beating of pulp, wet pressing, variation in

fiber orientation and drying conditions all affect internal stresses in the paper. Increasing the degree of beating increases the internal stresses. This is because beating increases the density which increases the potential to shrink during drying. Wet pressing also increases internal stresses. Because of nonuniform drying rate in the thickness direction and in-plane tension forces, internal stresses develop in paper during drying. Restrained drying increases internal stresses. Internal stresses in paper can also be caused by deformation. Internal stresses can be partly released by humidity cycling and cyclic mechanical straining. Internal stresses affect the mechanical properties of paper [30,31].

6.3.5 “Wire” and Felt Sides of Paper

On a fourdrinier table, the side of the sheet next to the forming fabric is called fabric side (also called ‘wire side’ from the metal wire days). The other side is usually referred to as “felt side”. A paper formed on a fourdrinier table may have the two sides different in structure. This phenomenon is called “two sidedness”. Structural differences may include sizing, pigments, fines and fibrous composition. Optical differences include finish and reflectance. The fabric topography may cause marks on the sheet which is commonly known as “wire mark”. Wire mark can penetrate up to 60% of the total sheet thickness [2].

Paper is coarser and more porous on the wire side than the felt side. Usually there is a high percentage of long fibers on the wire side. This is because fines are washed from the wire side before the larger fibers are laid down. A typical bond paper has 18% fines on the felt side and 11% on the wire side. Uniformity of fiber length decreases two-sidedness. Differences in two sides of a paper can be viewed best under a microscope. These differences may dictate some rules during application or end-use of the paper. For example, print quality is better on the felt side which is smoother. Laminated boards are glued together on the coarse sides. Stamps are printed on the fabric side and gummed on the felt side.

6.4 Paper Wet-End Chemistry

Various materials and additives are added to the pulp at the wet end to help with formation and improve final paper properties. Some of these materials become an integral part of the paper structure [2]. A brief summary of these materials and processes is given below.

6.4.1 Sizing

Sizing can be done externally (surface sizing or coating) and internally. External sizing is done after sheet is made by filling in pores on surface of paper with glutinous material such as starch, glue, PVA, wax and fluorochemicals. Internal sizing is done by additives at the wet end. Sizing makes the paper more hydrophobic by reducing the spread of fluids, thereby improving “holdout” in printing and coating.

Sizing systems on papermachines producing acid sheets normally utilize alum chemistry which can reduce paper life as a result of residual acid-forming salts. Sizing systems on alkaline sheets can minimize or eliminate alum, thereby enhancing sheet life while reducing the corrosive effects of low pH on the machine itself.

6.4.2 Dry and Wet Strength Agents

Dry strength agents such as starches, natural gums and emulsified latex are used to give sheets more tensile strength by improving bonding between fibers.

Wet strength agents provide paper strength in wet conditions. They prevent disruption of fiber to fiber bonds in water and therefore retain integrity of paper. Wet strength additives include Kymene, formaldehyde resins, and polyamines, which are all hydrophobic.

Dry and wet strength agents are added to the pulp before the headbox.

6.4.3 Flocculants and Deflocculants

Flocculants, such as cationic polymers, are the materials that bring particles together by

neutralizing negative surface charge. They enhance retention during formation of sheet. The primary flocculant used in papermaking is alum.

Deflocculants, such as anionic polymers, repel negative surface charges. They improve formation.

6.4.4 Fillers

Fillers are used to change the optical and strength properties of paper. There are various types of fillers including clays, calcium carbonate, titanium dioxide, hydrated alumina, talc or amorphous silica. Opacity, density, brightness and whiteness of a sheet are increased by addition of fillers. Fillers are sometimes used as low cost fiber substitutes since they close pores and provide smooth surface.

6.4.5 Drainage and Retention Aids

Drainage aids are used to prevent fines and small fibers from plugging the forming fabric which may slow down the speed. They attach fillers, fines and small fibers to large fibers. Polymers and additives that are used as drainage and retention aids include polyamines, polyamides, gums and cationic starches.

6.4.6 Formation and Dispersion Aids

These are additives that keep fibers from forming clots or flocs. Examples of these aids are gums and starches. Some of these materials make fiber surface repulsive or slippery to break up the flocs.

6.4.7 Other Agents

Several other materials are also used in the wet-end to help with formation and improve final paper properties. These include dimensional stability agents, biocides, pitch control agents and defoamers. Pitch comes from fatty acids and resin acids in the wood. Soaps, solvents and talc are used to control pitch.

6.5 Paper Properties and Testing

Paper testing can be done on-line or off-line. On-line measurements are done on the paper machine while the paper is being produced. These measurements are automated and computerized continuous measurements. They are in general indirect measurements which means that a particular property is determined based on the measurement of another related property. Examples of on-line testing are strength tests, caliper, grammage, moisture content, color, brightness, opacity and gloss [1,2].

Off-line measurements are more direct measurements which are usually done in the lab. Several institutions have established test methods for the paper industry. Main sources of test methods in the US are TAPPI (Technical Association of the Pulp and Paper Industry), ASTM (American Society for Testing and Materials) and ANSI (American National Standards Institute). TAPPI methods are the major source of test methods for the pulp and paper industry [32]. Various institutions in other countries also have established test methods such as CPPA (Canadian Pulp and Paper Association), SCAN (Scandinavian Pulp and Paper Association) and DIN (German Standards).

6.5.1 Tensile Properties

Tensile strength is probably the most important property of a paper. Tensile strength is especially important for printing papers, bag and wrapping papers and adhesive tape papers. For paper, tensile strength is expressed in breaking load/unit width. The unit is N/m in SI system, kg/m in metric system and lb/inch in English system.

Paper is not a homogeneous material. Therefore, there is tensile strength variation in paper which depends on the location. Stress in a local region can be calculated as follows [33]:

$$\sigma_1 = b\sigma_a \quad (6.7)$$

where σ_1 is the local stress, b is the stress

concentration factor and σ_a is the average stress acting on the cross-section of the sheet.

Sheet strength depends on two components: intrinsic fiber strength and bond strength. For well-bonded paper, sheet strength is directly related to fiber strength. Improving fiber bonding increases the strength contribution from the fibers. The specific strength of paper increases with density [6,34,35,36]. The relationship between fiber and sheet strength can be written as:

$$T_s = \Phi T_f \quad (6.8)$$

where

T_s = sheet strength

Φ = intensity function which shows the efficiency of fiber strength utilization in the structure

T_f = fiber strength

Sheet failure occurs when a local stress exceeds the fiber failure limit in that location. Paper physicists developed relatively simple models to predict the sheet stress based on individual fiber strength. Treating the sheet as a two phase structure made of fibers and void volume, sheet strength can be approximated as follows:

$$T_s = (1/3)V_f T_f \quad (6.9)$$

where V_f is the fiber volume fraction which is equal to the ratio of sheet density to fiber density. Theoretically it was found that maximum tensile strength of a random 2-D sheet is equal to one third of strength of individual component fibers [34].

In practical end use, sheet strength is judged by its strength per unit width for a given grammage which is called tensile index. The tensile index is equivalent to expressing the sheet strength as stress divided by density. Therefore, the tensile index or the specific sheet tensile strength is given by:

$$T_s^* = T_s / \rho_s \quad (6.10)$$

where ρ_s is the sheet density.

Similarly, the specific fiber strength can be calculated as:

$$T_f^* = T_f / \rho_f \quad (6.11)$$

where ρ_f is the fiber density.

As a result, the relation between sheet specific strength and fiber specific strength can be written as:

$$T_s^* = (1/3)T_f^* \quad (6.12)$$

This equation refers to the case where the full fiber strength is utilized in the sheet. Considering the efficiency function, Φ^* ,

$$T_s^* = (1/3)\Phi^* T_f^* \quad (6.13)$$

It was experimentally shown that for normal paper grades, Φ^* is between 0.2 and 0.7. The value increases with fiber beating until a certain point (Figure 1.8) and with the intensity of wet pressing of the wet web [37,38].

Figure 6.17 shows typical tension and compression curves of a paper made of bleached kraft board. The area under the tensile stress-strain curve is defined as the tensile energy absorption (TEA) of the sheet. It is a measure of toughness (J/m^2) which is important for grocery bags. Increasing stress or strain or both increases toughness.

Tensile strength of a paper is affected by the cellulose content (i.e. pulp yield) in the fibers. Increasing sheet density by beating increases the modulus and tensile strength of the sheet to a certain extent. This is because the fiber strength is better utilized towards the sheet strength. Increasing density also increases the strain to failure. However, it is important that fiber length loss is controlled during beating.

Paper stretch is an indication of toughness.

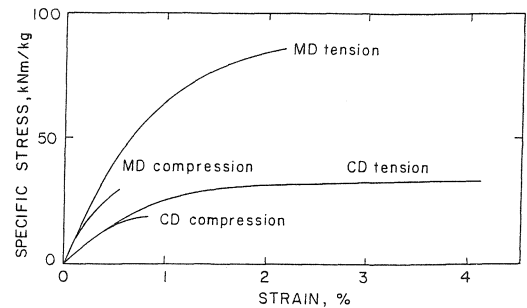


FIGURE 6.17. Tension and compression stress-strain curves of a bleached kraft paperboard [13].

Stretch becomes especially important for papers that are subject to stress during the end-use such as twisting papers, toweling, cable wrapping paper and corrugating papers. Stretch is increased by beating and reduced by short fibers, filling, heavy pressing and high tension during drying. Breaking strength of paper is approximately proportional to grammage.

Fiber orientation has a big influence on paper tensile properties such as elastic modulus and tensile strength (Section 6.3.1); however, it does not affect the strain to failure. In general, tensile strength is greater in MD than CD because of fiber alignment and strain developed during drying. Typical MD/CD tensile ratio is 2 for fine paper and 0.5–1 for tissue. Drying restraints basically influence the strain to failure and elastic modulus. For a freely dried sheet, the strain to failure is linearly related to the sheet shrinkage during drying and independent of the fiber orientation [16,39].

Wet Tensile Strength

Moisture affects tensile strength. Maximum tensile is obtained between 20–30% relative humidity (RH). Below this range stress intensification and above this range plastification make the paper weaker. Figure 6.18 shows

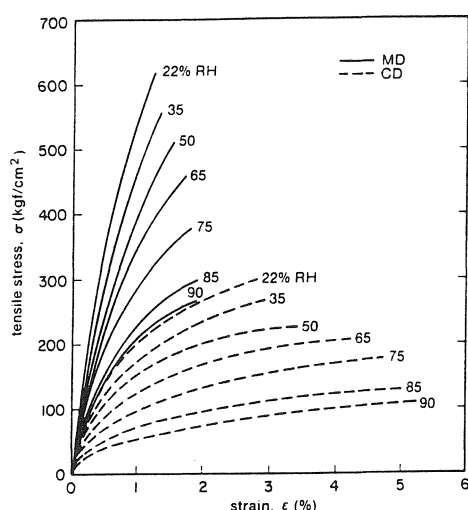


FIGURE 6.18. Effect of moisture on tensile strength of kraft linerboard [40].

the effect of moisture content on tensile strength of kraft linerboard. At complete saturation, the tensile strength of paper becomes practically zero. Sizing provides temporary wet strength to the paper. Some papers must have permanent wet strength due to their applications. Permanent wet strength is provided by wet-strength agents such as melamine-formaldehyde (MF), urea-formaldehyde (UF), Kymene (polyamide-polyamine-epichlorohydrin) and others. TAPPI T-456 is designed to measure the wet tensile breaking strength of paper and paperboard.

Z-Direction Tensile Strength (Internal Bond Strength)

Z-direction tensile test measures the internal bonding strength of paper. Internal bonding strength is especially important for paperboard for glue bonding at carton side seams, delamination on scoring and use of high tack coatings.

TAPPI T-541 is designed to measure the internal bond strength of paperboard. A one-inch square paper is attached to metal blocks of the jaws on each side by double sided tape. A tensile load is applied to the paper using a standard material testing system. This method can also be used to evaluate coated papers. Figure 6.19 shows typical stress-strain behavior of paper in Z-direction.

Zero-Span Tensile Strength

This test is basically done to determine a strength index of the longitudinal structure of individual fibers in a pulp test handsheet (TAPPI T-231). A zero-span jaw attachment is used in a regular tensile testing system (Figure 6.20). The ratio of normal to zero-span breaking lengths provides an index of fiber cohesiveness. The method is also useful to determine the relative degree of orientation of fibers in MD and CD.

6.5.2 Compression Behavior

Compression stresses may cause the paper to fail because of possible structural instabilities. Compression properties of paper are

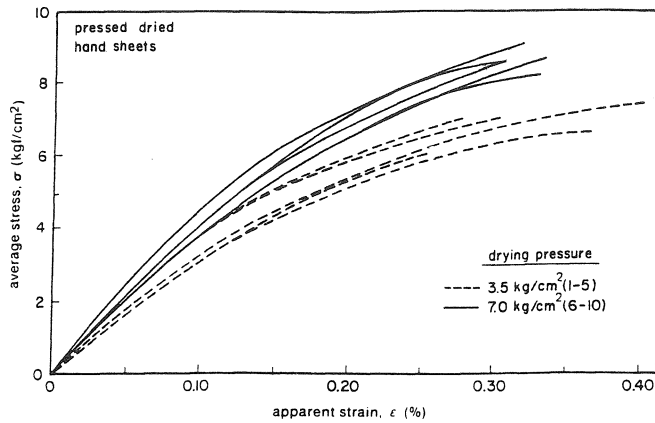


FIGURE 6.19. Typical stress-strain behavior of paper in Z-direction [40].

especially important for cartons and corrugated boxes since they are supposed to protect their contents against compression forces during packing, storage and distribution. Bending stiffness and edgewise compression strength are the two most important material properties for packaging boards [13,41–44].

Figure 6.17 shows the compression stress-strain curves of a typical bleached kraft paperboard. Since the elastic modulus is the same in tension and compression, the initial load bearing capacity is the same in both modes. However, the yield point, strain to failure and tenacity (stress at failure) are lower in the compression mode. Flaws in the paper structure are detrimental to compression strength. Compression failure usually does not cause any bond or disruptive fiber failure in the structure.

Increasing sheet density by beating increases the modulus and compression strength of the sheet due to better utilization of fiber strength towards the sheet strength. As the

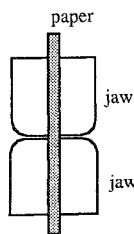


FIGURE 6.20. Arrangement for zero span test.

density increases, the bonding also increases which causes the failure to be fiber dominated. Compression strength is independent of the pulp yield which means that both cellulose and lignin bear the compression load. Compression strength increases with increasing wet pressing. Adsorption of moisture reduces the compression strength [13,42].

Different principles of compression testing have been developed including ring crush test, short span test, blade support test and plate support test. Edgewise compressive properties are critical for the performance of corrugated containers. Figure 6.21 shows a specimen, which is supported between two fixed parallel

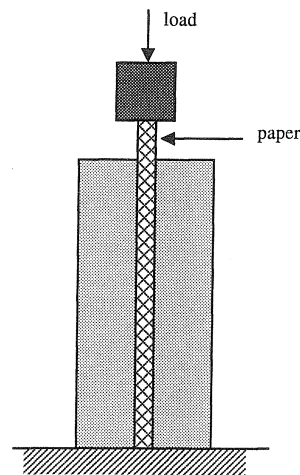


FIGURE 6.21. Schematic of edgewise compression test.

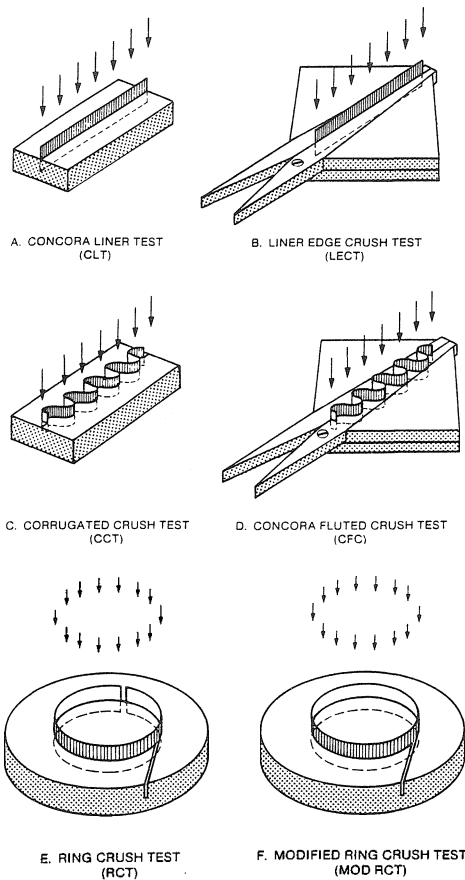


FIGURE 6.22. Various configurations for edge crush test [40].

plates, under load in edgewise compression testing. Other various test configurations are shown in Figure 6.22. Increasing basis weight increases edge crush resistance.

Flat Crush Tests

Flat crush test measures the flute rigidity of corrugated board. TAPPI has developed two test methods to measure the crush resistance of corrugated materials: T-808 (Flat Crush Test of Corrugated Board, Flexible Beam Method) and T-809 (Flat Crush of Corrugated Medium, CMT Test). A compressive load is applied normal to the plane of corrugating medium as shown in Figure 6.23. Maximum compressive force is measured for the failure which is defined as the collapse of the

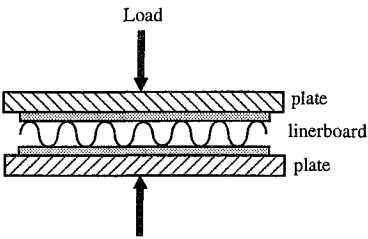


FIGURE 6.23. Schematic of flat crush test of corrugated linerboard.

side walls of the corrugations. Figure 6.24 shows the load-deformation curve for a corrugated linerboard.

Short Column Crush (SCC) Test

The SCC test method is developed to test the in-plane crush resistance of side panels of a corrugated container under a stacking load (Figure 6.25). This test gives an indication for the stacking strength performance of boxes.

6.5.3 Visco-Elastic Behavior

Like textile materials, paper is a visco-elastic material which means that its elongation-recovery properties are time dependent. For example, if a paper breaks in 4 seconds under a 10 kg load, it takes 11 min to break under 9 kg, 14 hours under 8 kg and 220 days under 4 kg [2].

Paper creeps with time under constant load. There are two types of creep (Figure 6.26): primary creep which is recoverable and secondary creep which is non-recoverable

A spring-dashpot model is representative

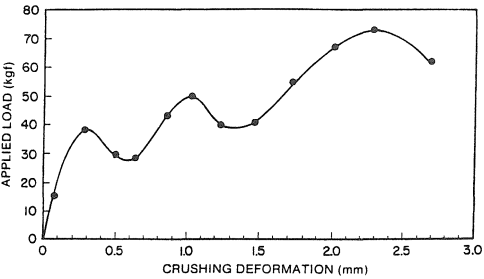


FIGURE 6.24. Typical flat crush load-deformation curve of a corrugated linerboard [40].

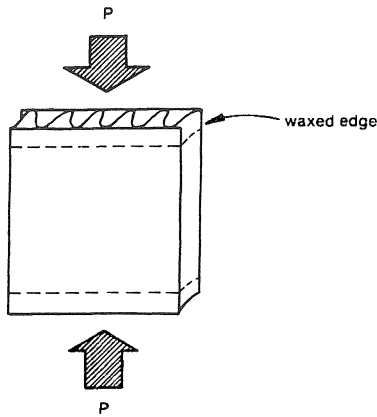


FIGURE 6.25. Short column crush test [40].

of the visco-elastic behavior of paper (Figure 6.27). Upon application of load, springs elongate and dashpot does not move. As the load is increased, the dashpot also starts moving. Due to the dashpot in the system, creep is completely recovered.

6.5.4 Bending Stiffness (Flexural Rigidity)

Bending involves both compression and tension (Figure 6.28). The bending moment for a given curvature is related to the stress-strain properties. Flexural rigidity for a material is the product of elastic modulus and moment of inertia. Paper stiffness is expressed in two ways [2]:

- (1) Paper stiffness: flexural rigidity/grammage

$$(2) \quad \text{Paper Stiffness} = \frac{ET^3c}{12d^2} \quad (6.14)$$

where

E = elastic modulus
 T = thickness of sample
 c = width of sample
 d = length of sample

Most stiffness measurements are based on either measurement of bending angle of a paper strip under a force or measurement of force required to bend the strip at a certain angle. TAPPI methods to measure stiffness are T-451 (Flexural Properties of Paper, Clark Stiffness), T-489 (Stiffness of Paper and Paperboard, Taber-type Stiffness Tester) and T-543 (Stiffness of Paper, Gurley Type Stiffness Tester).

Figure 6.29 shows the bending moment-curvature curves for bleached and unbleached sulfate board beams. The curves are analogous to tension and compression curves in Figure 6.17. Bending stiffness is represented by the initial slope. The maximum moment is the bending strength.

Bending stiffness is especially important in most packaging applications where paperboard is used. For these applications high bending stiffness both in MD and CD is required. Papers that must stand up during use, such as index Bristol, card, typing papers, playing cards, etc., require good stiffness. Corrugating medium and linerboard also need to have good flexural rigidity since they are used in structural constructions. Stiffness is

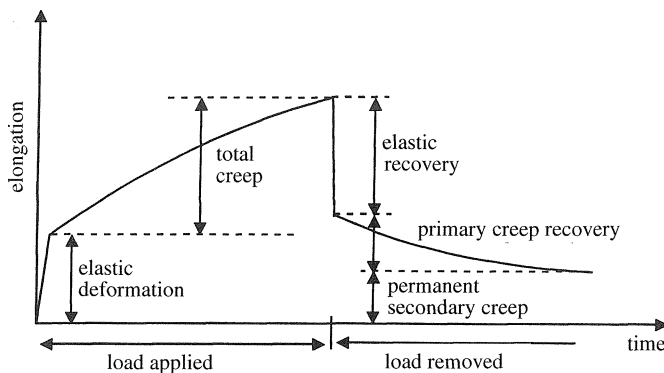


FIGURE 6.26. Primary and secondary creep.

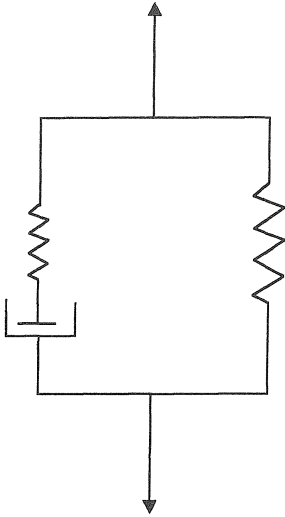


FIGURE 6.27. Spring-dashpot model for visco-elastic behavior of paper.

not wanted in tissue, toweling, printing papers and labels.

Bending stiffness depends on thickness, sheet construction, mass fraction of mixed pulps and degree of beating. Beating (refining) and additives, such as starch, increase stiffness. In general, multilayer sheet structures have greater bending stiffness than single layer structures [23,45].

6.5.5 Bursting Strength

Bursting strength is a measure of resistance against rupture of paper. It is defined as the hydrostatic pressure to rupture the paper as the pressure is increased at a constant rate. Bursting strength is expressed in kiloPascal (kPa) or psi (pounds per square inch).

Fiber length and bonding affect the bursting

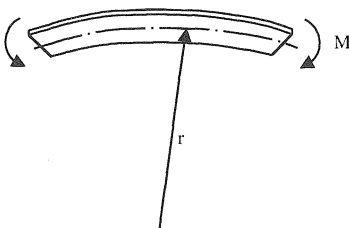


FIGURE 6.28. Bending of a paper strip; r = radius of curvature, M = bending moment.

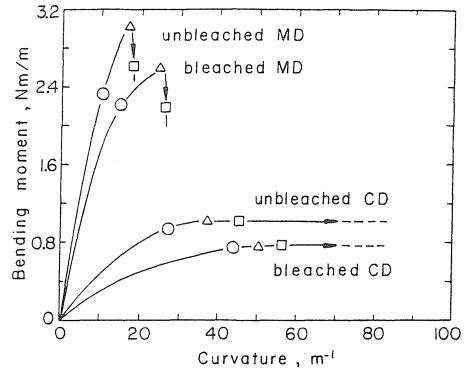


FIGURE 6.29. Bending moment versus curvature for bleached and unbleached paperboard [13].

resistance. Beating and pressing density increase burst strength. Loading conditions in bursting is similar to tensile loading. Therefore, if tensile strength increases, bursting resistance also generally increases. The failure occurs along the MD direction in bursting which is the direction with the least amount of stretch.

TAPPI has developed several test methods for bursting resistance of different papers: T-403 (Bursting Strength of Paper) which is also known as Mullen Burst test, T-807 (Bursting Strength of Paperboard and Linerboard), T-810 (Bursting Strength of Corrugated and Solid Fiberboard).

6.5.6 Tear Resistance

Long and coarse fibers increase tear resistance. During tearing a paper, fibers are broken and pulled out of the structure. Increasing grammage and flexibility increase tear resistance. Creped sheets also have higher tear strength.

TAPPI test method T-414 (Internal Tearing Resistance of Paper, Elmendorf Type Method) is designed to measure tear resistance of paper.

6.5.7 Folding Endurance

Folding endurance is defined as the number of folds that a paper will withstand before failure. Folding endurance is important for the papers that are subject to bending, folding and creasing. TAPPI method T-423 is designed to

measure folding endurance of papers with a thickness of 0.25 mm or less.

6.5.8 Formation

The term “formation” is used to describe the uniformity of fiber distribution in a paper sheet. The more uniform is the fiber distribution, the better is the formation. Formation is usually evaluated by inspecting the paper against a light source and estimating the variations in the transmitted light intensity. Formation testers, based on light transmission, are generally used for light weight grades. Formation generally correlates to other end use properties such as printability, smoothness or mechanical strength. Sheet formation affects the wet pressing and drying processing efficiencies.

A dimensionless Formation Number, N_f , is defined as an indication for formation. Formation number is based on local grammage variations and is defined as the coefficient of variation of local grammage W_A :

$$N_f = \frac{\sigma(W_A)}{\bar{W}} \quad (6.15)$$

where A indicates the size of the measuring area. As the uniformity in the sheet increases, the formation number decreases. A more accurate way of characterizing the formation is the formation spectrum (floc-size spectrum) which shows the contribution of different floc sizes to the variations in local grammage [46].

6.5.9 Dimensional Stability, Curl and Twist of Paper

Dimensional stability of paper is critical for most applications including papers for printing, maps, templates, wallboard tape, abrasion, etc. Dimensional instability can be reduced by several methods [2]:

- using fibers that do not expand
- running a bulky sheet
- reducing internal bonding
- using bonding agents
- coating the paper to make it

impervious to water

- restraining CD shrinkage during drying

Ambient moisture changes can cause the paper and paperboard to curl or twist by deviating from the flat structure (Figure 6.30). This may cause problems in converting or printing operations where a flat material is required for good runnability.

Curling may be due to a moisture gradient in thickness direction or an asymmetrical structure through the thickness. If a swelling agent like water or water vapor is absorbed, the sheet volume dilates. Restraining of swelling causes stress development in the material [23,47]. Curl free paper is necessary for photosensitive papers, printing papers, coated papers and adhesive papers such as labels. TAPPI methods to test for curl are T-466 (Degree of Curl and Sizing of Paper) and T-520 (Curl of Gummed Flat Papers).

Twisting of paperboard may take place due to nonsymmetrical orientation of the layers above and below the midplane as shown in Figure 6.30.

6.5.10 Grammage (Basis Weight)

“Grammage” is a relatively new term introduced in the metric system to replace the conventional “basis weight”. Grammage is the

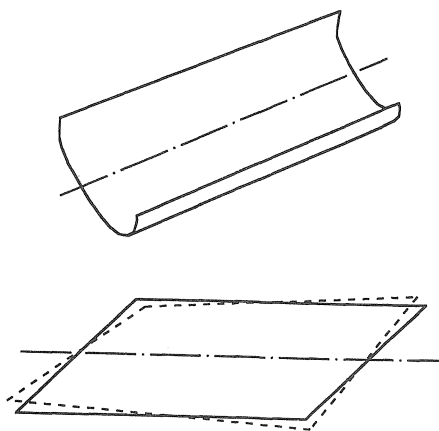


FIGURE 6.30. Paper curl (top) and twist (bottom).

weight of the paper (in grams) per unit area (one square meter). In the Standard system, the conventional basis weight (ream weight or substance) is based on the mass in pounds of a ream for a given sheet size (a ream can have 480 or 500 sheets). Different paper grades have different ream sizes. Different end use of each type of paper generally requires different grammage. Table 6.3 lists typical grammage values for different types of paper.

Grammage of paper is very important because all physical properties and many optical and electrical properties are affected by the change in grammage. Paper is sold by weight; therefore, the lowest grammage paper that will do the job is usually selected by the customer.

There are always grammage variations in a typical paper. If the fiber distribution in the paper were random, then the local grammage would be represented by a Poisson Distribution. However, the process of sheet formation causes deviations from such a distribution [46].

The grammage variation is determined by three major methods: weighing, absorption of light and absorption of beta radiation. Grammage measurement is done on-line with beta radiation. Grammage measurement is affected by the relative humidity. The standard conditions in the US are 50% relative humidity (RH) at 23°C. It is reported that a 10% change in RH around the standard RH can cause 1% change in grammage [2,46].

Grammage variations can cause less effective water removal and density fluctuations due to uneven wet pressing. Another effect of grammage variation is uneven drying conditions and variations in drying stresses.

TABLE 6.3. Values of Grammage for Different Paper Types [2,48].

Paper	Grammage (g/m ²)
Tissues and paper towels	16 to 57
Glassine	30 to 75
Newsprint	45 to 70
Grocery bags	49 to 98
Fine paper	60 to 150
Cement bag	120 to 195

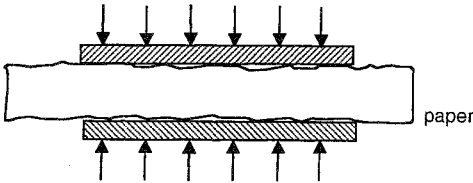


FIGURE 6.31. Measurement of sheet thickness.

6.5.11 Thickness (Caliper)

Uniformity of paper thickness is important especially for printing and writing. Sheet thickness is the perpendicular distance between the two surfaces of the sheet. The thickness is measured by placing the sheet between two parallel plates under a certain pressure (Figure 6.31). The pressure is 50 kPa in the standard TAPPI test method, T-411 Thickness (Caliper) of Paper, Paperboard and Combined Board. In Europe, 100 kPa pressure is applied. The unit for thickness is micrometer (μm) in the metric system and points (0.001 inch) in the Standard system. Table 6.4 shows typical caliper values for different papers. In general, as the grammage of paper increases, the thickness also increases as shown in Figure 6.32. Sheet thickness affects physical, optical and electrical properties.

6.5.12 Sheet Density and Specific Volume

Sheet density is a structural parameter that affects mechanical properties of paper. Sheet density is affected by both the total retention and the uniformity of retention of the fiber and fillers in the furnish. Total retention affects the finished optical properties of the sheet, and the

TABLE 6.4. Examples of Thickness Values for Different Papers [2,48].

Paper Sample	Thickness (μm)
Glassine	25–70
Facial tissue	55–75
Newsprint	75–95
Typing or ditto paper	80–120
Cement bag	230–635
Hardbound book cover	700–770

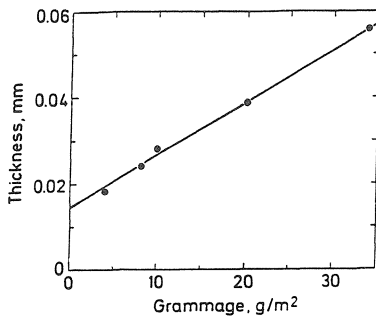


FIGURE 6.32. Thickness versus grammage for a typical paper [22].

uniformity of the fiber-fines-filler distribution can have an effect on top to bottom uniformity of density.

Sheet density is determined from the thickness and grammage measurements of the paper [49]:

$$\rho_s = W/t \quad (6.16)$$

where ρ_s is the density (g/cm^3), W is the grammage and t is the thickness. Increased beating of pulp increases the density of the sheet. The typical relation between sheet density and grammage is shown in Figure 6.33. In low grammage region, density is proportional to the grammage. In the higher grammage region, grammage does not have much effect on density. Short, flexible and highly fibrillated fibers give a dense sheet. Density is also affected by the amount of fiber bonding, amount of void volume and calendering.

Every optical and physical property are affected by sheet density. Sheet density is related to porosity, rigidity, hardness and

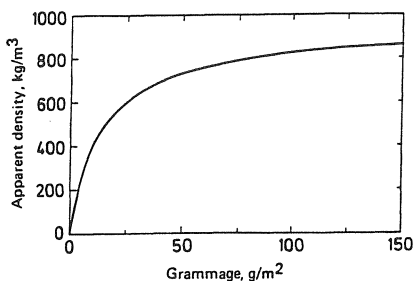


FIGURE 6.33. The relation between sheet density and grammage [22].

TABLE 6.5. Specific Volume (bulk) of Some Papers [2].

Paper Type	Specific Volume (cm^3/gr)
Bulky groundwood	3.0
Unbeaten sulfite	1.8
Glassine	1.0
Cellulose	0.65

strength of paper. Tensile and burst strengths are directly related to density; tear and porosity are inversely related. Specific volume or bulk is the reciprocal of density. Table 6.5 shows typical specific volumes of some papers.

6.5.13 Porosity and Permeability

Paper is a porous structure formed by a network of fibers. Therefore, paper is a two-phase structure in which pores and voids between the fibers are an important part of the structure. The total apparent pore volume can be calculated from the solid phase density and apparent density of the sheet as follows:

$$\begin{aligned} \text{Porosity} &= \frac{\text{pore volume}}{\text{total volume}} \quad (6.17) \\ &= 1 - \frac{\text{bulk density}}{\text{solid density}} \end{aligned}$$

Although pure cellulose has a density (specific gravity) of 1.5, paper has a low density of 0.5–0.8 due to its porous structure. Commercial papers contain about 70% air.

Pore volume is related to equivalent pore radius (EPR) which is used to compare paper porosities. Equivalent pore radius is defined as the radius of a single pore of length equal to the sheet thickness that would give the same flow as the average value for all the pores in a unit area of the paper [2]. The pore structure of a typical paper can be represented by a logarithmic normal distribution. This distribution is affected by beating, calendering and mixture of fibers in the pulp [12].

Pores affect the optical properties of the paper since they separate surfaces that controls light scattering. Pores also influence liquid absorption which is critical in many applications. Porosity is important for bag papers,

cigarette papers, filter papers and insulating papers.

Air Permeability of Paper

Flow of air through paper is directly proportional to the pressure difference, time, effective area of the specimen and the fourth power of the pore radius and indirectly proportional to sheet thickness. Air resistance is defined as the time necessary for a specific air volume to pass through the sheet, or, the air volume that passes in a specific amount of time. Increasing solid fraction increases air resistance. Air resistance depends on fiber type. TAPPI T-460 is developed to measure the air resistance of paper.

6.5.14 Surface Texture (Smoothness and Roughness)

Absence or presence of surface irregularities makes the paper smooth or rough, respectively. Gloss is related to smoothness both of which can be produced by supercalendering. Machine glaze (MG) in the paper can be obtained via Yankee dryer. Other factors affecting sheet smoothness are fiber source, pulping process, beating, shake of fourdrinier tables, forming and press fabric surface, wet pressing, filling, surface sizing and pigment coating. In general, use of short slim fibers in the furnish, promotes smoothness in the paper sheet.

Smoothness is important for quality printing, transfer of adhesives and non-slip packages, among other things. Smoothness can be measured in various ways including direct methods, optical contact methods and air leak methods. Most common methods are Sheffield (macro) and Parker Print (micro) smoothness. TAPPI T-479 is designed to measure the smoothness of paper with Bekk Method.

6.5.15 Absorbency and Swelling Properties (Water Resistance Tests)

Paper is a hygroscopic material which means that it reaches an equilibrium with the surrounding humidity by taking up or giving

off water with time. The sorptivity of paper comes from the fact that the paper structure is porous and the fibers can take up liquids by swelling. Thus, both phases in a paper structure are absorbent. The quantity of liquid absorbed is given by:

$$q = q_p + q_f \quad (6.18)$$

where q_p is the sorption into the pores and q_f is the sorption into the fibers. The liquid enters the pore network by capillary flow. The sorption of water into the fibers takes place by diffusion.

It has been shown that when the paper is immersed in a liquid, all the pore volume is accessible to the liquid through gaps between the fibers in the surface of the paper. The sorption of liquids by paper depends on the paper structure. In the case of aqueous liquids, chemical factors also become important. Liquid sorption increases the thickness of the paper due to swelling. The sorption mechanisms of sized and unsized papers are different. Paper sizing is done to make the paper hydrophobic. In unsized paper, there is a simultaneous pore and fiber take-up until the liquid penetrates to the reverse side of the paper with all the pores filled. After the liquids penetrate the paper, further sorption takes place due to continued swelling. In sized paper, water can not easily enter the pore system. Therefore, the only sorption mechanism is diffusion into and swelling of the fiber system [12].

There is a wetting delay before the water begins to be absorbed by the paper. It was reported that the wetting delay decreased with an increase in moisture content of the paper. Aging or heat treatment increases the wetting time. Pure cellulose has a wetting time of about 8 ms [50]. Results have also shown that the short time sorption of oils follows a square-root dependence on time.

Wettability

Measurements of contact angle of liquid drops on paper surface has been used to determine wettability of paper (Figure 6.34). TAPPI T-458 is designed to measure the surface wettability of paper (angle of contact

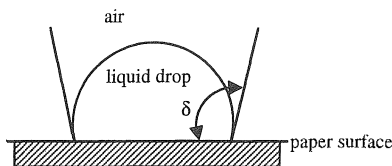


FIGURE 6.34. Contact angle between liquid drop and paper surface.

method). A small drop of water or ink is placed on the paper surface and contact angle after a certain amount of time is measured using a microscope. As the hydrophobicity of the surface increases, contact angle, δ , also increases. Wettable surfaces have smaller values of contact angle.

6.5.16 Optical Properties

Paper can reflect, transmit and absorb light (Figure 6.35). Optical properties of paper, such as opacity, reflectivity and transmittance, depend on type of pulp, amount of bleaching, fillers, coating, dyes and colored pigments, finishing operations and grammage. These in turn affect specific scattering coefficient and specific absorption coefficient of the paper. Increasing light scattering coefficient increases reflectivity and opacity and reduces transmittance.

Paper materials can be classified as optically thin or thick. A material is optically thin if most of the light is transmitted. If reflectivity is high (more than 50%) and the transmittance

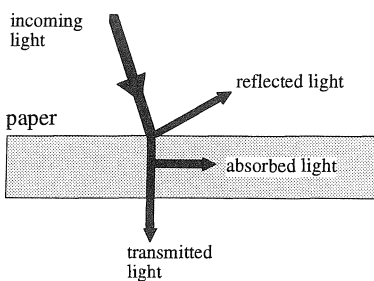


FIGURE 6.35. Interaction of light with sheet.

is low (less than 20%), then the material is said to be optically thick. Most papers belong to optically thick category [51,52].

For a single-layer structure, the reflectivity is increased with a high scattering coefficient or with a low absorption coefficient. Reflectivity is, by definition, independent of grammage. Opacity is increased by increasing grammage, light scattering coefficient and light absorption coefficient. However increasing absorption coefficient reduces reflectivity. Coating increases reflectivity and opacity. Type of filler also affects opacity.

A fine paper made of bleached chemical pulp has a high reflectivity and a low opacity due to its low scattering and absorption coefficients. To increase opacity of fine papers, usually a filler is included in the structure to increase light scattering coefficient. A paper with mechanical pulp has a higher opacity than a paper with bleached chemical pulp because of its high value of light scattering coefficient. However, its reflectivity is generally lower due to a higher value of absorption coefficient. Filler in a mechanical pulp would have a slight effect on opacity.

Coating is an effective way for improving both reflectivity and opacity. This is because a coating pigment has a high scattering coefficient and a fairly low absorption coefficient. Choice of pigment is important for the optical properties of the coating layer. It should be noted that not all coatings contain pigments.

Optical properties can be more readily altered by producing paper in layers. For example, a dark secondary or mechanical pulp may be placed in the center and covered with bleached chemical pulp for brightness. Playing cards contain a very dark center to ensure high opacity whereas the top layers have a high scattering coefficient for good reflectivity. In paperboard, a layered structure is generally used with a dark center and brighter top layer to increase reflectivity. In fact, paper is generally considered as a layered structure for the analysis of the optical properties. Kubelka-Munk Theory is a fairly accepted model to analyze the optical properties of multilayer paper structures. In

TABLE 6.6. Test Methods for Opacity and Brightness [53].

	Opacity	Brightness
International Organization for Standardization	ISO 2471	ISO 2470
USA	TAPPI T 425 T 519	TAPPI T 452
	ASTM D 589	ASTM D 985
Japan	JIS P-8138	JIS P-8123
Canada	CPPA E.2	CPPA E.1
Scandinavia	SCAN P8	SCAN P3

a multi layer structure, the optical properties are the result of contributions from each individual layer. Reflectivity of a multi layer structure depends on grammage [51].

Brightness

Brightness refers to the lightness of overall reflectivity of the paper. Brightness is a function of spectral reflectance, energy distribution of the illuminant and viewing conditions. TAPPI T-452 is used to measure the brightness of pulp, paper and paperboard. Table 6.6 shows standard test methods for opacity and brightness in different countries.

Opacity

Opacity is the property of the paper that prevents objects on the other side to be seen through the paper. The higher the opacity the less the objects can be seen through the paper. For example, if the print can not be seen through the page, the paper has high opacity. Opacity level is important especially for printing papers which require high opacity to prevent show-through. Bond, newsprint and writing papers also require high opacity. Some papers such as black photographic wrapping paper and paperboard are considered to be truly opaque. Low opacity is required in tracing and glassine papers. TAPPI T-425 (Opacity of Paper) and T-519 (Diffuse Opacity of Paper) are used to measure paper opacity. Opacity increases with grammage. The opposite of opacity is transparency which is defined as the ability of paper not to scatter light.

Gloss

Gloss is the lustrous or shiny appearance of the paper surface. Glossiness depends on the type of illumination, angle of incidence and reflection, and relative position of paper and observer. Rough surface may not seem as glossy. High gloss may make reading difficult because of glare. The opposite of gloss is matte. TAPPI T-480 (Specular Gloss of Paper and Paperboard at 75°) and T-653 (Specular Gloss of Paper and Paperboard at 20°) are used to measure gloss.

Color

Color for paper and paperboard is important for aesthetic purposes and marketing. Color can be characterized as a numerical value. TAPPI -527 is used to measure color of paper and paperboard.

6.5.17 Hand Properties (Softness and Hardness)

Softness is especially important for papers that come in contact with the skin such as sanitary and facial tissues and toweling.

There are several manufacturing parameters that affect softness. Minimum amount of bonding reduces stiffness and improves softness. High fiber strength is required to have adequate strength in soft sheets. Sulfite is considered a superior pulp for softness. Dry or wet creping methods are used in manufacturing to increase softness. Humectants (e.g. glycerin, diethylene glycol) and debonders (e.g. alkyl

amine salts, amides) are the additives that are used for softening. Sheets with nearly equal MD/CD tensile ratios also produce softer sheets.

TAPPI method T-451 (Flexural Properties of Paper) is used to measure softness. Softness is related to sheet density, rigidity, compressibility and surface smoothness.

Hardness can be defined as the resistance to indentation. Hardness is usually related to density. As a result, highly beaten pulps produce hard paper. Compressibility is the opposite of hardness. Compressibility is determined by the change in the thickness of the sheet.

6.5.18 *Electrical Properties*

Paper is used in a variety of electrical devices and related applications including capacitors, transformers, electronic circuit boards, wires, coil windings and cables. Paper has good dielectric properties. The electrical properties of paper are also important in direct electrophotographic and dielectric printing methods. Examples of other less known applications of paper are electrostatically assisted rotogravure, radio frequency or microwave dryers, and radio frequency or microwave moisture gages that involve the interaction of electromagnetic fields with paper and water systems [54].

Electrical properties of paper are anisotropic. The important electrical properties of paper and paperboard are volume and surface conductivities or resistivities, dielectric constants and loss, permittivities and dielectric breakdown strength. Dielectric constant is affected by the apparent density, fiber configuration and crystalline cellulose [55].

6.5.19 *Thermal Properties*

As the development and application areas of papers increase, the thermal properties of papers are also getting more attention. Important thermal properties of paper are specific heat, thermal conduction, thermal expansion,

combustion and thermal decomposition [56]. The specific heat of paper is similar to natural cellulosic fiber. Moisture content of paper affects these properties significantly. Thermal conductivity of paper depends on apparent density and absorbed moisture. Temperature and moisture changes affect paper expansion and shrinkage significantly. Of the two, moisture has more effect than temperature.

Paper and paperboard has a high flammability. This is an advantage for disposal by burning. However, flammability is a disadvantage for paper for applications where fire hazard is critical such as interior decoration. Methods to measure flame resistance of paper are TAPPI T-461 and ASTM D-777.

Heating changes the physical properties of paper leading to charring and pyrolysis. Increasing temperature increases the rate of deterioration and degradation. The thermal stability of paper can be improved by inclusion of inorganic materials.

6.5.20 *Finish*

Finish is a broad and relatively subjective term used to describe surface characteristics of paper that affect paper appearance and feel. It involves smoothness, softness, gloss and other properties that are explained above.

6.6 TAPPI Test Methods

The following is the breakdown of the TAPPI test methods [32]:

- Fibrous Materials and Pulp Testing
T 1-200 Series
- Paper and Paperboard Testing
T 400-500 Series
- Nonfibrous Materials Testing
T 600-700 Series
- Container Testing
T 800 Series
- Structural Materials Testing
T 1000 Series
- Testing Practices
T 1200 Series

Appendix E lists the major paper testing and research laboratories [32].

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6.8 Review Questions

- 1 What is the most important property of paper? Why?
- 2 What properties of wood fiber make it the most suitable fiber for papermaking? Explain.
- 3 What are the differences between hardwood and softwood tree fibers?
- 4 Calculate the maximum compacting force

between two fibers of 1 inch length (refer to Figure 6.16). Surface tension of water is 72 dynes per cm.

- 5** Are the fibers oriented randomly in the sheet? What are the factors that affect fiber orientation? Explain.
- 6** What is hydrogen bonding? How does it provide strength in the paper?
- 7** What are the possible solutions to the two sidedness of the sheet?
- 8** Which one is stronger: fiber or paper? Explain the strength relationship between them.
- 9** Explain visco-elasticity. What are the advantages and disadvantages of visco-elastic behavior of paper?
- 10** How does the grammage affect the other properties of paper?
- 11** What can be done to alter the optical properties of paper?

Total Quality Management (TQM)

Total Quality Management (TQM) is a management strategy that is designed to continuously improve performance in all areas of responsibility at every level in an organization. The very definition of TQM indicates three major components [1,2,3,4,5]:

- (1) The *totality* implies that the system should encompass all functions and groups in the organization such as manufacturing, engineering, research and development, sales and marketing and management at every level.
- (2) The *quality* is the end result that can be evaluated by two primary factors: performance and producer-user mutual satisfaction
- (3) The *management* implies the degree of communication among different units in the organization to provide the necessary advise and feedback which are essential to high quality

In a highly competitive environment, commitment to quality is not an option but an absolute necessity. In today's business environment, quality is a relentless pursuit without an end. To stay competitive, quality should come second only to health, safety and environmental factors. As in every industry, product quality has become one of the most critical factors in paper machine clothing and paper industries. Today, the same equipment and machinery are generally available to every

manufacturer. Yet, the quality of products from different manufacturers may differ. The reasons for this difference are various but include expertise and know how of the technical people as well as the management philosophy and style of the company.

7.1 Historical Development of TQM

The purpose of TQM is to ensure high quality products and services. Various methods and systems have been developed and implemented in the past to achieve this goal. The concept of quality dates back to 3000 BC Babylon where it was practiced to impose uniformity of units for weights and measures [6]. Throughout time, fitness for use of a product has been measured in regards to its uniformity and quality. Preliminary standards were used to ensure uniformity in goods produced which were measured by means of product physical attributes which were judged by the inspector's individual apprehension of these attributes. Prior to industrial revolution, quality was mostly managed by individuals that were involved in various levels of the product life cycle. In 1875, the principles of scientific management was introduced by F. W. Taylor. Around this time, assembly line concept was developed. Products were manufactured in several steps and usually different person(s) were involved at each step. Therefore, the workers no longer

had the opportunity to fabricate from beginning to end, personally monitoring and controlling the quality and productivity of their efforts. They became responsible only for a portion of the complete manufacturing cycle without seeing direct effects of their efforts on the final product and its required function. This caused the development of barriers or walls between the departments and workers.

During the early 20th century, written standards showing the product's critical statistics were used. In 1911, Taylor suggested "time" as the simplest measurement of performance. His efforts focused on development of time study analysis, work standards and wage incentives. Prior to 1920s, the implementation of quality control was still somewhat subjective. Product standards, as they are known, became more common after the World War I, and the only interaction between the standards and the product was the physical inspection. This era can be classified as small size business environment with low production and labor intensive manufacturing. There was an intimate relationship between the manufacturer and customer. International competition was relatively small.

In 1920s, Quality Control (QC) started to appear as a specific discipline and became one of the central activities in manufacturing. In-process and final inspection became two essential functions of quality control. Statistical quality control was developed for this purpose in this period. Inspection and sampling were evaluated using statistical control charts and sampling techniques such as random sampling.

Between 1930s and 1970s, quality control was mostly done by sampling inspection prior to shipping to customers. Off-line testing methods were developed for this purpose and statistics were used in sampling inspection. Government regulations and military standards during the World War II demanded sampling inspection. The American Society for Quality Control (ASQC) was established in 1945. Since the late 1970s, the emphasis has been on making quality products and delivering quality services. Quality control expanded to include evaluation of product

design, manufacturing and end-use [6]. Toward the end of this period, the international competition started to surface.

The 1980s witnessed the revival of quality control which was driven by global competition. Statistical Process Control (SPC) was introduced. The computer revolution has improved quality and quality control techniques.

In the mid 1980s, the concept of Total Quality Management (TQM) began to develop. Quality control and quality engineering were integrated with quality management. Efforts are being made to remove the barriers and walls among departments within the organization. Giant steps in information technology resulted in development of expert systems, integrated information systems, neural networking and fuzzy logic methods. The concept of concurrent engineering was developed. Continuous improvement became the way of life and different ways of wage incentives were developed.

7.2 Quality Philosophies

Quality today is measured in many different ways by many different experts. Webster's dictionary defines quality as "a. An inherent feature, b. Degree of excellence" [7]. Quality masters such as Dr. W. E. Deming, Dr. J. M. Juran and Dr. D. A. Garvin all have their definitions and explanations of causes associated with quality. Quality can be viewed as a matter of whether or not a product is useable and the extent to which that product is fit for use. This "fitness for use" [8] is gauged by the customer in different areas from price to ethics. Quality standards such as ISO give manufacturers assurance of minimized variance and adherence to specifications.

Quality is the main ingredient in a product that delights the customer by either meeting or exceeding expectations [9]. Thus, it should be paramount to any business, to meet or even exceed expectations of its customers. Another necessity for quality is the cost. Production costs can be very high when quality is having to be reworked into a satisfiable product. It is

reported that over 30% of production costs can be due to reworking or replacing defective parts or products [10]. If quality could be made into the product the first time, a company could save this 30%, which could improve quality without raising the price of the item out of the competitive range.

Customers today are more willing to pay for quality. It is reported that 87% of consumers said quality is their number one criterion when making a purchase and 84% of consumers said that they would be willing to pay more for top quality [11].

The benefits of quality to the companies are as follows [12,13]:

- (1) *Greater Market Share* is caused by the customer's perceived better quality of an item and their repeated buying and reference of the product to others.
- (2) *Higher Growth Rate* is directly proportional to the growth in the market share due to the role that a company's market share plays on the growth rate of that company.
- (3) *Premium Price* is a great benefit of perceived quality by the customer. A premium price can be charged for a product due to the users' willingness to pay more for what is perceived as better quality
- (4) *Loyal Customers* are produced when they see that a product has not let their expectations down and has fulfilled the promise of quality that the premium price sometimes dictates. This benefits the company greatly on the basis of advertising and reputation due to the loyalty of these repeat customers
- (5) *Highly Motivated Employees* come from having pride in the fact that the product they produce is perceived to have the highest quality.

These benefits translate into long lasting prosperity for these companies, as well as a healthy bottom line at the end of the financial quarter.

There are many dimensions of quality such as performance, features, reliability,

conformance, durability, service ability and perceived quality [14]. Aesthetic, which is often considered as a measure for quality, is secondary importance for industrial textiles such as paper machine clothing. These dimensions of quality are independent of one another while at the same time one may be sacrificed for the benefit of the other. A product or service may be high in one dimension while being low in another without the two being directly related. This is what gives these dimensions such an encompassing view of quality and its application.

Perceived quality is where the definition of quality gets somewhat vague. It seems to hold true that quality can be defined only in terms of the person assessing the item or service. This raises the question: who is the final judge of quality? Is it the customer or the manager? The manager would assume quality in his product if it meets manufacturing standards and production quotas. The customer has a different view of quality and its meaning. The factors that affect customer satisfaction for quality are price, technology, psychology, time orientation, contractual and ethical [9]. Customers tend to associate quality with higher price. There is some evidence that price is used by consumers in quality estimates and that for some products consumers' estimates of quality are affected by price [15].

In the manufacture of paper machine clothing, technology indicates factors such as fabric and seam strength and other properties that are affected by the state of technology in the industry. Psychological effect generally means aesthetic requirements of customers which have secondary significance in paper machine clothing. Time orientation includes durability. Contractual refers to a product guarantee, the refund policy, etc. Most major purchases made creates some type of anxiety about whether or not the choice was right. Refund policies also tell the customer that the producer stands behind his/her promise that the product will meet the customer's needs or the customer will receive a refund. Ethical refers to honesty of advertising, courtesy and responsiveness of sales/service personnel, etc.

7.2.1 *W. Edwards Deming's View of Quality*

According to Dr. Deming, quality begins with the intent, which is fixed by the management. The intent must be converted by engineers and others into plans, specifications, tests and production [16]. Deming supports this view with his 14 obligations of management which are listed in Table 7.1.

Deming's description of the steps to improve quality show mainly managerial techniques that might be ignored by companies who hope for quality to appear through such programs as management by objective or management by sheer numbers. This does not ensure that everyone in the organization is committed to quality or their intent is to build quality into the item itself. Without this intent for inherent quality, consistent perceived quality by both the customer and management will fall short.

Dr. Deming sees quality as a triangle of interaction among:

- (1) The product itself
- (2) The user and how he uses the product, how he installs it, how he takes care of it, what he was led to expect

- (3) Instructions for use, training of customer and training of repairmen, service provided for repairs and availability of parts

Dr. Deming supports standardization as a way of improving production. He clarifies regulation versus standards by explaining how standards made voluntarily keep federal regulations down to a minimum. With an industry overly regulated by the government there can be little room for that business to compete with unregulated businesses overseas.

7.2.2 *Joseph M. Juran's View of Quality*

Dr. Juran sees quality as "fitness for use" [8] which seems to have gained widespread acceptance as the short definition of quality. Dr. Juran describes quality as having a twofold meaning; that of product features that meet customer needs and a product's freedom from deficiencies. In the eyes of the customer, the fewer the deficiencies the higher the quality. Juran shows that product features mostly impact sales and increasing product features increases the cost. When product deficiencies are reduced, the cost of quality is decreased.

TABLE 7.1. Deming's 14 Points of Quality Management.

- | |
|---|
| <ol style="list-style-type: none"> 1. Create constancy of purpose for the improvement of product or service 2. Adopt the new philosophy 3. Cease dependence on mass inspection for quality control 4. End the practice of awarding business on the basis of price tag 5. Improve constantly and forever the system of production and service, to improve quality and productivity, and thus constantly decrease costs 6. Institute more thorough, better job-related training 7. Institute leadership 8. Drive out fear, so that everyone may work effectively for the organization 9. Break down barriers between departments 10. Eliminate slogans, exhortations and targets for the work force that ask for zero defects and new levels of productivity without a means of accomplishment 11. Eliminate work standards on the factory floor 12. Remove the barriers that rob employees at all levels in the company of their right to pride of workmanship 13. Institute a vigorous program of education and self improvement 14. Put everybody in the organization to work to accomplish the transformation |
|---|

7.2.3 David A. Garvin's View of Quality

Dr. Garvin, a Harvard expert on quality, states that quality has five definitions or approaches to definitions which are as follows [17]:

- *Transcendent* is a view of quality that believes in the “innate excellence” of a product. This view sometimes contends that quality is found in craftsmanship and not in mass production. This view is that of unanalyzable philosophical properties.
- *Product Based* is the view that quality is a precise and measurable variable, wherein differences in quality must be based on the components of that product.
- *User Based* definitions of quality take the cliché “Quality lies in the eyes of the beholder” approach.
- *Manufacturer Based* definitions are mainly based on the engineering and manufacturing side of the supply for the product.
- *Value Based* definitions are those that define quality in terms of costs and prices.

According to Garvin, these five definitions show all sides of quality. The definitions of Dr. Juran and Dr. Deming would fall into the user based and manufacturing based definitions, respectively.

7.2.4 Other Views of Quality

There have been many other academicians and scientists who have developed quality philosophies. Philip B. Crosby defined quality as the “conformance to requirements”. His method to achieve quality is based on prevention of defects. Performance standards are established for zero defects. According to him, the quality performance measurement is the cost of quality.

According to Genichi Taguchi, quality must be designed into every product and corres-

ponding process. He defines the quality as “the (minimum) loss imparted by the product to the society from the time the product is shipped.”

7.3 Core Concepts of TQM

TQM is an integrated quality improvement system. There are certain components of TQM that should be well defined, understood and implemented by the organization that is implementing it. First of all, the organization must define the quality either by adopting one of the existing quality definitions or by developing a new one to suit the company's needs. Before defining quality, a careful study of customer expectations, requirements and priorities would be helpful.

For the TQM to be successful, top management should be committed to the development and implementation of the quality program at every level of the company. Management is responsible for providing help and proper facilities, equipment, etc., for the company associates. Motivation and appreciation of the quality control activities are also crucial.

Training is an important part of TQM. Each person in the organization should get proper training in his/her area. The necessary tools of quality control including Statistical Process Control (SPC) should be taught in theory and practice. Training of employees is a long term investment for the company. Lack of training reduces the probability of success in quality manufacturing.

Job satisfaction encourages employees to make voluntary contributions to the company in the form of suggestions, initiatives, etc. Education, not criticism, should be the form of retribution in case of any problems.

Communication is one of the keys for the success of any company. Communication is especially critical between production and marketing. Optimization of product mix should be considered very carefully when planning production.

Modern textile manufacturing machinery minimize the labor involvement. They also require less frequent maintenance than in the

past. State of the art equipment can be a valuable asset for quality manufacturing. However, well run older equipment may be a better investment in the short term than poorly run high tech equipment. Computerized machinery and equipment increases accuracy and quality of production. Integrated automation involves automation of material flow and transportation, machine settings, machine operation, maintenance, data recording and analysis. Preventive maintenance reduces the risk of machine downtime. An effective plant housekeeping schedule should also be implemented.

Information systems provide on-line and real-time information about various production parameters including production rate, efficiency, speed and idle time. Mechanical, electronic and optical sensors can be installed for measuring several product properties such as thickness, uniformity, tension and surface characteristics. In addition to on-line information systems, off-line testing systems can provide detailed data analysis using computers and software packages [4].

The concepts and procedures of TQM are not industry specific, i.e. they can be applied in any industry, small or large. Quality improvements must continue even after a quality assurance program is implemented.

7.4 Basic Management Elements of TQM

Commitment, awareness, planning, accountability and recognition are the main managing elements in a TQM environment. Commitment level indicates where the management stands on quality. Awareness involves an ongoing education and training of people on the progress of quality improvement. Employees should be made aware of the relationship between the company and customer (partnership).

Good planning is one of the most critical managing elements in TQM. Quality improvement must be a result instead of a reaction. All quality improvement efforts must be distributed through different levels of the or-

ganization with careful planning. Levels of planning ranges from simple statements of goals and specific actions to more detailed approaches that include graphical and analytical tools such as flow diagrams, cause-and-effect diagrams, etc.

Accountability involves the measure, tracking and review of the performance of individuals and teams in improving quality. Recognition naturally follows the accountability. Proper recognition of the efforts of individuals and teams is an effective way of motivation.

7.5 Statistical Process Control

Statistics is a branch of mathematics dealing with the collection, analysis, interpretation and presentation of masses of numerical data. Statistical Process Control (SPC) is an analytical tool for TQM with a manufacturing focus on process variation. SPC is a method by which one identifies the capabilities of a given process and by using statistical methods, identifies certain aspects of this process that can be improved. The end result of a statistical process control program is reduced variability, increased productivity and cost reduction [18].

Applied statistics is a tool that can be used to answer a variety of questions and the use of statistical procedures is, therefore, always a middle step. To answer a question, first an investigation must be formed. The typical steps in an investigation are:

- (1) A substantive question is formulated and refined and a plan is developed to obtain relevant evidence.
- (2) When appropriate, a statistical model is chosen to assist in organizing and analyzing the data to be collected. At this point, a statistical question may be developed, the answer to which may be expected to throw light on the substantive question. A statistical question differs from a substantive question in that it always concerns a statistical property of that data such as the average of the set of measures.

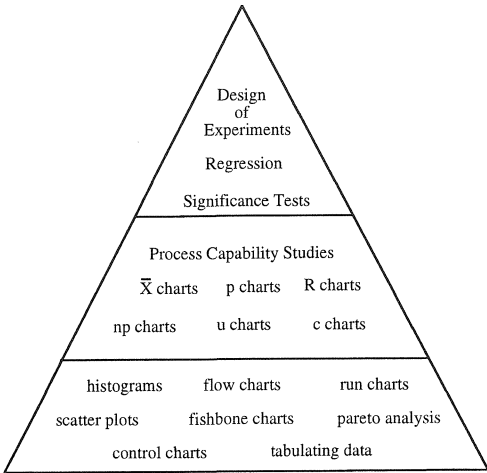


FIGURE 7.1. Statistical techniques.

- (3) Upon applying the statistical procedure, one arrives at a statistical conclusion.
- (4) Finally, a substantive conclusion is drawn.

There are numerous statistical techniques that have been available and well developed both in theory and in practice. Figure 7.1 shows some of the most common statistical techniques in use today. The degree of complexity of the technique increases towards the top. A detailed description of these methods is not among the purposes of this book. Therefore, only some of the more basic techniques will be described here. The reader is referred to the literature for the other techniques.

7.5.1 Data

Data are facts or figures from which conclusions can be drawn. Data are the raw materials of statistics. Data can be gathered from any number of sources. The term *population* refers to the complete set of observations or measurements about which we would like to draw conclusions. Single observations or measurements are called *elements*. An element is also referred to as a *sample*. Suppose we were interested in the average permeability of Type A dryer fabrics manufactured during a three month period. Our research reveals that 50

fabrics were manufactured during this time period and that a number of final permeabilities were produced. The results of our research are listed in Table 7.2.

The entire collection is referred to as the population and any individual final permeability as an element of that population. Assume that the target permeability was 75 cfm. Analysis of data shows that there are some number of permeabilities close to the target, some higher and some lower. Also note that there are some permeabilities in the 60's and some in the 90's, and that they are few in number, and that no permeabilities are lower or higher.

There are different purposes of collecting the data. When a particular method of doing a job is introduced, it is natural to consider whether the method is appropriate or not. The decision is usually based on past results and experience, or perhaps some conventional method. However, in the case of factory work where data are collected through the actual manufacturing process, the procedural methods are introduced on the basis of the information obtained. The manufacturing procedure will be most effective if a proper evaluation is made and on-the-job data are essential for making a proper evaluation. Data and subsequent evaluation will form the basis for actions and decisions. As factory operations will vary with the manufacturing procedure involved, data should be classified in terms of the various purposes. These purposes are:

- (1) Data to assist in understanding the actual situation: For example, these data are collected to check the extent of the dispersion

TABLE 7.2. The Final Permeability of Type A Dryer Fabrics Manufactured in One Quarter.

84	82	72	70	72
80	62	96	86	68
68	87	89	85	68
87	85	84	88	89
86	86	78	70	81
70	86	88	79	69
79	61	68	75	77
90	86	78	89	81
67	91	82	73	77
80	78	76	86	83

in part sizes coming from a machining process or to examine the percentage of defective parts contained in lots received. As the number of data increased, they can be arranged statistically for easier understanding. Estimates and comparisons can then be made concerning the condition of lots received as well as the manufacturing process, utilizing specified figures, standard figures, target figures, etc.

- (2) Data for analysis: Analytical data may be used, for example, in examining the relationship between a defect and its cause. Data are collected by examining past results and making new tests. In this case, various statistical methods are used to obtain correct information.
- (3) Data for process control: After investigating product quality, this kind of data can be used to determine whether or not the manufacturing process is normal and in control. Control charts are used in this evaluation and action is taken on the basis of these data.
- (4) Regulating data: This is the type of data used, for example, as the basis for raising or lowering the temperature of an electronic furnace so that a standardized temperature level may be obtained. Actions can be prescribed for each datum and measures taken accordingly.
- (5) Acceptance or rejection data: This form of data is used for approving or rejecting parts and products after inspection. On a basis of the information obtained it can be decided what to do with the parts or products.

Data serve as the basis for action. After evaluating actual conditions, as we have other data, proper action can be taken. After data are collected, they are analyzed, and the information is extracted through the use of statistical methods. Finally, data are collected because the population (the work situation) from where data are drawn is constantly changing.

To determine the magnitude of that change (in determining whether or not action must be taken) we statistically analyze that data to

determine if the variation is part of the natural variation for that process, or if the variation is due to some cause outside the normal variation of that process. After collecting data, it is possible to see that items seemingly produced in exactly the same way turn out differently. In many of the cases, this dispersion occurs because of differences in:

- the raw materials
- the tools, machinery or equipment
- the work method or process
- the measurement

Raw materials differ slightly in composition according to the source of supply and size differences will occur within accepted limits. Machines may seem to be functioning uniformly, but dispersion can arise from differences within parts of the machine itself. Likewise, a piece of equipment may be operating optimally only part of the time. Work methods, although programmed according to prescribed processes, can lead to perhaps even greater variations. Finally, a random task such as measurement is not always exactly achieved. Even slight differences can add up to a great deal of product quality dispersion.

7.5.2 Frequency Distributions

Statistical analysis can not proceed until the data of interest in an investigation are similarly organized in a useful manner. The simplest way to facilitate this kind of analysis is to put the permeabilities in order. To do so, locate the highest and lowest permeability values. Then, record all possible permeability values (including the two extremes) in descending order. Among the data in Table 7.2 the highest permeability is 96, and the lowest permeability is 61. The recorded sequence is 96, 95, 94, . . . , 61 as shown in Table 7.3 along with the number of occurrence (frequency) for each sample. Such an array showing the permeabilities and their frequency of occurrence is called a *Frequency Distribution*.

Thus organized, many features of interest are easily perceived. It is seen that the target permeability of 75 cfm is a little below the

TABLE 7.3. Permeabilities from Table 7.2, Organized in Order of Magnitude.

Permeability	Frequency	Permeability	Frequency
96	1	78	3
95	0	77	2
94	0	76	1
93	0	75	1
92	0	74	0
91	1	73	1
90	1	72	2
89	3	71	0
88	2	70	3
87	2	69	1
86	6	68	3
85	2	67	1
84	2	66	0
83	1	65	0
82	3	64	0
81	2	63	0
80	2	62	1
79	2	61	1

middle of the distribution. Although permeabilities range from 61 to 96, the bulk of the distribution lies between 67 and 91. There is one permeability near the top of the tolerance range and two permeabilities near the bottom of the tolerance range.

When data range widely, reducing individual data to a smaller number of groups makes it easier to display the data and to grasp their meaning. Table 7.4 shows two ways in which the permeabilities in Table 7.2 may be grouped into *class intervals*.

There are, however, two matters that should be kept in mind when considering grouped scores.

- (1) When scores are grouped some information is lost.
- (2) A set of raw scores does not result in a unique set of grouped scores.

Table 7.4 shows, for example, two different sets of grouped scores that may be formed from the raw data in Table 7.2. In converting raw data to grouped data there are several principles that should be kept in mind:

- A set of class intervals should be mutually exclusive. That is, intervals should be chosen so that one data point can not belong to more than one interval.
- All intervals should be the same width.
- Intervals should be continuous throughout the distribution. In part “A” of Table 7.4, there are no terms in an interval 64–66. To omit this interval and “close ranks” would create a misleading impression.
- The interval containing the highest value is placed at the top.
- The number of classes is dependent upon the amount of data.

Fewer class intervals mean greater interval width, with consequent loss of accuracy. The larger the number of classes in a frequency distribution, the more detail is shown. If the number of classes is too large, though, the classification loses its effectiveness for summarizing the data. Too few classes, on the other hand, condenses information so much as to leave little insight into the pattern of the distribution. The best number of classes in a frequency distribution often needs to be determined by experimentation. Usually an effective number of classes is somewhere between 4 and 20.

TABLE 7.4. Permeabilities from Table 7.2, Converted to Grouped Data Distributions.

A: Class Interval Width = 3		B: Class Interval Width = 5	
Permeability	Frequency	Permeability	Frequency
94–96	1	95–99	1
91–93	1	90–94	2
88–90	6	85–89	15
85–87	10	80–84	10
82–84	6	75–79	9
79–81	6	70–74	6
76–78	6	65–69	5
73–75	2	60–64	2
70–72	5		
67–69	5		
64–66	0		
61–63	2		

The choice of the class width (or class interval) is related to the determination of the number of classes. Many class intervals result in lesser convenience. It is generally best if all the classes have the same width. If the classes are not equally wide, one often can not tell readily whether the differences in class frequencies are due mainly to differences in the concentration of items, or to differences in the class width. For choice of class limits, it is usually suggested as good statistical practice that the class limits be chosen so that the midpoint of each class is approximately equal to the arithmetic average of the values falling in that class. The class limit is often referred to as the class boundary.

The next step is to translate a set of raw scores to a grouped data frequency distribution. We can illustrate the procedure with the data in Table 7.4. The first step is to find the lowest score, 61, and the highest score, 96. The difference between these two figures is the range of scores. Therefore, the range is $96 - 61 = 35$. Since there are to be from 4 to 20 class intervals we must find an interval width satisfying this condition. Often, this can be done by inspection. Next, we must determine the starting point of the bottom class interval. Since the lowest permeability is 61, the lowest interval should be 59–61, 60–62 or 61–63. Often, it is convenient to make the lower score limit a multiple of the class interval width. Next, refer to the collection of raw data in whatever order they may be, and tally their occurrence one by one against the list of class intervals. Note that when it comes to the selection of width of the class interval a choice is usually available. The number of intervals desired is one factor in choosing, but convenience is sometimes another. Choosing a width of five or ten or some multiple thereof makes both the construction and interpretation of a distribution easier.

The steps in constructing a frequency distribution are summarized below:

- (1) Find the value of the lowest score and the highest score.
- (2) Find the range by subtracting the lowest score from the highest.

- (3) Determine the width of the class interval needed to yield four to twenty class intervals.
- (4) Determine the point at which the lowest interval should begin.
- (5) Record the limits of all class intervals placing the interval containing the highest value at the top. Intervals should be continuous and of the same width.
- (6) Using the tally system enter the raw data in the appropriate class intervals.

A frequency distribution can be presented as a table or a graph. A graph is based entirely on the tabled data and therefore can tell no story that can not be learned by inspecting the table. However, graphic representation often makes it easier to see pertinent features of a set of data. Graphic representation is one way of presenting all kinds of quantitative information. There are many different kinds of graphs. For the representation of frequency distributions, there are three main types of graphs: the *histogram*, its variant, the *bar diagram*, and the *frequency polygon*.

7.5.3 Histogram

The histogram consists of a series of rectangles, each of which represents the frequency of data in one of the class intervals in a tabled distribution. A sample histogram is illustrated in Figure 7.2. The rectangle is erected so that

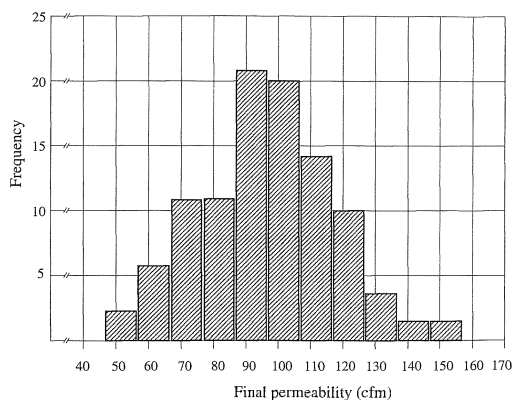


FIGURE 7.2. Histogram: Final permeability of Type A dryer fabric manufactured in a year.

its two vertical boundaries coincide with the exact limits of the particular interval, and its height is specified by the frequency of scores for that interval. Either frequencies or proportionate frequencies may be represented by the histogram. There are some details in construction that apply to the histogram.

- The graph has two axes: horizontal and vertical. The intersection of the two axes represents the 0 (zero) point on both scales.
- It is customary to represent scores (the measured data) along the horizontal axis and frequency along the vertical axes.
- Convenient units should be chosen to identify position along the axes. Limits of class intervals or interval midpoints usually do not form the most convenient frame of reference for the horizontal axis.
- Whether the graph of a frequency distribution appears squat or slender depends on the choice of scale used to represent position along the two axes.

Since it is desirable that similar distributions should appear similar when graphed, it is customary to choose a relative scale such that the height will be no less than about half or no more than about 3/4's of the width. Some trial and error may be necessary to create a graph suitable in size and convenient in scale. Construction of a histogram is very similar to that of a frequency distribution. The details are given below:

- (1) Calculate the range of all the data: $X_1 - X_s$ where X_1 is the largest value and X_s is the smallest value in the group.
- (2) Determine the number of classes (the number of histogram bars) as follows:

Number of Data	Approximate Number of Classes (k)
under 50	7–10
50–100	10–15
100–250	15–20
over 250	20–25

- (3) Calculate the class interval (width of class)

$$h = (X_1 - X_s)/k \quad (7.1)$$

(round h to a convenient number)

- (4) Determine class boundaries (class limits)
- (5) Tally points falling into each class
- (6) Construct histogram

7.5.4 Bar Diagram

The bar diagram is very similar to the histogram, except that space is inserted between the rectangles, thus properly suggesting the essential discontinuity of the several categories. An example of a bar diagram is given in Figure 7.3. Since the categories have no necessary order they may or may not be arranged in order of magnitude of frequency. Bar diagrams can be vertical or horizontal.

7.5.5 Frequency Polygon

The same data plotted as a histogram in Figure 7.2 has been represented by frequency polygon in Figure 7.4. In this type of graph, a point is plotted above the midpoint of each class interval at a height commensurate with the frequency of the scores in that interval. These points are then connected with straight lines.

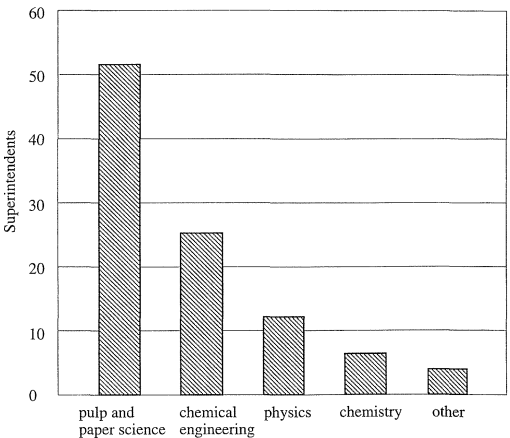


FIGURE 7.3. Bar diagram: comparative frequency of academic majors of machine superintendents in fine paper mills in Alabama.

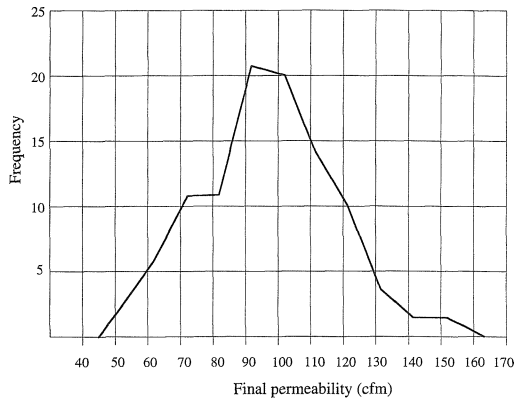


FIGURE 7.4. Frequency polygon: final permeability of Type A dryer fabrics manufactured in one year.

7.5.6 Pie Chart

In pie charts, the data are presented in a circle (Figure 7.5). The entire circle represents 100% of the data. The circle (pie) is divided into percentage slices that show the shares of data. This is useful in the same way as a Pareto Chart.

7.5.7 Flow Chart

A flow chart is used to identify the steps of a process. Flow charts are useful in showing the sequence and relationship of the process steps. Flow charts allow to uncover loopholes which are potential sources of problems in the process. Flow charts can be applied to any type of process. Figure 2.26 shows the flow chart for forming fabric manufacturing.

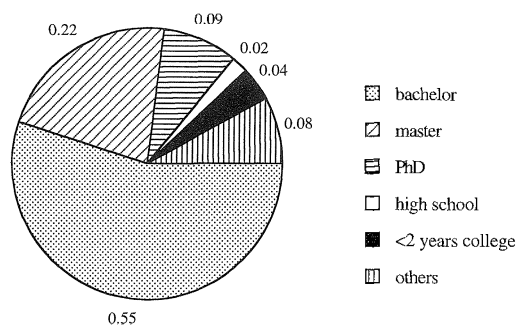


FIGURE 7.5. Pie chart: highest level of education in paper industry.

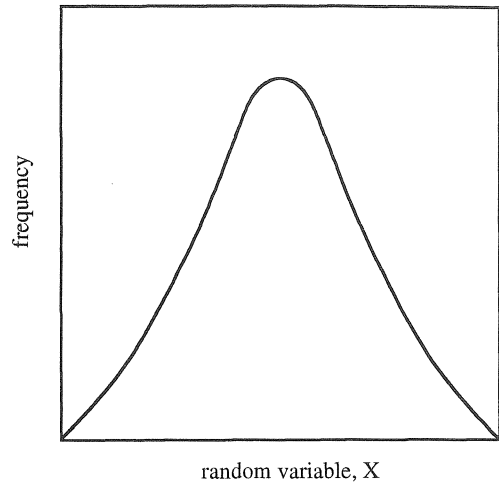


FIGURE 7.6. Normal curve.

7.5.8 Normal Curve

The “normal curve” is shown in Figure 7.6. It represents a distribution of any random variable that is symmetrical.

7.5.9 Measures of Central Tendency

There are several measurements that summarize important characteristics about the data sets that are being considered. These measurements fall into two broad categories: measures of central tendency and measures of dispersion (variability).

Measures of central tendency (also known as measures of position) are intended to provide a single summary figure which describes the level (high or low) of a set of observations. The three common measures of central tendency are arithmetic mean, mode and median.

In statistical terminology, the capital letter X is used as a collective term to specify the particular set of data being considered. An individual score in a set may be identified by a subscript such as X_1 , X_2 , etc. For example, consider the set of three scores 31, 21 and 41. We may identify them as follows:

$$X_1 = 31, X_2 = 21, X_3 = 41$$

A set of scores in a sample may be represented as follows:

$$X : X_1, X_2, X_3, \dots, X_n$$

When the scores in a set are to be summed, the capital Sigma (Σ) indicates that this operation is to be performed. It should be read “the sum of” (whatever follows).

Mean

The arithmetic mean (or average) $\bar{X}$ of a set of values $X_1, X_2, \dots, X_n$, is the sum of the values divided by the number of items, n :

$$\bar{X} = \frac{\sum_{i=1}^n X_i}{n} \quad (7.2)$$

Example: Find the mean of the following group of numbers.

$$\begin{aligned} 17, 9, 23, 18, 6, 42, 15 \quad n &= 7 \\ \bar{X} &= \frac{\sum_{i=1}^7 X_i}{n} \\ &= \frac{(17 + 9 + 23 + 18 + 6 + 42 + 15)}{7} \\ &= \frac{130}{7} = 18.57 \end{aligned}$$

The mean has a number of unique properties that should be considered when deciding on an appropriate measure of position.

- The sum of the values of a set of items is equal to the mean multiplied by the number of items.
- The sum of the deviations of the X_n values from their mean $\bar{X}$ is 0.

Mode (Mo) and Median (Mdn)

The *mode* is a score that occurs with the greatest frequency. In grouped data, it is taken as the midpoint of the class interval that contains the greatest number of scores. The symbol is Mo. The median (Mdn) of a set of items is the value of the middle item when all items are arranged by magnitude.

Mdn: — Arrange the items (numbers) in numerical order (ascending or descending)

— For an odd-numbered group the median is the middle value

— For an even-numbered group the median is the average of the two middle numbers

Example: Find the median of the following groups of numbers.

A. 17, 9, 23, 18, 6, 42, 15 becomes
6, 9, 15, 17, 18, 23, 42.
17 is the median.

B. 17, 9, 23, 18, 6, 42, 15, 25 becomes
6, 9, 15, 17, 18, 23, 25, 42.
The median is $(17 + 18)/2 = 17.5$.

Properties of Mode, Median and Mean

The mode is easy to obtain, but it is not very stable. Further, when the data are grouped, the mode may be strongly affected by the width and location of class intervals. Another problem is that there may be more than one mode for a particular set of scores.

In distributions that are perfectly symmetrical, that is, the left half is a mirror image of the right half, the mean, median and mode will yield the same value. It is important to note that the widely occurring “normal distribution” falls in this category. We have earlier referred to this phenomenon as the normal curve. Figure 7.7 shows what happens to

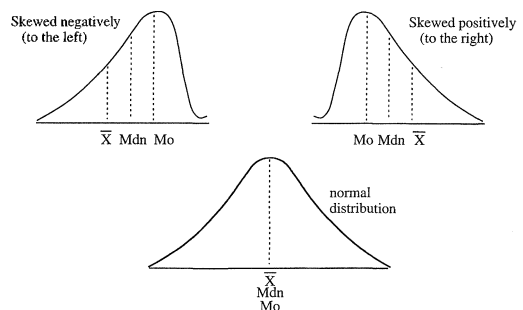


FIGURE 7.7. $\bar{X}$, Mo and Mdn in the normal distribution and in skewed distributions.

mean, median and mode in skewed distributions, as compared with the normal distribution.

Equality of mean and median does not guarantee that the distribution is symmetrical, although it is not likely to depart very far from that condition. On the other hand, if the mean and median have different values, the distribution can not be symmetrical. Furthermore, the more skewed, or lopsided the distribution is, the greater the discrepancy between these two measures. In a smooth negatively skewed distribution the mode has the highest score value and the median falls at a point about 2/3's of the distance between that and the mean. The mean, as might be expected, has been specially affected by the fewer, but relatively extreme values in the tail. In a positively skewed distribution, exactly the opposite situation is obtained. As a consequence, the relative position of the median and the mean may be used to determine a direction of skewness in the absence of a look at the entire distribution. At a rough descriptive level, the magnitude of the discrepancy between the two measures will afford a clue to the degree of departure from a state of symmetry.

The median is the point that divides the upper half of the scores from the lower half. In doing so, the median responds to how many scores lie below or above it, but not to how far away the scores may be. A little below the median, or a lot, both count the same in determining its value. Since the median is sensitive to the number of scores below it, but not to how far they are below, the median is less sensitive than the mean to the presence of a few extreme scores. Consider this distribution:

$X : 5, 6, 7, 8, 24$

The median is 7, the mean is 10. The value of the top score, 24, is quite different from that of the remainder of scores. It strongly affects the total, and hence the mean, but it is just another score above the median. If the score had been 9 rather than 24, the median would be the same.

Sometimes one encounters a distribution that is reasonably regular except for several

quite deviant scores at one end. If it is feared that these scores will carry undue weight in determining the mean, a possible solution is to calculate the median. The median will respond to their presence, but no more than to others that lie on that side of its position. Similarly, in distributions that are strongly asymmetrical, the median may be the better choice if it is desired to represent the bulk of the scores and not give undue weight to the relatively few deviant ones. Of the measures of central tendency under consideration, the median stands second to the mean in ability to resist the influence of sampling fluctuation in ordinary circumstances.

The mean, unlike any of the other measures of central tendency, is responsive to the exact position of each score in the distribution. The mean may be thought of as a balance point of the distribution. The mean is more sensitive to the presence (or absence) of scores at the extremes of a distribution than are the median and the mode. When a measure of central tendency should reflect the total of the scores, the mean is the choice since it is the only measure based on this quantity. When further statistical computation is to be done, the mean is likely to be the most useful of the measures of central tendency.

A characteristic of great significance concerns sampling stability. Suppose that from a large population of scores samples were completely drawn at random. Means of such samples would have similar but not identical values. This would also be true for the medians of the same samples and for their modes. However, for distributions encountered most frequently in statistical work, the means vary least among them. Thus, under ordinary circumstances, the mean best resists the influence of sampling fluctuation.

7.5.10 Measures of Dispersion (Variability)

Measures of variability express quantitatively the extent to which the data in a set scatter about or cluster together. Whereas a measure of central tendency is a summary

description of the level of performance of a group, a measure of variability is a summary description of the spread of performance. Measures of variation include the range, deviation, variance and the standard deviation.

Range

The range is the difference between the largest and smallest values in a set of items. The range indicates the maximum extent of variation in a set. A limitation of the range as the measure of variation of a set of items is that it depends only on the extreme items, and does not consider the others. Thus, a data set might contain items quite close to each other with the exception of one extreme value.

Despite the concentration of almost all the items, the range should be large since it is based only on the extreme values. Another limitation of the range as a measure of variability is that it depends on the number of items in the set. The larger the number of items, the larger the range tends to be. Like other measures of variability, the range is a distance and not, like measures of central tendency, a location. In a frequency distribution, the range may be figured as the difference between the value of the lowest raw scores that could be included in the bottom class interval and that of the highest raw scores that could be included in the upper-most class interval.

$$\text{Range: } R = X_l - X_s \quad (7.3)$$

where l denotes the largest number in the group, and s denotes the smallest number in the group.

Example: Determine the range, R , for the following group of numbers

17, 9, 23, 18, 6, 42, 15

$$X_l = 42$$

$$X_s = 6$$

$$R = 42 - 6 = 36$$

The range is a rough and ready measure of variability. In most situations, including those involving samples drawn from a normal distribution, the range varies more with

sampling fluctuation than other measurements do. Only the two outermost scores of a distribution affect this value. The remainder could lie anywhere between them. It is thus not very sensitive to the total conditions of the distribution. In addition, a single errant score is likely to have a more substantial effect on the range than on the standard deviation, which is particularly sensitive to conditions at the extremes of a distribution. Like other measures, with the exception of the standard deviation, it is of little use beyond the descriptive level. In many types of distributions, including the important normal distribution, the range is dependent on sample size, being greater when sample size is larger.

Deviation

A deviation score expresses the location of a score by indicating how many score points it lies above or below the mean of the distribution. The deviation score is symbolized by the lowercase letter x to distinguish it from a raw score which is symbolized by the upper case letter X . In symbols, the deviation score is defined as:

$$x = (X - \bar{X}) \quad (7.4)$$

The deviation score states the position of scores relative to the mean.

Variance

A commonly used measure of dispersion is called the *variance*. It is a measure that takes into account all the values in a set of items. It can be defined as the mean of the squares of the deviation scores. The defining formula for the variation of a sample (variance) is:

$$\text{Variance} = S^2 = \frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1} \quad (7.5)$$

The variance is a measure which provides quantified information about the variability in a data set. The first step in calculating the

variance is to square the deviations around the mean. Then, to obtain an overall measure of variability per item, the average of the squared deviation is taken. This mean squared deviation is called the variance. The units of the variance is the square of the variable units.

The wider the dispersion of the values around the mean, the greater the variance. If there is no dispersion of the values (if all are equal and hence all are equal to the mean), then the variance is equal to zero.

For calculation purposes, equation (7.5) can be written as:

$$S^2 = \frac{\sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2}{n - 1} \quad (7.6)$$

Standard Deviation

The most important and widely used measure of variability is the standard deviation, s . It is the square root of the variance.

$$s = \sqrt{\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{n - 1}} \quad (7.7)$$

Example: Determine the standard deviation for the following group of numbers.

17, 9, 23, 18, 6, 42, 15

Calculation:

$$\begin{aligned} \bar{X} &= \frac{\sum X_i}{n} \\ &= \frac{(17 + 9 + 23 + 18 + 6 + 42 + 15)}{7} \\ &= \frac{130}{7} = 18.57 \\ \sum X_i^2 &= 17^2 + 9^2 + 23^2 + 18^2 \\ &\quad + 6^2 + 42^2 + 15^2 \\ &= 3248 \\ \frac{(\sum X_i)^2}{n} &= \frac{130^2}{7} = 2414.28 \end{aligned}$$

$$\begin{aligned} S^2 &= \frac{3248 - 2414.28}{7 - 1} = 138.95 \\ s &= \sqrt{138.95} = 11.78 \end{aligned}$$

The standard deviation, like the mean, is responsive to the exact position of every score in a distribution. If a score is shifted to a position more deviant from the mean, the standard deviation will be larger than before. If the shift is to a position closer to the mean, the magnitude of the standard deviation is reduced. Consequently, we may expect that the standard deviation is more sensitive to the exact condition of the distribution than measures that do not share this property. Because of this characteristic sensitivity, the standard deviation may not be the best choice among measures of variability when the distribution contains a very few extreme scores, or when the distribution is barely skewed. One of the most important points favoring the standard deviation is its resistance to sampling fluctuation. In repeated sampling of the same population, the numerical value of the standard deviation tends to jump about less than would other measures computed from the same samples.

Coefficient of Variation (CV)

$$\begin{aligned} CV\% &= \frac{\text{Standard Deviation}}{\text{Mean}} \times 100 \quad (7.8) \\ &= s \times \frac{100}{\bar{X}} \end{aligned}$$

Measures of position are concerned with the location around which items are concentrated, whereas measures of dispersion consider the extent to which the items vary. Measures of skewness are still another kind of measure summarizing the extent to which the items are symmetrically distributed. One way to determine the nature of skewness of a frequency distribution is to compare the values of the mode, median and mean. The three frequency polygons in Figure 7.8 show the comparative positions of the mean, median and mode in unimodal (one mode) frequency distributions with differing degrees and directions of asymmetry.

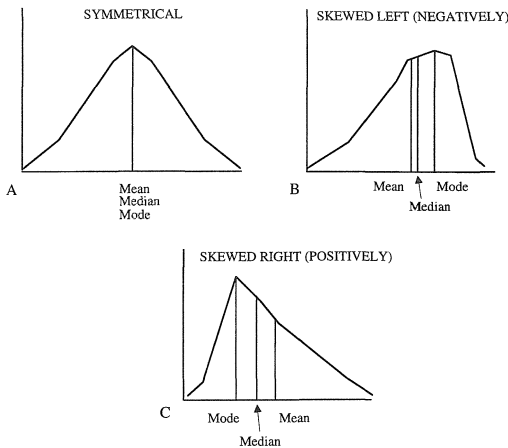


FIGURE 7.8. Examples of symmetrical and skewed unimodal frequency distributions.

Figure 7.8(A) is an example of a symmetrical unimodal frequency distribution. The values of the mean, median and mode in any section distribution are identical. Figures (B) and (C) show, respectively, an example of a unimodal frequency distribution skewed to the left, or skewed negatively, and one skewed to the right, or skewed positively. These figures indicate the relationship typically existing between the mean, median and mode in unimodal frequency distributions that are moderately skewed: the mean is furthest out toward the tail of the distribution and the median is between the mean and the mode.

Let us now consider measures of variability and the normal distribution. Since the measures of variability are defined in different ways we would expect them to yield different values when calculated for the same distribution. Their relative magnitudes can not be exactly specified in terms of a generalization true for all distributions, because special characteristics of some distributions have a different effect on the different measures. The normal distribution, however, is so frequently useful that properties of these measures in that distribution ought to be given attention.

In the ideal normal distribution, the interval $\pm 1s$ contains about 68% of the scores. The interval $\pm 2s$ contains about 95% of the scores and the interval $\pm 3s$ contains about 99.7% of the scores. These relationships are pictured in Figure 7.9.

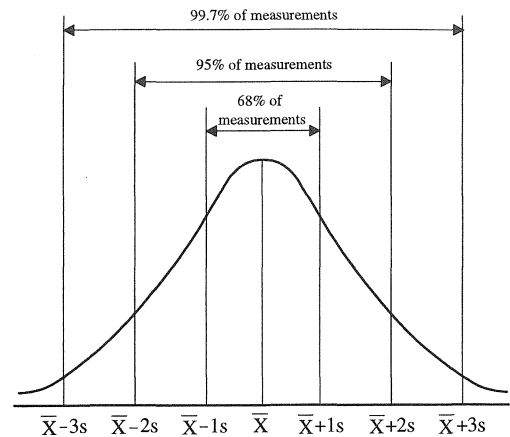


FIGURE 7.9. Relative frequency of cases contained within certain limits in the normal distribution.

Of course, in any sample drawn from a normal distribution, there will be a certain degree of irregularity, and these statements will be only approximately true. Figure 7.9 also shows that the more extreme the score the greater the rarity of its occurrence in the normal distribution. Consequently, when a sample of limited size is drawn at random from a large number of scores that are normally distributed, very extreme scores may not be encountered.

It clearly would be advantageous if substantially all the information contained in a frequency distribution could be packed into two figures, such as a measure of central tendency and a measure of dispersion. Under favorable circumstances it is possible to come close enough to this ideal for many purposes. Mathematical statisticians have discovered that the best measure of central tendency for this purpose is the arithmetic mean. A useful measure of dispersion for this purpose is the standard deviation. The ideal would be to be able to say, given any average and standard deviation, just what proportion of the measurements fell within any specified limits. These two numbers sufficiently define the normal curve that has been mentioned earlier. Many observed frequency distributions of measured qualities of manufactured products and many other frequency distributions found in nature, do correspond roughly to this normal curve. Although

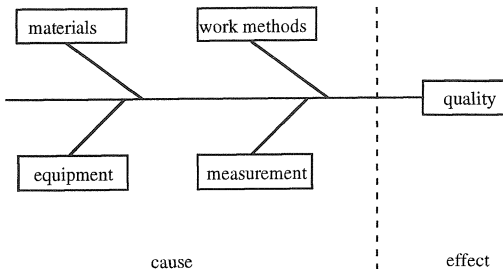


FIGURE 7.10. Cause-and-effect diagram.

most of the area under this bell shaped symmetrical form is included within the limits $\bar{X} \pm 3s$, the curve extends from minus infinity ($-\infty$) to plus infinity ($+\infty$). The curve is fully defined by $\bar{X}$ and s .

Many of the useful actions that may be taken in manufacturing are related to future production rather than completed lots of items. The knowledge that must be gained from the sampling procedure is knowledge of the pattern of variation of the production process from which the sample was drawn. We have already defined population as the complete set of observations or measurements about which we would like to draw conclusions. In order to draw conclusions about this population it is necessary to rely on numerical values derived

from samples drawn from that population. Such numerical values, which include the mean, median, standard deviation and range, summarize the information contained in the sample data.

7.5.11 Cause-and-Effect Diagram

The causal factors of dispersion diagrammed in Figure 7.10 point out the cause-and-effect relationship. The objective of improving the quality of the output first must be approached by an analysis of the causal factors. But it is necessary to know both cause and effects in greater detail and in more concrete terms in order to illustrate their relationship on a diagram and make them more useful. The variables which can cause the dispersion can be called factors. Quality characteristics describe the effect. A cause-and-effect diagram is useful in sorting out the causes of dispersion and organizing mutual relationships. These same cause-and-effect diagrams can assist in determining from which areas the data should be collected.

Figure 7.11 shows a cause-and-effect diagram for print quality variables. Because of their shapes, cause-and-effect diagrams are

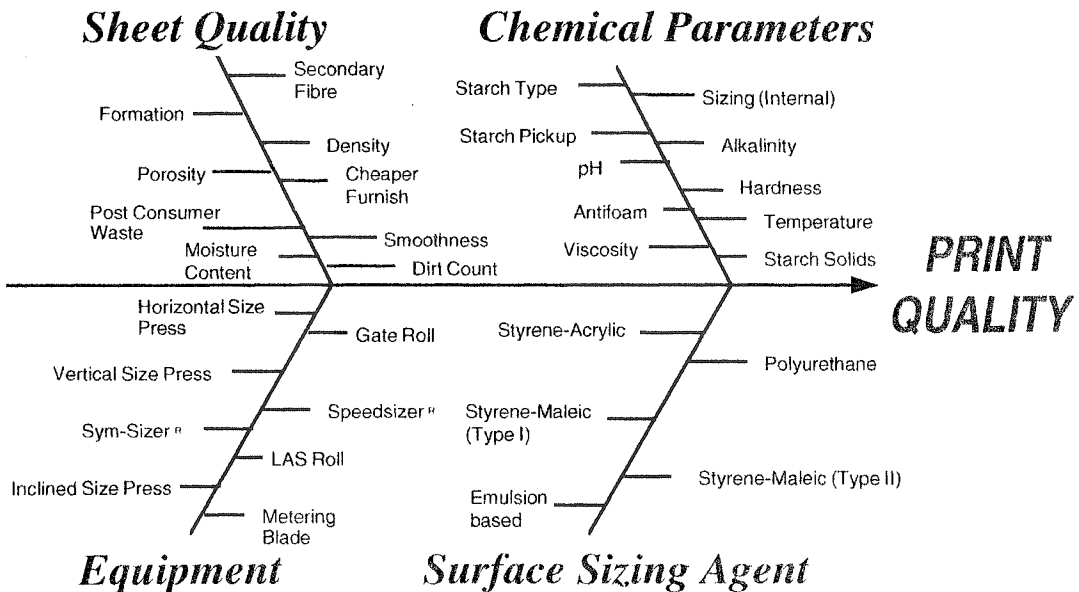


FIGURE 7.11. A cause-and-effect diagram for print quality variables of paper [19].

also called fishbone diagrams. To construct a cause-and-effect diagram, first determine the quality characteristic which is what to be improved and controlled. Second, write the main factors which may be contributing to the quality characteristic. It is recommended to group the major possible causal factors of dispersion into such items as raw materials, equipment (machines or tools), method of work (workers), measuring method (inspection), etc. Each individual group will form a branch. Third, onto each of these branch items, write in the detailed factors which may be regarded as the causes. Continue to write—in even more detailed factors. Defining and linking the relationships of the possible cause of factors should lead to the source of the quality characteristic.

Cause-and-effect diagrams are drawn to clearly illustrate the various causes affecting product quality by sorting out and relating the causes. There are several ways to use a cause-and-effect diagram, but primarily, the cause-and-effect diagram is a guide to concrete action. The cause-and-effect diagram reveals those areas or aspects which influence quality.

7.5.12 Pareto Diagram

Once the causes or factors producing a particular result are determined, the next step is to determine which factor to work on first. A Pareto diagram is a useful tool for just this purpose. A pareto diagram is a special form of vertical bar diagram. As a hypothetical example, Table 7.5 lists specific complaints encountered in joining pin seams.

TABLE 7.5. Pin Seam Joining Problems (Hypothetical).

Type Problem	Number of Complaints	Percent Distribution
Wrong wire	4	10.5
Zip tape failure	5	13.2
Missing loops	3	7.9
Loops too small	15	39.5
Irregular loop size	11	28.9

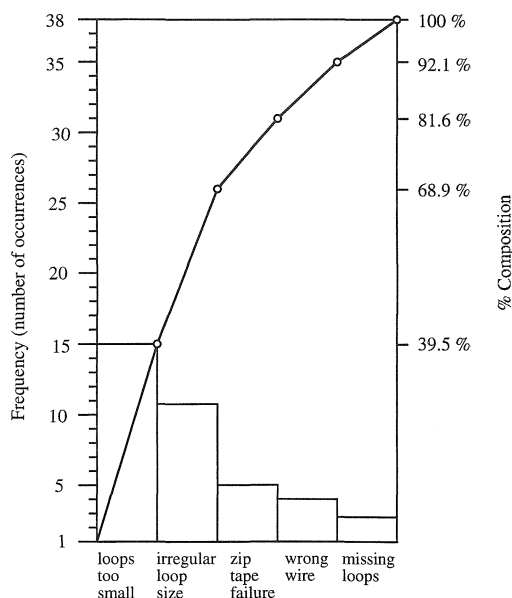


FIGURE 7.12. Pareto diagram.

The number of specific complaints is shown as is the percent distribution of that specific complaint. Data from this table has been made into a bar graph as shown in Figure 7.12.

In the Figure, the left vertical axis shows the number of defectives for each defective category and the right vertical axis shows the percent distribution for each category over the total defectives. The horizontal axis lists the defective items starting with the most frequent one on the left progressing to the least frequent on the extreme right and the rest arranged by order of magnitude. The cumulative total of the number of defectives for each category is shown by the line graph.

A Pareto diagram, such as this, indicates which problem should be solved first in eliminating defects and improving the operation. According to the graph “loops too small” should be tackled first because it forms the tallest bar. The next most significant defective item is the second tallest bar. This may appear very simple, yet bar graphs are extremely useful in factory quality control. It is much easier to see which defects are most important with the bar graph than by using only a table of numbers.

Establishment of a Pareto Diagram

- (1) Step 1. Ascertain the classifications that will be used in the graph. For example, graphs could list items according to kinds of defectives, defects, work groups, products, size, damage, etc.
- (2) Step 2. Decide on the time period to be covered on the graph. There is no prescribed period of time, so naturally the period will vary according to the situation.
- (3) Step 3. Total the frequency of occurrence for each category for the period. Each total will be shown by the length of the bar.
- (4) Step 4. Draw horizontal and vertical axes on graph paper and demarcate the vertical axis in the proper units.
- (5) Step 5. Draw in the bars, beginning on the far left with the most frequent defective items. The height of the bar will correspond to the value on the left vertical axis. Keep the width of the bars the same and in contact with its neighbor. If there are several categories with limited frequencies, they can be grouped together as "others" and placed at the right most bar on the graph.
- (6) Step 6. Under the horizontal axis label each of the bars.
- (7) Step 7. Plot a line showing the cumulative total reached from the addition of each category.
- (8) Step 8. Title the graph and briefly write the source of the data on which the graph is based.

The Pareto diagram clearly and distinctly exposes the relative magnitude of defects and provides a base of common knowledge from which to operate. At a glance one can see that the two or three taller bars account for the majority of problems, the smaller bars being lesser causes.

Experience has shown that it is easier to reduce a tall bar by half than to reduce a short bar to zero. If we can reduce the tallest bars, the ones which account for the most defectives, it will have been a considerable accomplishment. If it requires the same effort to

reduce the tall bar by half as to reduce a short bar by half, there is no doubt which should be selected as the target.

The value of Pareto diagrams is that they indicate which factors are most prevalent and therefore deserve concentrated efforts for improvement.

7.5.13 Run Charts and Control Charts

Histograms consolidate the data to show the overall picture. Pareto diagrams are used to indicate problem areas. These methods group the data for a specified period and express them in static form. However, in a manufacturing environment, one also wants to know more about the nature of the changes that take place over a specified period of time, that is, the dynamic form. This means that we not only have to see what changes in data occur over time; but we must also study the impact of the various factors in the process that change over time.

Thus, if the materials, the workers, or the working methods or equipment were to change during this time, we would have to note the effect of such changes on production. One way of following these changes is by using graphs called run charts.

As an example, Table 7.6 lists coil diameter measurements taken over a 20 hour period. Four measurements were taken each hour. The mean ($\bar{X}$) and the range (R) for each subgroup have been calculated. Additionally, an "average of the average" ($\bar{\bar{X}}$) and an "average of the range" ($\bar{R}$) have been determined.

Using this data, a run chart (Figure 7.13) was drawn indicating the average hourly value and the hourly range.

This run chart shows that the values were low at the outset but showed a tendency to rise over time. Table 7.6 does not reveal this trend. In other words, new information is discovered by looking at the movement of the data.

Now the problem is to find out whether the points on the run chart are abnormal or not. For example, the first 4 points of $\bar{X}$ could be normal or they may be below normal. Such determination can not be made

TABLE 7.6. Spiralmesh Coil Diameter Measurements (in mm) Taken at Coiling Machine over a 20-hour Period.

Sampling (Subgroup)	X_1	X_2	X_3	X_4	$\bar{X}$	R
1	19.8	21.4	19.4	20.1	20.2	2.0
2	19.5	18.7	20.3	18.8	19.3	1.6
3	19.8	19.3	20.5	21.2	20.2	1.9
4	18.0	19.0	21.0	20.0	19.5	3.0
5	20.4	21.0	20.8	21.0	20.8	0.6
6	20.0	20.9	20.6	19.7	20.3	1.2
7	21.8	21.3	20.3	19.3	20.7	2.5
8	20.2	19.1	21.4	19.4	20.0	2.3
9	20.6	19.1	20.4	20.6	20.2	1.5
10	18.2	19.1	20.1	20.0	19.4	1.9
11	20.4	20.0	19.0	21.0	20.1	2.0
12	20.6	20.0	19.5	21.4	20.4	1.9
13	18.9	21.4	20.4	20.2	20.2	2.5
14	19.1	18.7	20.6	21.1	19.9	2.4
15	20.4	18.9	19.1	20.2	19.7	1.5
16	21.3	19.7	20.7	19.6	20.3	1.7
17	20.5	18.7	20.3	20.1	19.9	1.8
18	22.2	20.7	20.9	20.0	21.0	2.2
19	19.9	20.2	20.9	20.2	20.3	1.7
20	21.5	21.0	19.0	19.9	20.4	2.5
					$\bar{\bar{X}} = 402.8/20 = 20.14$	
					$\bar{R} = 38.7/20 = 1.9$	

unless standards of evaluation are set. Without such standards, one is liable to make arbitrary judgment of the ones favorable to oneself and the graph can not be meaningful. When irrational evaluations are made, necessary action may be missed or unsuitable action may be taken in haste, thus causing confusion. This will result in inappropriate conclusions being drawn, thus lowering quality and efficiency.

Limit lines can be drawn on run charts to indicate the standards for evaluation. These lines indicate the dispersion of data on a statistical basis and indicate if an abnormal situation occurs in production. If we add limit lines to Figure 7.13, the graph in Figure 7.14 is obtained which is called a control chart.

The upper control limit corresponds to $+3s$. The lower control limit is equivalent to $-3s$.

The purpose of drawing a control chart is to detect any changes in the process, signaled by any abnormal points on the graph from which the data has been collected. Therefore

each point on the graph must correctly indicate from which process the data were drawn.

For example, in making control charts the daily data are averaged out in order to obtain an average value for that day. Each of these values then becomes a point on the control chart which represents the characteristics of that given day. Or, data may be taken on a lot by lot basis. In this case, data must be collected in such a way that the point represents the given lot.

The points on a control chart represent arbitrary divisions in the manufacturing process. The data broken down into these divisions are referred to as subgroups. In Figure 7.13, the four measurements made in one hour constitute one subgroup. In other words, the production process is divided into units of one hour, and daily production (the 20 hour period) has been presented by points on a control chart. We can now determine whether the process is in a controlled state or not.

Control charts for $\bar{X}$ and R supply basis

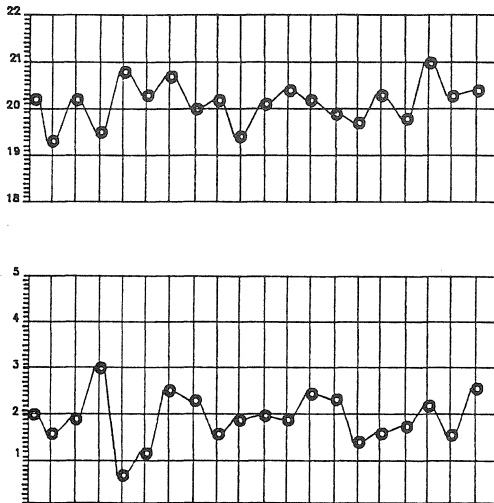


FIGURE 7.13. Run chart.

for judgment on a major question of practical importance. This question might be phrased in different ways, such as, “Does this graph indicate a stable pattern of variation?” or “Is this variation the result of a specific cause that must be corrected?”

Any rule that might be established for providing a definite “yes” or “no” answer to these questions is bound to give the wrong answer part of the time. The decision where to draw the line between a “yes” and a “no” answer must be based on the expected action to be taken if each answer is given.

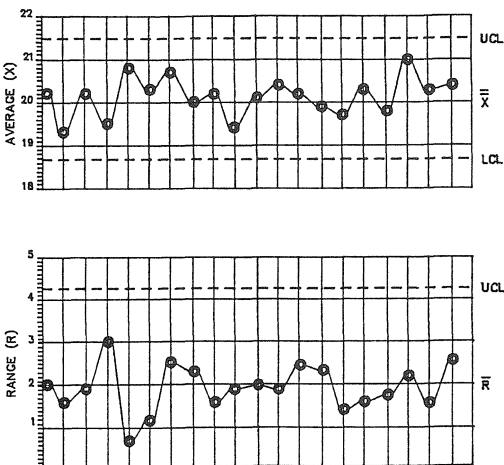


FIGURE 7.14. Control chart with limit lines.

In quality control in manufacturing, the answer, “No, this is not normal variation”, leads to a hunt for an assignable cause of variation in an attempt to remove it, if possible. The answer, “Yes, this is normal variation”, leads to leaving the process alone, making no effort to hunt for causes of variation. The rule for establishing the control limits that will determine the “yes” or “no” answer in any case should strike an economic balance between the cost due to two kinds of errors—the error of hunting for trouble when it is absent, and the error of leaving the process alone when an error really is present.

Any rule for establishing control limits for use in manufacturing should be a practical one based on this point of economic balance. It is common practice to use the $3s$ limits. Experience indicates that in most cases $3s$ limits do actually strike a satisfactory economic balance between the two questions.

Measured quality of manufactured product is always subject to a certain amount of variation as a result of chance. Some stable system of chance causes is inherent in any particular scheme of production and inspection. Variation within the stable pattern is inevitable. The results of a variation outside this stable pattern may be discovered and corrected. The power of the control chart lies in its ability to separate out these assignable causes of quality variation. This makes the diagnosis and correction of many production troubles possible, and often brings substantial improvements in product quality and reduction in rejects and rework. Moreover, by identifying certain of the quality variations as inevitable chance variations, the control chart tells when to leave a process alone, and thus prevents unnecessarily frequent adjustments that tend to increase the variability of the process rather than to decrease it.

Control charts provide more information than mere data plotted in a chronological sequence to indicate how the influences of various factors change over a period of time. Assume that the process analysis has been made and that a control state has been achieved. Standardization of working methods is necessary to obtain this state. Control

charts with control limit lines enable us to see if the standardization applied was correct, and if it is being maintained. If so, then all points in the chart thereafter should be within the control limit lines. If points appear on the control chart outside these limits, then some change must have occurred on the assembly or manufacturing line. The cause must be investigated and the proper action taken. This use of charts is called process control. By setting standards of evaluation to define a range of acceptability, we can observe abnormal or unacceptable values, investigate the probable cause and initiate action for process correction.

Making an $\bar{X}$ - R Control Chart

An $\bar{X}$ - R control chart is one that shows both the mean value, $\bar{X}$, and the range, R . This is the most common type of control chart using continuous measurements. The $\bar{X}$ portion of the chart mainly shows any changes in the mean values of the process, while the R portion shows any changes in the dispersion process.

This chart is particularly useful because it shows changes in mean value and dispersion of the process at the same time, making it a very effective method for checking abnormalities in the process. The steps to make an $\bar{X}$ - R control chart are as follows:

- (1) Collect the data
- (2) Divide the data into subgroups. The data should be divided into subgroups in keeping with the following conditions.
 - a. The data obtained under the same technical conditions should form a subgroup
 - b. A subgroup should not include data for a different lot or of a different nature.
- (3) Record the data in such a manner that it allows for easy calculation of the mean and the range for each subgroup.
- (4) Find the mean value, $\bar{X}$, for each subgroup.
- (5) Find the range, R .
- (6) Find the overall mean, $\bar{\bar{X}}$.
- (7) Compute the average value of the range, $\bar{R}$.
- (8) Compute the control limit lines ($\pm 3s$). The formulas for $3s$ control limits on charts for $\bar{X}$ are given below.

Central Line CL = $\bar{\bar{X}}$
 Upper control limit UCL = $\bar{\bar{X}} + A_2\bar{R}$
 Lower control limit LCL = $\bar{\bar{X}} - A_2\bar{R}$

The equations for the upper and lower control limits for the Range are given below.

Central Line CL = $\bar{R}$
 Upper control limit UCL = $D_4\bar{R}$
 Lower control limit LCL = $D_3\bar{R}$

The constants A_2 , D_3 and D_4 are derived from statistics, the mechanics of which are beyond the scope of this book. Table 7.7 gives these constants for various subgroups. The use of the above equations

TABLE 7.7. Constants of A_2 , D_3 and D_4 .

Subgroup Size (n)	A_2	D_3	D_4
2	1.880	0.0	3.267
3	1.023	0.0	2.574
4	0.729	0.0	2.282
5	0.577	0.0	2.114
6	0.483	0.0	2.004
7	0.419	0.076	1.924
8	0.373	0.136	1.864
9	0.337	0.184	1.816
10	0.308	0.223	1.777
11	0.285	0.256	1.744
12	0.266	0.283	1.717
13	0.249	0.307	1.693
14	0.235	0.328	1.672
15	0.223	0.347	1.653
16	0.212	0.363	1.637
17	0.203	0.378	1.622
18	0.194	0.391	1.608
19	0.187	0.403	1.597
20	0.180	0.415	1.585
21	0.173	0.425	1.575
22	0.167	0.434	1.566
23	0.162	0.443	1.557
24	0.157	0.451	1.548
25	0.153	0.459	1.541

provides a quick and easy method of calculation of the $3s$ control limits.

- (9) Construct the control chart. Draw in the control lines and label them with their appropriate numerical values. The central line is a solid line and the limit lines are dotted lines.
- (10) Plot the $\bar{X}$ and R values as computed for each subgroup.

Reading and Using Control Charts

The purpose of making a control chart is to determine, on the basis of the movements of the points, what kind of changes have taken place in the production process. Therefore, to use the control chart effectively, we have to set the criteria for evaluating what we consider an abnormality. When a production process is in a controlled state, it means that:

- (1) All points lie within the control limits
- (2) The point grouping does not assume a particular form

We would therefore know that an abnormality has developed if:

- (1) Some points are outside the control limits (including points on the limits) which means that the process is not under control.
- (2) The points assume some sort of particular form even though they are all within the control limits. Four terms describe the patterns that might appear on a control chart.
 - When several points line up consecutively on one side of the center line, this is called a *run*. The number of points in that run is called the *length of run*. In evaluating runs, if the run has a length of seven points, we conclude that there is an abnormality in the process. Even with a run of less than 6, if 10 out of 11 points or 12 out of 14 points lie on one side, we consider that there is an abnormality in the production process.

- If there is a continued rise or fall in a series of points, it is considered a *trend*. In evaluating trends, we consider that if 7 consecutive points continue to rise or fall, there is an abnormality. Often, however, the points will extend beyond the control limits before reaching 7.
- If the points show the same pattern of change (for example, rise or fall) over equal intervals, then there is *periodicity*. Unfortunately, when it comes to evaluating periodicity there is no simple method as with runs and trends.
- When the points on the control chart stick close to the central line or to the control limit line, it is called *hugging of the control line*. Often, in this situation, a different type of data or data from different factors have been mixed into the subgroup. It is therefore necessary to change the subgrouping, reassemble the data, and redraw the control chart.

Some of the patterns described above are illustrated in Figure 7.15. A sample control chart showing points grouped near the control limits is also included.

When points fall outside the control limits we say that the process is “out of control.” This is equivalent to saying that factors outside normal variation are present. With $3s$ limits, we can make the statement with considerable confidence. In contrast to this, when all points fall inside the control limits we can not say with the same assurance, “No assignable causes of variation are present.” No statistical test can give us this positive assurance. When we say, “This process is in control,” the statement really means, “For practical purposes, it pays to act as if no assignable causes of variation are present.”

7.5.14 Process Capability

Any process in total quality management must be repeatable and consistent in addition to being under control. This is characterized

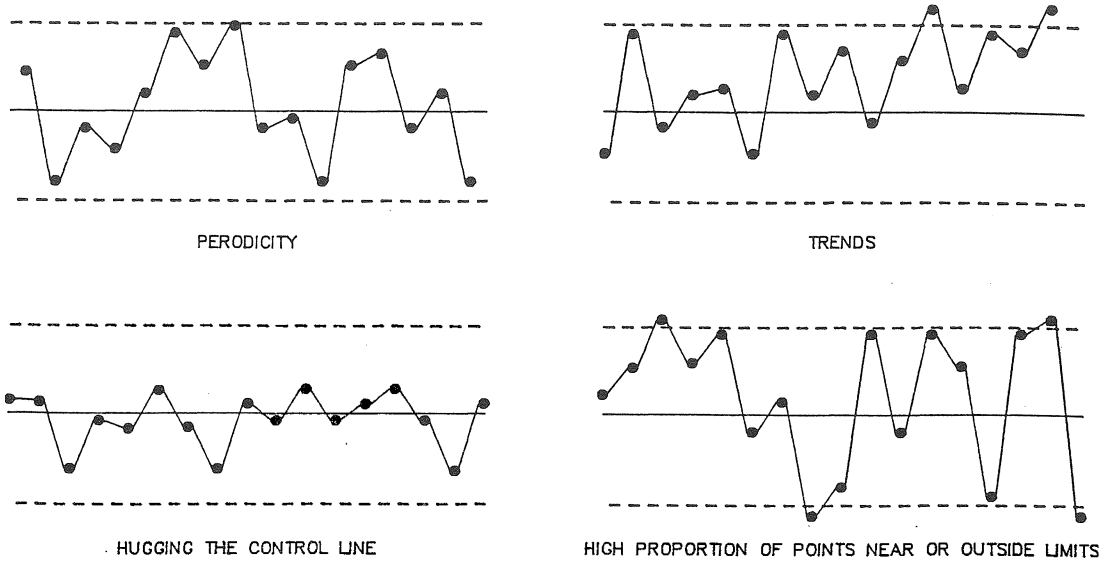


FIGURE 7.15. Patterns indicative of abnormal situations that might appear on a control chart.

by process capability which is indicated by the process capability index, C_p :

$$C_p = (USL - LSL)/6s \quad (7.9)$$

where

USL = upper specification limit

LSL = lower specification limit

s = standard deviation of the process

If C_p is equal to or greater than 1.0, then it is concluded that the process is capable of operating within the process tolerance ($6s$) or within specifications (Figure 7.16). If C_p is less than 1.0, then the process variation exceeds process tolerance.

The process capability index C_p , does not show how well the process average, $\bar{X}$, is

centered to the target value. Therefore, in practice, C_{pk} is used to predict process capability:

$$C_{pk} = \min \{C_{pl}, C_{pu}\} \quad (7.10)$$

$$C_{pu} = (USL - \bar{X})/3s \quad (7.11)$$

$$C_{pl} = (\bar{X} - LSL)/3s \quad (7.12)$$

where

C_{pu} = upper half capability index

C_{pl} = lower half capability index

$\bar{X}$ = process average (target value).

Although it is arbitrary, in general, if C_{pk} is equal to or greater than 1.33, then the process is considered as "capable" [20, 21, 22].

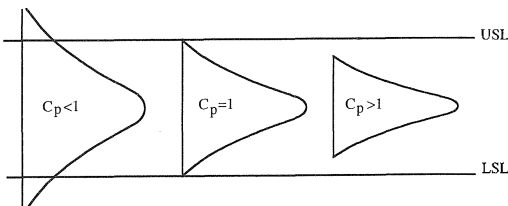


FIGURE 7.16. Process capability index in relation to USL and LSL.

7.6 Regulations and Standards

7.6.1 ISO 9000 Quality Standards

ISO 9000 is a series of standards prepared by the International Organization for Standardization (ISO) headquartered in Geneva, Switzerland. ISO formed a technical committee on quality (TC 176) in 1980 which

authored five standards that were released to the world in 1987: ISO 9000-9004. ISO 9000 developed from the British Standards, BS 5750. The European equivalent to ISO is the EN 29000 series and the US equivalent is ANSI/ASQC Q90 series.

The ISO 9000 series provides a set of good practice procedures in order to gain consistency in production, design or provision of services. It is a set of international standards designed to be used for establishing and maintaining quality management and systems for company use or to satisfy outside contracts. It provides standards for registering quality systems and is accepted worldwide. Numerous standardization organizations have adopted the ISO 9000 series bringing most of the world under a universal standardization system.

The ISO 9000 series is not product specific, but instead is a generic quality management system model that supplements specifications. ISO may be used by both larger and smaller companies as a tool to help insure quality throughout the organization.

ISO 9000 series is reviewed and revised on a periodic basis to insure that it is updated. The ISO Technical Committee TC 176 works on a five year cycle extending the scope and detail for the series.

Documentation is at the center of the ISO 9000 series. Management must document all changes and decisions in order to be consistent. The ISO 9000 series operates under the assumption that if all personnel were suddenly replaced, the new people, properly trained, could use the documentation to continue making the products or providing the services as before [23].

Contents of ISO 9000 Series

The ISO series of standards consists of five standards (US equivalents are shown in parenthesis):

- ISO 9000 (ANSI/ASQC Q90)
- ISO 9001 (ANSI/ASQC Q91)
- ISO 9002 (ANSI/ASQC Q92)

- ISO 9003 (ANSI/ASQC Q93)
- ISO 9004 (ANSI/ASQC Q94)

ISO 9000, the first standard in the series, is the roadmap for choosing which contractual standard (9001, 9002 or 9003) is best suited for which type of business organization. It also provides key definitions for the use of the standards. Table 7.8 lists the specifications for 9001-9003.

ISO 9001 specifies a quality system model for a contract between two parties that requires the demonstration of a supplier's capability to design, produce, install and service the product. This certification is the highest an organization can acquire.

ISO 9002 is specifically designed for process and manufacturing business. ISO 9002 is distinguished from ISO 9001 due to the fact that it does not cover service and development and/or design of a product. ISO 9002 covers the remaining of the 9001 elements.

ISO 9003 deals mainly with documentation, detection and controlling non-conforming materials.

ISO 9004 addresses the elements to be used by an organization in order to develop and implement a quality management system. Selection of the appropriate elements contained in ISO and the extent to which these elements are adopted and applied by an organization depends upon the factors such as markets being served, nature of products, production processes and customer needs.

An organization using total quality management uses all resources to control processes and services that affect the quality of the end product. It is important that management systems be documented when variation occurs in order to analyze the cause and implement a possible solution to correct the variation.

Procedure for ISO Certification

A company that wants to register for ISO 9000 must be certified by an accredited third party registrar agency. A national accreditation body gives accreditation to each registrar. In the US, the "American National

TABLE 7.8. Specifications of ISO 9001–9003 Standards.

	ISO 9001	ISO 9002	ISO 9003
1. Management responsibility	x	x	x
2. Quality system	x	x	x
3. Document control	x	x	x
4. Product identification and traceability	x	x	x
5. Inspection and testing	x	x	x
6. Inspection, measuring and test equipment	x	x	x
7. Inspection and test status	x	x	x
8. Control of nonconforming product	x	x	x
9. Handling, storage, packaging and delivery	x	x	x
10. Quality records	x	x	x
11. Training	x	x	x
12. Statistical techniques	x	x	x
13. Contract review	x	x	
14. Process control	x	x	
15. Purchasing	x	x	
16. Purchaser supplied product	x	x	
17. Corrective action	x	x	
18. Internal quality audits	x	x	
19. Design control	x		
20. Servicing	x		

Accreditation Program for Registrars of Quality Systems“ is established by the American National Standards Institute (ANSI) and the Registrar Accreditation Board (RAB) for accrediting registrars.

The commitment from the top management of an organization is crucial for the successful implementation of any ISO quality system. The main steps to ISO certification are as follows:

- comparison of existing quality procedures against the requirements of ISO 9001-9003 standards
- determination of the corrective action needed to conform with ISO 9000
- preparation of a quality assurance program
- preparation of definitions and documentation of new procedures and implement them
- preparation of a quality manual
- initial audit by the registrar. Take corrective measures based on findings by the registrar's audit team

- final visit by the registrar and official registration

Benefits of ISO Certification

The ISO certification process in the US is relatively new and the long term benefits of ISO certification is yet to be judged. However, from the early results, it is reported that the firms employing ISO quality systems experienced a decrease in seconds, as well as in returns. These same firms reported a marked decrease in rejects of raw materials from suppliers due to the strict regulations that suppliers must meet before doing business with an ISO registered firm. Companies that implemented ISO system reported that their total costs went down with regard to maintaining quality [24].

Although ISO does not guarantee quality itself, it can help a company keep up with its quality. It gives guidelines for keeping quality constantly high and improving the processes that are essential for the desired

quality standards. When management becomes responsible for quality, which is required by ISO, then everyone in the organization is usually made aware of the significance of the quality for their job. ISO quality system can also become a good marketing tool for a company. Many customers look at ISO certified companies as having already taken care of quality.

Reduction of the number of suppliers to a business reduces the chance for variation in raw materials which causes problems in mass-production industries. The ability to use just one supplier is feasible due to the ISO certification since the buyer will know the rigorous standards and auditing that are encompassed in the ISO series. This is not the case however with most companies since the reliance on one supplier is rather risky. More than one certified supplier may be used by a company, yet one may be used more than the other. ISO certification, for suppliers and production organizations alike, may lead to the competitive advantages in markets and the improved productivity which is vital for international survival in today's economic market.

7.6.2 ISO 14000 Environmental Standards

ISO 14000 is an international standard that is developed for Environmental Management System (EMS). What ISO 9000 is to quality management is what ISO 14000 is to environmental management. Therefore, there are some similarities between ISO 9000 and 14000 in concept. In fact, some of the elements of both systems, such as document control and statistical methods, are the same. ISO 9000 and 14000 can exist simultaneously to form one uniform management system for quality and the environment. ISO 14000 has the potential to apply to every manufacturing company. It is supposed to help any company in any country to meet the goal of "sustainable development" and environmental friendliness [25].

As in the case of ISO 9000, ISO 14000 is

a series of standards. The family of ISO 14000 standards are listed in Table 7.9.

The members of the ISO approved ISO 14001 and 14004 as official international standards in the summer of 1996. ISO 14001 is the registration standard for an Environmental Management System (EMS) and ISO 14004 is the guideline standard for more guidance on implementation. These two new standards have also been approved by the European Union as a European standard. The official approval of the complete ISO 14000 family of standards is still a couple of years away. However, many countries already adopted the complete series. ISO 14000 was adopted by the US as a national standard in early 1996. ISO Technical Committee TC 207 is responsible for the development and revision of the ISO 14000 standards.

Implementation of ISO 14000

The following steps are useful for the implementation process of ISO 14000.

- Senior management must support the implementation of the system.
- Identify the external environmental regulations to which your company must comply
- Internal audit of the environmental system
- Identify the environmental regulations and issues
- Develop and write a strategic environmental plan
- Start implementing the strategic plan
- Continually monitor the progress

ISO 14000 can be used as a powerful marketing tool since environmental friendliness is popular with individuals, companies, state and federal agencies. However, it should be noted that ISO 14000 does not guarantee an environmentally friendly company as ISO 9000 does not guarantee quality. Moreover, ISO 14000 does not automatically provide compliance with the local, state or federal regulations.

TABLE 7.9. ISO 14000 Series of Standards.

ISO 14001	Environmental Management Systems—Specification with guidance for use
ISO 14004	Environmental Management Systems—General guidelines on principles, systems and supporting techniques
ISO 14010	Guidelines for Environmental Auditing—General principles on environmental auditing
ISO 14011/1	Guidelines for Environmental Auditing—Audit Procedures, Auditing of Environmental Management Systems
ISO 14011/2	Guidelines for Environmental Auditing—Compliance audits
ISO 14011/3	Guidelines for Environmental Auditing—Statement audits
ISO 14012	Guidelines for Environmental Auditing—Qualification criteria for environmental auditors
ISO 14013	Guidelines for Environmental Auditing—Management of environmental audit programs
ISO 14014	Guidelines for Initial Reviews
ISO 14015	Guidelines for Environmental Site Assessments
ISO 14020	Principles of Environmental Labeling
ISO 14021	Environmental Labeling—Terms and definitions of self-declaration environmental claims
ISO 14022	Symbols of Environmental Labeling
ISO 14023	Testing and Verification of Environmental Labeling
ISO 14024	Environmental Labeling—Guiding principles, practices and certification procedures for multiple criteria
ISO 14030	Environmental Performance Evaluation
ISO 14040	Principles and Guidelines of Life Cycle Assessment
ISO 14041	Life Cycle Assessment—Life cycle inventory analysis
ISO 14042	Life Cycle Assessment—Impact Assessment
ISO 14043	Life Cycle Assessment—Evaluation and interpretation
ISO 14050	Terms and Definitions

7.6.3 Associations for Standards, Regulations and Specifications

—American National Standards Institute
(ANSI)

11 West 42nd Str., 13th Floor
New York, NY 10036
USA

Phone: (212) 642-4900
Fax: (212) 302-1286
<http://www.ansi.org>

—American Society for Quality Control
(ASQC)

P.O. Box 3005
Milwaukee, WI 53201
USA

Phone: 800-248-1946
<http://www.asqc.org>

—American Society of Testing and
Materials (ASTM)

100 Barr Harbor Drive
West Conshohocken, PA 19428-2959
USA

Phone: (610) 832-9585
Fax: (610) 832-9555
<http://www.astm.org>

—American Association of Textile
Chemists and Colorists (AATCC)
P.O. Box 12215
Research Triangle Park, NC 27709
USA

—Environmental Protection Agency (EPA)
National Headquarters
401 M-Street, SW
Washington, DC 20460
USA
Phone: (202) 260-5917
Fax: (202) 260-3923
<http://www.epa.gov>

—Industrial Fabrics Association
International (IFAI)
345 Cedar St., Suite 800
St. Paul, MN 55101-1088
USA

Phone: (612) 222-2508
Fax: (612) 222-8215

—International Standards Organization
(ISO)

Rue de Varembe 1
CH-1211 Geneva 20
Switzerland
Phone: 41 22 749 0111
Fax: 41 22 733 3430
<http://www.iso.ch>

—National Institute of Standards and
Technology (NIST)
US Department of Commerce
Administration Building, Room A629
Gaithersburg, MD 20899
USA
Phone: (301) 975-5923
<http://www.nist.gov>

—National Standards Association
1200 Quince Orchard Boulevard
Gaithersburg, MD 20878
USA

—Occupational Safety and Health
Administration (OSHA)
401 M Street SW
Washington, DC 20460
USA
Phone: (202) 219-4667
<http://www.osha.gov>

• Region IV (Alabama, Florida, Georgia,
Kentucky, Mississippi, North Carolina,
South Carolina, Tennessee)
1375 Peachtree Street, NE, Suite 587
Atlanta, GA 30367
USA
Phone: (404) 347-3573

—Registrar Accreditation Board (RAB)
611 East Wisconsin Avenue
P.O. Box 3005
Milwaukee, WI 53201-3005

USA
Phone: (414) 272-8575
Fax: (414) 765-8661

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7.8 Review Questions

- 1 Explain what you understand from the term *TQM*.
- 2 Can you make your own definition of quality?
- 3 Does ISO guarantee quality products and services? Explain why.
- 4 Is there a relation between "consistency" and quality in products and services. Explain.
- 5 Explain the core concepts and basic elements of TQM.
- 6 The following data indicate the air permeabilities of a single layer forming fabric type and its variations.

605, 680, 675, 550, 590, 580, 740, 665, 650, 630, 625, 675, 660, 560, 500, 545, 580, 660, 700, 725, 605, 695, 665, 650, 680, 550, 600, 610, 590, 595.

Calculate:

- a. mean
 - b. mode
 - c. median
 - d. range
 - e. standard deviation
 - f. coefficient of variation (CV%).
- 7 Establish a cause-and-effect diagram for seam quality in dryer fabrics.

- 8 What are the characteristics of a normal curve?
- 9 Among the product quality, safety and environment, which one should have the first priority in a manufacturing environment? Which is second? Explain your reasons.
- 10 Summarize the major steps in ISO qualification process.